Libera Università Maria SS. Assunta

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MEASURING DIGITAL PERFORMANCE IN THE SELF-ADHESIVE LABEL INDUSTRY

The Auroflex SRL case

Relatore
Ch.ma Prof.ssa Francesca Costanza

Correlatore
Ch.ma Prof.ssa Letizia Lo Presti

Candidato
Riccardo Randazzo
Matr.P5360/401

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ABSTRACT

This thesis analyzes performance measurement systems in the digital transformation of the self-adhesive label sector through the case of Auroflex S.r.l., an Italian manufacturing company with significant digital investments. This research investigates how traditional B2B manufacturers can reconcile traditional financial indicators with emerging digital indicators within a performance architecture. The importance of this study derives from the need to reduce the misalignment between traditional management control and digital performance, particularly critical for manufacturing SMEs in niche sectors like self-adhesive labels, characterized by high technological intensity but limited academic attention. The study relies on an empirical case study with triangulated empirical design: financial statement analysis, digital investment tracking, customer satisfaction survey, and e-commerce analytics, despite limitations due to absence of complete transactional data. Findings indicate that Auroflex presents a significant digital maturity gap, while maintaining financial solidity. Through a Digital Balanced Scorecard adapted to digital manufacturing, the thesis develops an integrated dashboard of operational and digital KPIs across the four classic perspectives (financial, customer, internal processes, learning and growth), enriched by transversal digital variables. The theoretical contribution enriches literature on performance measurement in niche B2B contexts undergoing digitalization, demonstrating BSC adaptability while maintaining its four-perspective structure. Managerially, it provides Auroflex a systemic vision mapping connections between financial and non-financial variables, facilitating organizational learning with transferable implications for manufacturing SMEs in similar contexts.

INDEX

INTRODUCTION 7

CHAPTER 1 17

Background and goals of the thesis 17

1.1 Digital Transformation and Performance Measurement Systems 17

1.1.1 Implications for the design of KPIs 20

1.1.2 Performance Measurement Systems: from traditional control to data-driven control 21

1.1.3 Performance measurement tools: state of the art and application limits 22

1.2 Self adhesive - label landscape in Italy 24

1.3 Reasearch Aim and Questions 27

1.4 Thesis Structure 30

CHAPTER 2 34

Literature review 34

2.1 Performance Measurements Systems 34

2.1.1 Definition and Strategic Role of PMS 34

2.1.2 The Foundational studies. 35

2.1.3 PMS Maturity and Effectiveness 37

2.1.4 Performance Measurement Systems in the Digital Era 38

2.1.5 Implications for the Self-Adhesive Label Sector and the Auroflex Case 39

2.2 Economic-Financial Metrics and Business Performance 41

2.2.1 Foundations of Economic-Financial Measurement 42

2.2.2 Fundamental Financial Metrics for B2B Manufacturing Companies 43

2.2.3 A second group of metrics pertains to margins and operational profitability. 44

2.2.4 A third and final group of metrics concerns operational efficiency. 45

2.2.5 DuPont Analysis as a Performance Decomposition Framework ... 45

2.2.6 Supply Chain Metrics and Impact on Economic Performance 46

2.2.7 EBITDA and Operating Margins as Efficiency Indicators in Manufacturing 48

2.2.8 The Gap Between Traditional Financial Metrics and Digital Transformation 49

2.3 Non-Financial Digital Metrics in B2B Manufacturing 51
2.3.1 Digital Servitization and Relational Value Propositions 52
2.3.2 Digital KPIs for B2B Manufacturing E-commerce 53
2.3.3 Specific Metrics for the Self-Adhesive Label Sector 54
2.3.4 Digital Customer Experience and B2B Touchpoints 55
2.3.5 Integration into the Performance Measurement System 56
2.4 Adapting the Balanced Scorecard to digital B2B manufacturing 58
2.4.1 Financial perspective: beyond traditional economic and financial indicators 60
2.4.2 Customer perspective: digital experience and relationship performance in B2B 61
2.4.3 Internal processes perspective: digital operational efficiency and automation 62
2.4.4 Learning & growth perspective: digital competencies and data governance 64
2.4.5 Digital and non-financial KPIs within the Balanced Scorecard framework 66
2.4.6 Implications for SMEs and self-adhesive label converters 68
2.4.7 Towards a Digital Balanced Scorecard for Auroflex Srl 70
CHAPTER 3 75
Methodology 75
3.1 Research Design 75
3.2 Criteria for the choice of the company under investigation 84
3.3 Data-collection protocol and analysis techniques 93
3.3.1 Calculation of financial KPIs 96
3.3.2 Digital investment data 97
3.3.3 Customer satisfaction survey 98
3.3.4 E-commerce platform analytics: absence of transactional data .. 102
3.3.5 Implications for Digital Balanced Scorecard 103
3.4 Reliability, Validity, and Ethics 123
CHAPTER 4 128
Findings 128
4.1 Company Profiles 128
4.2 History and governance 129
4.3 Product portfolio & market positioning (self-adhesive labelS) 130

4.4 Digital-maturity gap (website, CRM, analytics) 131
4.5 Financial performance 133
4.6 Snapshot implementation roadmap 141
4.7 Operational & digital KPI dashboard 142
4.8 Customer-behaviour insights 145
4.9 Digital balanced scorecard 147
DISCUSSION AND CONCLUSIONS 152

BIBLIOGRAPHY 161
WEBLIOGRAPHY 166

LIST OF TABLES

1. Revenue costs and equity……133
2. Financial variations……..134
3. ROI……...134
4. ROS……..135
5. EBITDA……...135
6. Working capital management indicators……..139

LIST OF FIGURES

1. Main tools……. …23
2. Top 10 Key account…… :… 102
3. Monthly production volumes……...107
4. Job level time breakdown……...108
5. Key findings……115
6. Tool adoption and usage intensity……..115
7. Self-assessed digital competencies…….116
8. BSC…….151

INTRODUCTION

Background and Motivation

Digital transformation has emerged as a defining strategic imperative for manufacturing enterprises in the twenty-first century, fundamentally reshaping competitive dynamics, organizational capabilities, and performance measurement paradigms across industrial sectors. The integration of advanced digital technologies - including artificial intelligence-based configurators, automated production systems, e-commerce platforms, and real-time analytics - into traditionally analogue business processes is reconfiguring how firms create, deliver, and capture value in increasingly complex B2B markets. This transformation extends beyond mere technological adoption; it encompasses profound changes in business models, customer relationships, operational processes, and strategic management practices.

Against this backdrop of industrial digitalization, performance measurement systems play a central role for navigating the uncertainties and opportunities inherent in digital transformation journeys. Traditional performance measurement frameworks, historically anchored in financial indicators such as ROI, ROS, and EBITDA, struggle to capture the multifaceted impacts of organizational performance. Digital transformation requires more adaptive measurement systems capable of integrating classic financial metrics with emerging digital and operational performance indicators – such as website traffic, conversion rates, customer lifetime value, digital engagement metrics, and operational efficiency measures derived from Manufacturing Execution Systems (MES). The ability to systematically track and interpret performance across these diverse dimensions is progressively becoming a key factor for organizations seeking to understand whether their digital investments are generating tangible economic returns and sustainable competitive advantages.

The self-adhesive label industry represents a particularly relevant yet significantly underexplored context for research on digital transformation and performance measurement. This sector occupies a critical position within broader packaging value chains, playing essential roles in branding, product differentiation, regulatory

compliance, and consumer experience. Self-adhesive label manufacturers combine traditional printing craftsmanship with increasingly sophisticated digital technologies, creating a distinctive operational environment characterized by high product mix variability, customization demands, and rapid turnaround requirements. Recent industry analyses indicate substantial growth potential for the European self-adhesive label market, projected to expand from USD 3.92 billion in 2025 to USD 13.09 billion by 2033, driven by regulatory mandates regarding packaging traceability, anti-counterfeiting requirements, demand for digitally printed labels with high customization capabilities, and increasing adoption of sustainable materials.

Although the sector holds clear strategic relevance , the self-adhesive label sector - particularly small and medium-sized enterprises (SMEs) operating within it - remains relatively underexplored in academic literature concerning digital transformation and performance measurement. Much of the existing literature tend to prioritize large-scale firms in high-technology sectors such as for example automotive, aerospace, and electronics, often neglecting the distinctive characteristics and operational dynamics of SMEs in traditional manufacturing contexts. In addition, the limited research addressing digital transformation in the label industry tends to adopt generic digital maturity models replicated from other sectors without sufficient customization to the specific competitive requirements and technological configurations characteristic of label converting operations.

For Italian self-adhesive label manufacturers, these challenges acquire urgency. Italy's label industry comprises a highly fragmented landscape dominated by small, often family-owned enterprises concentrated in Emilia-Romagna, Veneto, and Lombardy. These firms excel in artisanal production and customization responsiveness but confront a dual challenge: technological evolution requiring investment in digital printing automation and production management systems, alongside imperatives to control increasingly digital-oriented sales and communication channels. Research in Sinergie (Italian Journal of Management, 2020) demonstrates that Italian SMEs embracing digitalization exhibit enhanced internationalization propensities and profitability growth through processing time reductions and commercial process optimization. However, sector analyses indicate

many label converters adopt digital technologies fragmentarily, implementing digital printing without integrating complementary management systems with robust performance analysis capabilities, substantially limiting the strategic potential of digital investments.

European regulation adds an additional layer of pressure. EU Regulation 2024/825 imposes demanding packaging requirements for recyclability thresholds and labeling standards, requiring label converters to guarantee compliant materials and implement digital traceability systems capable of tracking product origins and life cycles. Studies by CONAI in collaboration with Bocconi University (2022) document how digitization of ESG reporting processes improves regulatory compliance and reduces waste management costs. For label manufacturers, this implies the development of integrated information systems and ESG performance analytics – unevenly implemented in most Italian SMEs' strategies.

Research Gap and Contribution

Despite the acknowledged importance of performance measurement for digitally transforming manufacturing SMEs, current literature still lacks clarity regarding how such organizations should design and utilize measurement systems that link digital initiatives to financial and operational result. Existing scholarship on Performance Measurement Systems (PMS) in manufacturing predominantly addresses large enterprises in high-technology sectors, providing limited guidance for resource-constrained SMEs operating in traditional industries undergoing digital transitions. Moreover, while substantial literature exists separately on performance measurement frameworks (including multidimensional tools such as the Balanced Scorecard) and on digitalization processes in manufacturing, these two research streams remain insufficiently integrated, particularly concerning their application to niche B2B sectors characterized by high customization and relational complexity.

In the self-adhesive label industry, academic research addressing digital maturity, performance measurement, and strategic management remains limited. The sector's organizational and production features - including high-mix/low-volume production, intensive customer co-design processes, short product development

cycle turnover, and integration challenges between e-commerce and production lines – call for customized performance frameworks are insufficiently covered by standardized models. Beyond this, empirical guidance remains scarce concerning which KPIs are truly meaningful for measuring digital transformation impacts in this context, how to align financial metrics with digital and operational indicators, what internal resistances limit implementation of advanced measurement systems, and how SMEs can translate strategy into dashboard solutions despite resource and competency constraints.

This thesis addresses these theoretical and empirical gaps by investigating performance measurement systems in the context of digital transformation with specific focus on the B2B self-adhesive label manufacturing industry. The research contributes to scholarly literature by bridging two distinct academic stream performance measurement and management (encompassing multidimensional frameworks such as the Balanced Scorecard) and digitalization studies in manufacturing-with particular attention to SME contexts and niche industrial sectors. Theoretically, the study advances understanding of how foundational performance management tools can be adapted by permeating their original structures with variables specifically related to digital transformation processes while maintaining their fundamental architectural integrity. The prototype Digital Balanced Scorecard developed through this research generates important insights regarding synergies between digital analytical tools and traditional financial-accounting instruments for comprehensive performance monitoring in digitally evolving organizations.

From a practical standpoint, the thesis generated applied insights by developing systematic approaches for measuring and managing performance in digital-first manufacturing SMEs. Concretely, the study results in: a multidimensional KPI system; tools for assessing digital maturity gaps; structured implementation guidelines; and a compact managerial dashboard design in the form of a Digital Balanced Scorecard enabling management to visualize performance across dimension and support strategic planning. These findings are not limited to the specific empirical case, offering applicable insight for managers in similar contexts regarding the design and implementation of performance measurement systems

aligned with digital transformation efforts in manufacturing SMEs.

Research Aims and Questions

This work seeks to conduct a comprehensive investigation of performance measurements concerning digital transformation processes, focusing specifically on digital-first business models in the B2B self-adhesive label sector. This general objective is pursued through empirical investigation of a representative case study - Auroflex S.r.l., an Italian self-adhesive label manufacturer that has recently initiated digital transformation projects including opening new digital sales channels, implementing production automation and AI-based configuration tools for label design, integrating online platforms with manufacturing systems, and developing customer-facing digital touchpoints.

This general research objective is articulated into four specific sub-objectives:

1. To define what "performance" comprehensively means for a digital-first self-adhesive label manufacturer, considering both traditional manufacturing excellence dimensions and emerging digital service delivery capabilities;
2. To develop a comprehensive KPI set that effectively blends classic financial ratios-including ROI, ROS, EBITDA, and Cash Conversion Cycle-with operational and digital metrics such as website traffic, conversion rates, customer lifetime value, digital engagement indicators, and operational efficiency measures;
3. To map Auroflex's current digital maturity status, identifying critical capability gaps such as the website's beta development stage, absence of structured Customer Relationship Management (CRM) systems, and limited analytics capabilities, while simultaneously proposing quick-win improvement actions that management can implement in the short term;
4. To deliver a concise, actionable one-page performance dashboard in the form of a Digital Balanced Scorecard that integrates financial, customer, internal process, and learning and growth perspectives, enabling management to monitor comprehensive performance at a glance and support strategic decision-making processes.

To guide research execution effectively, these objectives are translated into specific research questions that structure data collection and analytical activities throughout the investigation:

RQ1: What are the Key Performance Indicators of greatest relevance for measuring the impact of digital transformation on the economic and operational performance of an enterprise operating in the Italian B2B self-adhesive label sector?

To answer this question, reviewing existing literature on digital and financial KPIs, engaging directly with company management and heads of commercial, administrative, and production functions, and examining how and when benefits emerge in terms of margins and revenue. This requires developing analytical models capable of isolating effects of digital interventions from other external and internal factors affecting corporate performance.

RQ2: How can an integrated dashboard system be designed that allows management to monitor the digital transformation evolutionary process in real time and make timely strategic decisions?

To tackle this problem, it becomes necessary to structure an information architecture able to combine data from different systems such as ERP, web analytics platforms, MES, and accounting databases, like CERM, into actionable management visualizations.

RQ3: What organizational barriers hinder the adoption of advanced performance measurement systems in self-adhesive label manufacturing SMEs, and what strategies should be adopted to overcome these obstacles?

This question focuses on implementation challenges, including organizational constraints and skill gaps needed to ensure sustainability and effectiveness of measurement systems in the medium to long term.

Overall, they shape data collection and analysis, guiding data collection and subsequent analysis employed throughout the study.

Methodological Approach and Case Study

The research employs an empirical single-case study design grounded in established literature on case study research methodology, interventionist research approaches, and performance measurement system design. The case study method proves particularly appropriate for investigating complex, contextually situated

phenomena such as digital transformation in small and medium-sized manufacturing enterprises, where understanding requires deep engagement with organizational processes, strategic decision-making dynamics, and the interplay between technological investments and performance outcomes. The OECD, through its report *The Digital Transformation of SMEs* (2021), identifies case study analysis as a methodological tool particularly effective for understanding dynamics of technology adoption and mechanisms through which digital investments translate into performance improvements, especially in resource-constrained organizations highly dependent on proximity relationships with customers.

The study focuses on Auroflex S.r.l., an Italian self-adhesive label manufacturer selected following defined methodological criteria of the research. As explained in Chapter 3, include: sectoral representativeness of the firm within the Italian self-adhesive label industry; active engagement in digital transformation processes spanning production automation and an e-commerce platform development; a transitional stage of digital maturity – developed thought not fully mature to benefit from systematic performance measurement system design; access to a balance sheet and production data spanning multiple years; and managerial openness to research involvement.

The research design incorporates multiple data sources to ensure methodological rigor and enable triangulation of findings. Primary data sources include:

- **Financial statement analysis:** Five years of balance sheet and income statement data (2020-2025) enabling calculation of economic-financial KPIs including ROI, ROS, EBITDA margins, Cash Conversion Cycle components, and profitability trend analyses.
- **Digital investment tracking:** Documentation of digital technology investments, implementation timelines, and associated cost structures.
- **Customer satisfaction surveys:** Structured questionnaires assessing service quality perceptions, digital experience evaluations, and customer relationship dynamics, providing insights into customer-facing performance dimensions.
- **E-commerce platform analytics:** Website traffic data, user behavior metrics, and digital engagement indicators, though with noted limitations

regarding absence of complete transactional data due to the platform's ongoing developmental stage.

- **Operational data:** Production efficiency metrics, order fulfillment cycle times, and manufacturing performance indicators.

The use of multiple data sources increases the robustness of the results of findings and considers confidentiality concerns linked to company information involving proprietary company information. Methodologically, the study combines evaluation of the current performance configuration with formulation of a new measurement architecture, aligned with an intervention – oriented research approach where academic analysis translates into practical tools addressing real organizational challenges.

Thesis Structure is divided into five chapters, each contributing to a different phase of the analysis and following a progression that moves beginning with theory, followed by empirical analysis, and concluding with discussion.

Chapter 1 establishes the background and objectives of the research by examining digital transformation's strategic implications for performance measurement systems in manufacturing contexts. The chapter introduces the self-adhesive label industry landscape in Italy, highlighting market dynamics, competitive challenges, and regulatory pressures shaping the sector. Section 1.1 explores in depth the phenomenon of digital transformation in B2B manufacturing, analyzing technological, demand-side, regulatory, and organizational drivers inducing SMEs to adopt innovative digital solutions while presenting main types of Performance Measurement Systems tools available-including self-service Business Intelligence platforms and web analytics systems-and their potential alongside operational limitations. Section 1.2 describes the European and Italian self-adhesive label market context, emphasizing the growing strategic role of digital innovation. Section 1.3 articulates the research aims and formulates the specific research questions guiding the investigation. Section 1.4 outlines the logical structure of the thesis, explicating the argumentative progression connecting different components of the work.

Chapter 2 provides a comprehensive literature review constituting the theoretical foundation of the research, organized into four principal sections. The first section

explores conceptual models of Performance Measurement Systems, examining foundational contributions from scholars such as Neely, Gregory, and Platts on PMS design, alongside Kaplan and Norton's seminal work on the Balanced Scorecard, and analyzing evolution toward more complex adaptive frameworks including Digital Balanced Scorecard variants. The second section examines financial and economic metrics traditionally employed for corporate performance evaluation - including ROI, ROS, EBITDA, DuPont analysis frameworks, and supply chain financial indicators - while discussing their limitations and adequacy in contexts of digital transformation. The third section focuses on non-financial and digital metrics, concentrating on indicators such as web traffic, conversion rates, customer lifetime value, Net Promoter Score, and operational efficiency indices derived from MES systems, highlighting how these function as leading indicators capable of anticipating variations in financial performance. The fourth section proposes an adapted Balanced Scorecard model tailored to digital B2B manufacturing contexts, integrating perspectives addressing digital maturity in terms of data governance, technological evolution capacity, and digital service delivery capabilities, with specific implications for SMEs in the self-adhesive label sector.

Chapter 3 outlines the research methodology, discussing the reasoning behind the research design and explaining the choice of a single case design. It explain the criteria guiding company selection, ensuring Auroflex represents an appropriate and a case providing rich empirical data for investigating performance measurement in digital transformation contexts. The chapter further describes data collection procedures and method applied for both financial KPI calculation and evaluation of digital indicators, also discussing methodological challenges, such as the absence of complete e-commerce transactional data due to ongoing development phase. Methodological rigor and ethical responsibility are explicitly considered.

Chapter 4 presents empirical findings structured around multiple analytical dimensions. The chapter begins with Auroflex's company profile, historical development trajectory, and governance structure, establishing organizational context for subsequent analyses. Product portfolio and market positioning within

the self-adhesive label sector are examined, followed by digital maturity gap analysis identifying current capability deficiencies across website functionality, CRM system absence, and analytics infrastructure. Financial performance evaluation presents calculated KPIs spanning the 2020-2025 period, revealing trends and relationships between digital investments and economic outcomes. A snapshot implementation roadmap proposes prioritized improvement initiatives addressing identified digital maturity gaps. The operational and digital KPI dashboard design integrates multiple performance dimensions into coherent visual management tools. Customer behavior insights derived from survey data illuminate service quality perceptions and digital touchpoint effectiveness. The chapter culminates in presentation of the proposed Digital Balanced Scorecard, synthesizing financial, customer, internal process, and learning and growth perspectives into a comprehensive, actionable performance measurement framework specifically adapted to Auroflex's digital transformation context.

Discussion and conclusion synthesize the main research findings and discusses managerial implications. It also acknowledges the study’s limitations and outlines possible directions for future research. The chapter offers a critical reflection on how the results contribute to a deeper understanding of performance measurement in digital transformation processes, with particular attention to SMEs operating in traditional manufacturing sectors. It further highlights insights that, while grounded in the Auroflex case, may be relevant to comparable organizational contexts.

Through this progression from theoretical foundations to empirical analysis and practical tool development, the thesis aims to provide both an academic contribution to literature and concrete managerial guidance for performance measurement in manufacturing SMEs undergoing digital transformation.

CHAPTER 1

Background and goals of the thesis

1.1 Digital Transformation and Performance Measurement Systems

Digital transformation has, throughout modern history, shed its role as a managerial aphorism to become with increasing force a true strategic variable capable of shaping the economic approach of enterprises and, in a broader perspective, even the industrial morphology itself.

Amit Zavery[1], stated: *"Think of digital transformation not so much as a technology project to be completed, but as a state of perpetual agility, always ready to evolve to meet customer needs, and you will be directed on the right path."* (Zavery, 2021)

This assumption finds its theoretical origin in studies on organizational change Research in Organizational Behavior, and on technology-enabled change by (Brynjolfsson, & Hitt 2000*)* , but it is with the advent of cloud computing, the industrial Internet of Things (IoT) and predictive analytics that the literature has begun to recognize its structural nature.

Digital transformation, in this analysis, must be considered as an evolutionary process founded on the modernization and incorporation of innovative digital technologies. It implies, for the enterprise, a redesign of assumed business models and a profound recalibration of value creation strategies. (Vial, G. 2019), (Kane, & al. 2015, 2022). .*MIT Sloan Management Review*.

In order to achieve said transformation, a path is followed that passes through two defined phases:

- **Digitization**, namely the process of converting analogue information into digital format.
- **Digitalization**, that is, the integration of digital technology into business operations.

Reaching, finally, a phase characterized by the use of Information and Communication Technology (ICT) tools, useful for identifying targets and

[1] Amit Zavery, former Vice President and Head of Google Cloud Computing Platform, today Chief Operating Officer of ServiceNow.

measuring corporate performances and arriving at a true Digital Transformation.
The current literature on the subject identifies 4 drivers:

1. **Technological drivers**: data processing tools such as Power Bi and Salesforce, business intelligence and analytics platforms that visualize and analyze corporate data providing effective and modular reports.
2. **Demand drivers**: created to study and analyze Customer behavior, having regard to an increasingly demanding and evolved consumer in need of a Customer experience centered on the emotional sphere.
3. **Regulatory and sustainability drivers**: according to the new EU regulation on packaging, (Reg.2024/825) minimum thresholds are imposed for recyclability and traceability of label & package solutions. Data capture technologies and ESG reporting platforms therefore become fundamental.
4. **Organizational drivers**: in the presence of scarcity of specialized skills, work environments capable of absorbing digital evolution and equipped with knowledge retention tools such as digital twin and AR remote assistance are necessary (Schuh et al .2023).

Therefore, corporate performance cannot disregard customer satisfaction; the "*result*" in fact must not be identified with a simple pursuit of mere profit but pertains to a more complex sphere that incorporates the aforementioned requirements in a perspective of achieving financial balance. From this emerges the concept of corporate performance and similarly of its measurement, understood as a trait d'union between intuitive measures such as "*customer's side*" and quantitative measures such as "*supplier's side*". A link that involves the ideal design of the process and its executive management aimed at planning all activities in order to improve performance.

The Performance Measurement System (PMS) is a set of metrics and processes that aim to transform strategic theory into action; the adoption of digital PMS tools has assumed a crucial role in the current period in which efficiency, sustainability and decision-making speed are fundamental competitive levers. These tools, ranging from business intelligence systems to real-time operational dashboards, allow the analysis of heterogeneous data (ERP, MES, CERM) converting them into information suitable for guiding strategic business choices.

Business intelligence (BI) platforms such as Microsoft Power Bi or Qlik Sense allow non-specialist users to build interactive reports and process more detailed "*drill down*" data and hypothetical scenarios with "*what if*" analyses.

Although in use by almost 45% of Italian companies, these tools reveal certain critical issues substantially connected to the culture of "*data-modeling*" and to the absence of formalized data governance understood as well-defined rules, roles and processes.

Having regard to a more operational level, the MES/OEE dashboards of BOBST or ENLYZE, two companies that offer digital solutions, prove extremely useful for instantly monitoring certain indicators: nevertheless, the high cost of the initial investment and the difficulty of interacting with other "*legacy*" management systems still widespread limit their diffusion to a contained 30%.

To the aforementioned tools must be added "*Web & Marketing Analytics*" such as Matomo and Google Analytics 4.0 which allow tracking user behavior and simultaneously analyzing commercial funnels from lead to sale. However, in the industrial B2B sector these latter technologies still have enormous growth margins, currently standing at 18%, probably due to the absence of dedicated roles.

Recent literature highlights some important implications for the structure of Key Performance Indicators (KPIs) in the context of digital transformation. They can *vary according to the organization and specific objectives;* Kotarba [2] in 2017 *"proposed a multi-level taxonomy of digitization indicators ranging from macroeconomic impacts to micro-corporate efficiency parameters*". In this study, the fundamental importance of aligning KPIs with the enterprise's digital strategy and of increasing measures that serve not only to project technological capabilities but also organizational performances to manage the transformation is emphasized. Some examples among the most common KPIs pertain to Customer Experience such as the "*retention rate*" and the "*net promoter score*" (NPS). If instead we focus on Revenue and Growth, we cannot fail to pay attention to the increase in online sales and revenue growth.

[2] Kotarba. is a professor associated at Warsavia University of Tecnology.

Finally, it is useful to focus on Data-driven decision Making understood as the quality of the data produced, the use of the same to make decisions and the final analysis to optimize processes.

Applying the above to the self-adhesive label sector, the digitization KPIs for process and product become fundamental, although according to what Kotarba writes *"they still prove neglected in Italian empirical studies"* among them we include the line automation index and the percentage of digital jobs on the total orders (Kotarba, 2017).

This sector, which we can consider as a strategic niche in the larger universe of packaging, has an estimated European turnover of approximately 17 billion euros and growth rates exceeding 5% annually, (FINAT RADAR 2024**)**, with increasing digital verticalization of production activities. However, technological adoption often proves fragmented; the LabelTech Observatory (Politecnico di Milano, 2025) reports that "*su un campione di 150 imprese italiane del comparto, il 62% ha installato almeno una linea di stampa digitale, ma solo il 28% integra i dati di macchina con il sistema gestionale; tale percentuale scende al 12% se si considera anche l'allineamento con i dati di canale"*.

Having regard to the scarcity of literary corpus useful for quantifying and measuring how digital maturity impacts the economic and financial performance of companies active in the self-adhesive label sector, the purpose of this thesis is to fill the cognitive gap pertaining to the link between investments in digital transformation, in terms of data and process integration, and the strengthening of corporate results.

The objective of the present research is intended to be pursued, also through the application of conceptual frameworks such as those proposed by Kotarba, to analyze the multi-level impact of digital transformation in manufacturing SMEs in the sector.

1.1.1 Implications for the design of KPIs

(Kotarba 2017) proposes a multi-level taxonomy of digitization indicators ranging from macroeconomic impacts to micro-corporate efficiency parameters. Applying this classification, in the self-adhesive label sector *the process & product*

digitalization KPIs prove crucial (e.g. line automation index, percentage of digital jobs on the total), hitherto neglected in Italian empirical studies.

In the B2B manufacturing context, DT is not configured as simple digitization of documentary flows, but rather as a transition from linear pipelines to ecosystem platforms in which product, process and customer data converge in a continuous feedback cycle (Porter & Heppelmann 2015). It is therefore appropriate to speak of "*digital continuum"* rather than a concluded design phase.

The self-adhesive label sector is facing two parallel trends: consolidation (through acquisitions of local converters by global groups) and digital verticalization.

According to the Label Tech Observatory (Politecnico di Milano, 2025) indicate that, from a sample of 150 Italian companies, 62% have installed at least one digital printing line, but only 28% integrate machine data with the management system; the percentage drops to 12% if channel data alignment (CRM, webtoprint) is considered. This suggests an adoption of "*isolated*" technologies, not yet part of an integrated Industry 4.0 architecture.

Analogous sectors – In flexible packaging and folding cartons, the penetration of end-to-end workflows is more advanced, thanks to the push from food multinationals that impose serialization and traceability standards. These sectors have already experienced the positive impact of OEE dashboards based on edge analytics (heidelberg direct 2024), demonstrating how digital maturity can translate into waste reductions exceeding 15%.

Identified gap – There is a lack of a body of literature that quantifies how, and to what extent, such maturity impacts the economic and financial performance of label converters, especially in district contexts with a high incidence of SMEs.

1.1.2 Performance Measurement Systems: from traditional control to data-driven control

Theories on management control have placed at the center the concept of *Performance Measurement System* **(**PMS**)** as a set of metrics, processes and infrastructures tasked with translating strategy into action. (Ittner & Larcker 2003).

The contribution of the Digital Balanced Scorecard – (Fabac et al. 2022) expand

the Kaplan & Norton [3] framework by formalizing the *Digital BSC*, which introduces two additional perspectives - *digital readiness* and *data & technology* -bridging the gap between digital transformation and competitive strategy. This extension proves particularly relevant for label converters, where the integration of traditional KPIs (ROS, OEE) with IoT maturity metrics and data analysis intensity still represents unexplored territory.

Conceptual evolution – From the classic Balanced Scorecard *(*Kaplan & Norton, 1992) the field has moved to live models fed by real-time big data flows **.** The literature refers to concepts of *data-driven PMS* and *dynamic capabilities of contro*, emphasizing the need to integrate operational and digital KPIs - such as click-through rate (CTR) or IoT maturity index - with financial indicators.

Enabling technologies – Business Intelligence platforms (Power BI, Tableau, Qlik) and Manufacturing Execution Systems (MES) enriched with analytics modules today allow the automation of data collection and the construction of multi-level performance dashboards. In the self-adhesive label context, the ability to track downtime, ink consumption and print quality in a single view represents a tangible competitive advantage.

Identified gap – Most research has focused on large manufacturing groups (automotive, aerospace), with scarce attention to the micro-dynamics of SMEs and to low-tech/high-mix industries such as label printing, where the variability of orders and batches requires adaptive metrics.

1.1.3 Performance measurement tools: state of the art and application limits

The comparative analysis of the main tools reveals a plurality of solutions at different technological depths, often adopted with specific and non-synergistic purposes.

[3] Kaplan and Norton are the creators of Balanced Scorecard. Kaplan is a professor at the Harvard Business School and Norton a business consultant and co-founder of the "*Balanced Scorecard Collaborative*".

Instrument Class	Key Vendors	Key Functionalities	Penetration Level Among Italian Converters	Critical Issues / Gaps
Business Intelligence (BI) Self-Service	Microsoft Power BI, Qlik Sense	ERP/MES/CRM data mashup; Job-level drill-down analysis; What-if simulations	Medium (≈45%)	Lack of data modelling skills; Unstructured data governance
MES/OEE Real-Time Dashboards	BOBST Connect, ENLYZE, Screen PressWise	Downtime monitoring; Waste tracking; Energy consumption monitoring	Low–Medium (≈30%)	High initial investment; Integration challenges with legacy ERP systems
Web & Marketing Analytics	Google Analytics 4, Matomo	Lead–quote–order funnel tracking; CTR monitoring; Cost-per-lead analysis	Very Low (≈18%)	Limited marketing culture in B2B firms; Lack of dedicated digital roles

Figure nr. 1 Main tools. Source: own elaboration.

The table highlights the fragmentation of instrumentation and the lack of horizontal integration.

In the field of B2B web analytics, (Saura et al. 2017) demonstrate that combining metrics such as Click-Through Rate (CTR), Average Session Duration and Customer Lifetime Value with financial indicators increases the predictive capacity of performance models compared to ROI alone. Connecting these KPIs to order data and print quality parameters allows obtaining an integrated dashboard capable of supporting both marketing and production decisions in the Auroflex context. This results in information silos that hinder a holistic view of performance.

Towards a research contribution: the questions still open:

1. How does the relationship between digital maturity and economic results manifest itself in Italian label converters? The few available sources offer qualitative data but lack statistically valid measures.
2. Is it possible to adapt the Balanced Scorecard by incorporating digital KPIs specific to label printing (e.g. artwork-to-print lead time, digital metric waste)? The literature suggests generic digital BSC models, but none has been empirically tested in the label industry.
3. What skills and organizational structures favor the integration of BI systems with shopfloor platforms? Studies on smart factories highlight the role of data translators; however, it is not clear how such figures can be sustainable

in SMEs.

4. How do new traceability regulations such as Digital Product Passport, reshape performance measurement? Existing research suggests the alignment between sustainability and operational performance indicators, however, common implementation standards are still missing..

Examining these questions has both theoretical and managerial relevance, because it refine the application of DT and PMS concepts to a relatively underexplored sector - and practical, since it provides Auroflex managers with structured decision-making framework to manage multiple technological initiatives such as digital printing, data analysis, CRM platforms, in a way aligned with strategic objectives.

1.2 Self adhesive - label landscape in Italy

Within the packaging supply chain, the European self-adhesive label market is characterized by strong dynamism and strategic relevance. From an analysis derived from the report (Europe Self-Label Market Outlook 2025-2030)[4] the entire sector appears driven by three main guidelines:

- The expansion of the food & beverage and pharmaceutical sectors
- The growth in demand for digitally printed labels equipped with high customization
- The increasing use of sustainable materials.

In this context, Italy – equipped with a highly fragmented industrial fabric consisting predominantly of small and medium-sized enterprises that are highly specialized – is considered among the main producers of the continent. The companies that are part of it are concentrated mainly in Emilia Romagna, Veneto and Lombardy and are often small family-run realities. They excel in the production of artisanal products capable of reporting customer requests in detail like tailor-made garments, but currently find themselves managing a dual challenge consisting of technological evolution for the automation of digital printing processes and the need to control sales and communication channels strongly oriented toward

[4] The Europe Smart Label Market is projected to grow significantly from 2025 to 2033, driven by regulatory mandates, anti-conterfeiting needs, and rising consumer demand for interactive product information.

digitization.

It can therefore be stated that the transformation underway affects the system not only technologically but embraces the entire business model.

In the study (*Sinergie, Italian Journal of Management* 2020**)**,[5] it emerges that Italian companies that have chosen digitization prove to be particularly inclined toward internationalization, receiving growth in profitability to be attributed to the reduction of processing times and an optimization of all commercial processes.

In the self-adhesive label sector, this innovation is achieved through the integration of web-to-print platforms, through the use of MES systems and systematic real-time observation of production lines as well as with the use of CRM solutions for customer portfolio management.

It must, in truth, be reported that according to a recent sector analysis by Metaprintart, the above is fully adopted only in a fragmentary manner by SMEs that often limit themselves to the use of digital printing without integrating them with adequate management systems equipped with performance analysis, thus limiting in fact the strategic potential of the investment.

The European regulatory framework regarding sustainability and packaging traceability constitutes another critical factor. EU Regulation 2024/825 on packaging lays down the law regarding minimum recyclability thresholds and imposes stringent requirements for labeling. The above implies consequences for label converters who necessarily must guarantee compliant materials and equip themselves with digital traceability systems capable of documenting the origin and life cycle of the product.

Conai, collaborating with Bocconi University (Green research center of Bocconi University 2022) provided a study on Screening the efficiency of packaging waste in Europe in which it emerges how the efficiency improvement of recovery and recycling of packaging materials among member states and the digitization of ESG reporting processes serves to improve regulatory compliance and at the same time reduce the operational costs of waste management. It appears clear that for label manufacturers this translates into the essential need to use information systems, but

[5] Sinergie, SIMA Italian Society of Management is a peer-reviewed academic publication focusing on the main trends in management studies. Printed from 1983.

also to carry out an analysis of data strictly pertaining to environmental sustainability, a topic – the latter – currently lacking in the business strategies of most Italian SMEs.

For the self-adhesive label sector, academic research on the topic of digital maturity proves scarce today. The current literature on Performance Measurement Systems in the manufacturing sector focuses mainly on segments with high technological intensity such as automotive and aerospace, in fact neglecting high-mix/low-volume industries such as labels where the standardization of operational KPIs appears decidedly complex.

The study Business *Digital Maturity in Performance*, published in (Science Direct 2022) provides an analysis of the connection between the level of digitization and economic performance on a sample of European sector companies. It was possible to demonstrate that high degrees of digitization correspond to operating margins 12-18% higher than more traditionalist competitors. It must, in truth, be said that label converters are not included in the analyzed sample and the specificity of the great variability among orders, small-scale batches and rapid delivery cycles is not explored in detail.

The manifestation of a significant knowledge gap is now real, in fact, empirical studies capable of systematically and solidly quantifying how and to what extent digital maturity impacts the economic and financial performance of self-adhesive label converters are lacking. More specifically, it is not clear whether investment in B2B e-commerce platforms constitutes a flywheel for customer development and reduction of sales costs, and likewise whether the use of CRM systems can anticipate and prevent any variations in corporate profitability. Currently, there exist no benchmarks suitable for evaluating the level of digital readiness of an enterprise compared to competitors nor Balanced Scorecard models adapted to the label sector.

The purpose of the present research is to fill this gap through the analysis of the Auroflex case study, an Italian SME that has operated for years in the sector of self-adhesive label production for the food, wine, spirits, beer and cosmetics sectors.

The company, in fact, currently finds itself in an emblematic digital transition phase. While on one hand it has already proceeded to make investments in digital

printing technologies and has underway the launch of a B2B e-commerce platform, on the other it is not yet equipped with a well-structured CRM system nor with an integrated framework suitable for measuring digital performance.

Studying the Auroflex case therefore means observing in practice and in the field the dynamics, opportunities and critical issues that characterize the digital transition of a representative enterprise of the Italian manufacturing fabric in the sector, with the intent of providing a small contribution both to the academic debate and to managerial practice through direct and replicable empirical evidence.

1.3 Reasearch Aim and Questions

For small and medium-sized manufacturing enterprises, digital transformation currently represents one of the main vectors of change. It must be understood not as mere technological transformation but rather as profound reengineering of the operational model, value creation strategies and market approach.

However, despite the growing academic attention on the subject, the cognitive gap remains broad regarding the measurement of economic, financial and operational effects with respect to such change, especially in the niche manufacturing sector characterized by strong competitive dynamics.

The sector that concerns us, therefore, of self-adhesive labels, can represent an emblematic example; in fact, while constituting a strategic sector for the packaging supply chain with a European market value exceeding 17 billion euros and consolidated annual growth rates, is in effect particularly underrepresented in the literature on "*Performance Measurement Systems*" in digital contexts.

In an article by (Naeem & Garengo, 2022)[6] the relationship between degrees of digital maturity and effectiveness of performance measurement systems is analyzed within the production processes of small European SMEs. What emerged served to demonstrate that enterprises with a high degree of integration between Industry 4.0 technologies and management control systems obtain better market response times, increase the quality of the final product and optimize operating margins. The

[6] Naeem and Garengo have conducted a study titled "*The interplay between industry 4.0 maturity of manufacturing processes and performance measurement and management in SMEs*" published in the International Journal of Productivity and Performance Management.

authors – in this analysis – introduce the concept of digital-operational alignment, namely the need to align technology investments with the structure of corporate Key Performance Indicators in order to guide and support management action. The shortcoming of this strategy remains the difficulty in identifying internal data analytics and data governance skills, leading to reduced capacity to fully exploit the potential of applied innovative technologies.

In the study by (Elsevier 2025) on the matter under analysis, the results demonstrate that there exists a statistically significant and positive correlation between the level of digital maturity and balance sheet indicators such as Return on Sales, EBITDA margin and revenue growth rate, although this relationship is mediated by certain variables such as, as previously mentioned, the presence of internal skills capable of developing Data-Driven models or the adoption of integrated Business Intelligence platforms. The study then reveals that there exists a measurable time gap between 18 and 36 months between the initial investment for digitization and the positive effects in terms of balance sheet analysis, demonstrating, in fact, the need to equip oneself with measurement systems suitable for monitoring leading and not only lagging indicators, in order to anticipate the effects of transformation before they impact economic results.

Wishing to focus attention on the self-adhesive label sector, it is necessary to report that it is experiencing a period of strong technological and competitive transformation, linked not only to digital printing and product customization but above all to the strong regulatory pressures toward sustainability and traceability. Unfortunately, as already reported, there exists no solid quantitative evidence of the relationship between investments and performance, nor have predictive models been developed for the sector capable of providing reliable estimates of the impact of specific technological interventions such as the adoption of B2B e-commerce platforms or the use of CRM.

Moreover, analyzing a methodological approach, a performance measurement framework specifically designed for the sector suitable for integrating traditional financial indicators such as ROI and EBITDA with more technological ones such as the automation index of printing lines, the percentage of jobs managed in digital mode, the conversion rate of web-to-print platforms and the customer lifetime value

of digital channels is completely lacking.

By virtue of what has been stated above, we can list the research aims to be considered:

1. To design and validate an integrated performance measurement framework adapted to the needs of the self-adhesive label segment; this tool must be constructed as an extension of the Digital Balanced Scorecard model, integrating the traditional perspectives of Kaplan and Norton with specific dimensions such as digital readiness, data maturity and digital customer experience, testing everything through the empirical case of Auroflex.
2. To quantify the impact of digital maturity on the economic and operational performance of a self-adhesive label converter through the analysis of financial, production and channel data in order to identify and define the causal relationships between technological investments, evolution of digital KPIs and variation of economic and financial indicators.
3. To define operational guidelines for the implementation of measurement systems, with particular regard to organizational constraints, necessary skills and governance processes required to ensure the sustainability and effectiveness of the system in the medium-long term.

This being stated, the first question to be asked is what are the Key Performance Indicators of greatest relevance for measuring the impact of digital transformation on the economy of an enterprise operating in the Italian B2B sector, and to provide this answer one must certainly analyze the existing literature and confront in the field with company management and heads of commercial and administrative production lines. The comparison will also be useful for understanding to what extent and with what time delay advantages can be appreciated in terms of margins and revenue. To do this, it is useful to construct a time-series analysis model capable of isolating the effect of digital interventions from other exogenous variables that influence corporate performance.

A further question concerns the method of designing an integrated dashboard system that allows management to monitor the evolutionary process in real time and make strategic decisions in time. For this purpose, an information architecture must be defined that integrates data from heterogeneous sources, such as ERP

systems, web analytics platforms, MES systems and accounting databases.

It will be useful at this point to identify the organizational barriers hindering the adoption of advanced measurement systems and what strategies should be adopted to bypass them. Internal dynamics must in fact be thoroughly analyzed in the case study with particular regard to resistance to change and the capacity to invest in new infrastructures.

Analyzing these questions contributes to understanding a relatively underexplored sector by the literature, providing empirical insights into the applicabilty of models such as the Digital Balanced Scorecard in B2B contexts with high operational complexity.

At the same time, the research will be useful for equipping the management of Auroflex and more generally self-adhesive label converters with a practical analytical tool capable of allocating technological and human resources in a manner consistent with strategic targets, anticipating the impact of operational choices on future economic results.

1.4 Thesis Structure

The present work is articulated in four chapters, each of which responds to a specific function within the overall research design, following a sequential logic that begins with the definition of the theoretical-conceptual framework and empirical validation, to conclude with the synthesis of results and the discussion of academic and managerial implications.

The adopted structure reflects the canons of qualitative research in the managerial field, with the particular case study method as the privileged investigation strategy for the analysis of complex and contextually situated phenomena, such as, precisely, digital transformation in small and medium-sized manufacturing enterprises.

The OECD, through the report The Digital Transformation of SMEs published in 2021, considers the study of company cases to be a valid methodological tool particularly effective for understanding the dynamics of technology adoption and the mechanisms by which investments in digital evolution lead to performance upgrades, especially if the analysis uses realities with limited resources lacking

specific skills and highly dependent on proximity relationships with customers.

The Introduction presents the context and rationale of the study, illustrating why digital transformation is now a strategic priority for manufacturing SMEs in the twenty-first century. It identifies the principal research gap: although performance measurement plays a crucial strategic role for digitally transforming enterprises, academic literature specifically addressing the integration of digital KPIs and traditional financial metrics in the self-adhesive label sector remains limited. This perspective clarifies the contribution and guides the t analytical and methodological sections.

Chapter 1, Background and Goals of the Thesis, fulfills the function of the general contextual framework of the research. Section 1.1 explores the phenomenon of digital transformation in B2B manufacturing contexts, examining the technological, demand, regulatory, and organizational drivers that induce SMEs to adopt innovative digital solutions. Three subsections articulate, respectively, the implications of digital transformation for KPI design (1.1.1), the evolution of Performance Measurement Systems from traditional retrospective control instruments toward data-driven management frameworks (1.1.2), and the current state of the art of available performance measurement tools including self-service Business Intelligence platforms and web analytics systems with their potential and operational limits (1.1.3). Section 1.2 describes the European and Italian self-adhesive label market context, emphasizing the growing strategic role of digital innovation. Section 1.3 articulates the research aims and formulates the specific research questions guiding the investigation. Section 1.4, the present section, outlines the logical structure of the work.

Chapter 2, Literature Review, constitutes the theoretical foundation of the research, organized into four principal sections. Section 2.1 examines the conceptual models of Performance Measurement Systems, drawing on foundational contributions from Neely, Gregory, and Platts on PMS design and from Kaplan and Norton on the Balanced Scorecard, and tracing the evolution toward more complex and adaptive frameworks. Section 2.2 examines the financial and economic metrics traditionally employed for corporate performance evaluation including ROI, ROS, EBITDA, DuPont decomposition analysis, and supply chain financial indicators discussing

their limitations and adequacy in contexts of digital transformation. Section 2.3 focuses on non-financial and digital metrics such as web traffic, conversion rates, customer lifetime value, Net Promoter Score, and operational efficiency indices derived from MES systems, highlighting how these function as leading indicators capable of anticipating variations in financial performance. Section 2.4 proposes an adapted Balanced Scorecard model tailored to the digital B2B manufacturing context, integrating financial, customer, internal process, and learning and growth perspectives across seven subsections that culminate in a Digital Balanced Scorecard specifically designed for Auroflex Srl.

Chapter 3, Methodology, describes the research design and data collection strategy. Section 3.1 justifies the adoption of an interventionist single-case study design (Yin, 2018; Jönsson & Lukka, 2006) as the most appropriate strategy for the analysis of complex and contextually situated phenomena such as digital transformation in manufacturing SMEs. Section 3.2 specifies the criteria for selecting Auroflex Srl as the unit of analysis, characterizing it as an information-rich case at a significant threshold of digital transition. Section 3.3 describes the data collection protocol and analytical techniques, encompassing the calculation of financial KPIs from balance sheet data, the collection of digital investment data, the administration of a customer satisfaction survey, and the handling of e-commerce platform analytics given the pre-commercial status of the platform; five subsections detail each data source and its methodological implications. Section 3.4 addresses reliability, validity, and research ethics, discussing the triangulation strategy adopted and the measures taken to ensure data quality, analytical rigor, and confidentiality of company information.

Chapter 4, Findings, presents the empirical results of the research, structured around multiple analytical dimensions. Sections 4.1 through 4.3 establish the organizational context by profiling Auroflex's company history, governance structure, product portfolio, and competitive positioning in the self-adhesive label sector. Section 4.4 maps the company's current digital maturity gaps across website functionality, CRM system absence, and analytics infrastructure. Section 4.5 presents the analysis of Auroflex's economic and financial performance over the 2020–2025 period, including calculated KPIs and the application of DuPont

analysis to trace relationships between operational decisions and return on equity. Section 4.6 proposes a snapshot implementation roadmap that prioritizes improvement initiatives according to identified gaps and strategic dependencies. Section 4.7 presents the design of the operational and digital KPI dashboard, integrating multiple performance dimensions into coherent visual management tools. Section 4.8 reports customer behaviour insights derived from the satisfaction survey. Section 4.9 presents the Digital Balanced Scorecard co-designed with Auroflex management, establishing explicit and quantified linkages across all four perspectives and translating strategic objectives into measurable targets.

The Discussion and Conclusions synthesize the key insights emerging from the case study, discussing both the practical and theoretical contributions of the research. The primary practical contribution is the Digital Balanced Scorecard itself, which transforms Auroflex's previously fragmented performance monitoring into a systemic and multidimensional architecture. The chapter further examines the managerial implications for SMEs in the self-adhesive label sector, reflects on how the co-design methodology catalyzed organizational learning at strategic, managerial, and operational levels, and acknowledges the principal limitations of the study in particular those inherent to the single-case design and to the difficulty of isolating the effects of digital interventions from exogenous variables. It concludes by proposing directions for future research, including the extension of the framework to larger samples of label converters and the investigation of specific digital interventions such as web-to-print platforms and predictive maintenance systems.

CHAPTER 2

Literature review

2.1 Performance Measurements Systems

An effective digital transformation strategy requires a well-organized and coherent performance measurement system capable of translating strategic targets into concrete actions.

Fora a B2B manufacturing company such as Auroflex, marked by high product customization and complex production flow, the design and adoption of a PMS (Performance Measurement System) is a crucial step in the digital transformation process, in order to ensure that digital investments generate tangible financial and operational results.

This section therefore seeks to examine the theoretical framework of PMS by analyzing the transformation path from traditional concepts such as retrospective economic indicators toward integrated multidimensional predictive models capable of encompassing digital transformation. The objective is to develop in subsequent sections a measurement framework calibrated to the needs of the self-adhesive label industry.

2.1.1 Definition and Strategic Role of PMS

A PMS can be defined as a coordinated and organic set of metrics, processes, methodologies, and technological infrastructures that an economic entity uses to quantify the effectiveness and efficiency of its actions in relation to predetermined strategic objectives. Unlike individual indicators considered in isolation, a PMS constitutes a complex architecture that connects different dimensions of business performance and creates causal relationships among various metrics (Neely et al., 1995). It defines the criteria for data collection and analysis and supports decision-making mechanisms at every organizational level.

In light of what has already been analyzed in the previous chapter, the concept of performance measurement, beyond being mere reporting of results, must necessarily include other components such as customer satisfaction, operational

process efficiency, innovation capacity, and fruitful organizational learning. This multidimensional vision gains greater strength in the context of digital transformation, where simple profitability criteria give way to broader concepts such as the effectiveness of innovations on business processes.

Current PMS must therefore be capable of assessing the digital maturity of an economic entity as well as the impact of emerging technologies on business processes and customer relationships.

Thus a company such as the one under consideration, characterized by:

- a “high-mix/low-volume” production model,
- the need to ensure rapid delivery times with high quality standards,
- growing regulatory pressure regarding sustainability and material traceability imposed by EU Regulation 2024/825,
- and the managerial complexity of B2B customer relationships in the food & beverage and pharmaceutical markets,

requires a performance measurement system capable of simultaneously balancing these heterogeneous aspects.

It should be noted that purely financial indicators do not lose their centrality in the overall assessment of a company's economic soundness, but in light of what has been stated, they may prove insufficient to capture certain aspects of the analysis. The design of an effective PMS in a digital transformation context must account for the dynamism of technological change. Unlike investments in traditional tangible assets, whose results are often predictable and linear over time, everything pertaining to the digital sphere produces impacts that manifest with variability and through complex and often non-evident causal chains. A PMS adequate to this reality must include both lagging indicators that quantify results already achieved and measurable ex-post, and leading indicators capable of anticipating future trends and predicting corrective measures before problems lead to economic losses.

2.1.2 The Foundational studies.

The contribution that Neely, Gregory, and Platts[7] offered in the mid-1990s

[7] Neely, Gregory and Platts are prominent researchers in the field of performance measurement and operations management.

represents a watershed moment in the theoretical conceptualization of PMS.Their 1995 paper "*Performance Measurement System Design: A literary review and Research Agenda*" published in the *International Journal of Operations & Production Management* is a foundational work that has significantly influenced subsequent research.

They proposed a model articulated across three hierarchical levels that allows for analyzing systems in their organizational complexity and their connection to competitive strategy (Neely et al., 1995).

1. The first level of analysis focuses on individual performance indicators. Each indicator must possess specific and variable attributes, definitional clarity that makes them easily interpretable by all stakeholders, objective measurability, direct relevance to the strategic objectives it aims to monitor, and economic sustainability in terms of costs and complexity for data collection. This level of analysis leads to certain difficulties for Italian SMEs in the label sector, related to limited availability of human and economic resources dedicated to management activities and the operational simplicity typical of smaller enterprises.
2. The second level treats the measurement system as an organic and interconnected entity, analyzing the relationships among different indicators and their capacity to provide a complete representation of business performance. A weakness at this level is the uncontrolled proliferation of metrics, generating what is defined in management literature as "measurement overload," or information overload (Neely et al., 1995). For the label sector, it is necessary to identify a limited but significant number of key metrics capable of capturing the critical dimensions of performance without burdening monitoring processes.
3. The third level positions the PMS within the strategic competitive context of the firm and aims to optimize the selected indicators and their systemic integration with the organization's strategic priorities in relation to its industry sector. For Auroflex in particular, this must be combined not only with production characterized by high levels of quality and dimensional precision but also with the expectations of a certainly loyal yet particularly

demanding customer base.

In a B2B manufacturing context, customer satisfaction is of paramount importance, and the ability to manage highly customized requests constitutes a primary driver often more relevant than pure cost advantage. PMS designed for repetitive and standardized mass-production contexts will not suffice; flexible and adaptive measurement systems are therefore required.

2.1.3 PMS Maturity and Effectiveness

The temporal evolution of PMS can be analyzed through the concept of maturity, which describes the degree of refinement, systemic integration, and strategic alignment of a system. The model proposed by Van Aken and colleagues identifies various phases ranging from mere collection of financial data to complex systems capable of supporting real-time decision-making processes (Van Aken et al., 2005). In many Italian manufacturing SMEs, early-stage maturity is marked by a predominant focus on historical economic-financial data, mainly for external reporting to banks, tax authorities, or for regulatory obligations. Measurement typically follows rigid reporting schedules and is often disconnected day-to-day operations.

With the evolution of system maturity achieved through investments, PMS tend to incorporate dimensions beyond financial ones, develop verifiable causal links between operations and strategic targets, and integrate with managerial decision-making processes (Van Aken et al., 2005). In doing so, the technological infrastructure evolves toward integrated platforms that automate data collection, reducing burdens in favor of business intelligence and data visualization tools. At advanced maturity levels, typical of entities with strong analytical culture, the PMS becomes predictive as well as descriptive and utilizes machine learning systems to identify hidden patterns or anomalies in data and supports continuous improvement processes through rapid operational feedback cycles.

In the context of self-adhesive labels, the analysis of PMS maturity shows important limitations related to the specific production logic of the sector. Indeed, when machine-level data are stored in MES systems that are only partially integrated with the central ERP system, when commercial and digital marketing information

is managed through separate web platforms managed by external providers, and financial data are handled in different accounting software with their own logic, creating a real-time integrated dashboard required additional resources and increases the risk of data inconsistencies.. Consequently, problems may arise concerning data reliability issues and the credibility of the measurement system itself, leading managers to rely less on performance metrics.

It should be noted, however, that digital transformation offers concrete opportunities to accelerate evolution toward higher levels of maturity. Indeed, modern cloud-based business intelligence platforms accessible even to SMEs and data analytics systems based on artificial intelligence algorithms overcome traditional information silos and enable the construction of robust, reliable, and accessible systems.

For the Auroflex case, as will be seen in coming chapters, the path toward a mature and digitally enabled PMS must necessarily pass through targeted investments in technological integration of existing systems and simultaneously in the development of internal competencies.

2.1.4 Performance Measurement Systems in the Digital Era

The advent of digital technology has given rise to what recent academic contributions define as a new generation of data-driven PMS, capable of processing significant and growing volumes of operational data in real-time or near-real-time and producing actionable insights through advanced analytical techniques and interactive visualizations (Kotarba, 2017).

The slowness and cumbersomeness of traditional PMS is now being overcome through digitalization. Contemporary PMS, fed by continuous and automated data flows from integrated ERP systems, MES systems and IoT sensors installed directly on production lines for continuous monitoring, B2B e-commerce platforms, and advanced web analytics tools that track customer digital behavior, enable continuous and granular performance monitoring (Kotarba, 2017). Technical integration between centralized management systems and production equipment with embedded intelligence enables real-time tracking of order progress through different phases of the production process and dynamic calculation of operational

efficiency indicators such as OEE (*Overall Equipment Effectiveness*), as well as automatic detection of anomalies or unknown casualties through pattern recognition algorithms.

In the self-adhesive label sector, what enables optimization of management and strategic pricing is the ability to measure with extreme precision setup times between consecutive processing of different products, the capacity to systematically analyze patterns of raw material usage and waste, as well as monitoring the dimensional and aesthetic quality of finished products through artificial vision systems integrated directly into converting lines, and finally to robustly correlate this operational data with precise economic data regarding profitability by individual customer or individual order.

The operational connection between digital PMS and the previously introduced Digital Balanced Scorecard now becomes evident and strategically relevant. The digital BSC does not simply represent an extension of the traditional scorecard through the addition of some digital KPIs, but rather configures a methodologically integrated approach to measurement. It exploits the potential of digital technologies to operationalize and dynamize the multidimensional strategic model originally proposed by Kaplan and Norton (Kaplan & Norton, 1996,); (Kotarba, 2017).

The true challenge of the digital shift lies in the organizational and cultural changes it entails. According to Kotarba's current literature, it is indeed crucial to develop widespread competencies within the organization not limited to IT or finance functions traditionally responsible for management control systems but extended also to operational managers who must daily utilize the insights generated by PMS. Structured training of personnel must be conducted at all levels in order to systematically incorporate quantitative data analysis into operational routines.

Although, according to some scholars, data abundance can paradoxically generate decision-making confusion and analytical paralysis if not accompanied by clarity in frameworks.

2.1.5 Implications for the Self-Adhesive Label Sector and the Auroflex Case

The design of an effective PMS for the self-adhesive label sector must take into consideration multiple interdependent dimensions:

1. The high-mix/low-volume nature of customized label production implies measurement systems capable of operating under conditions of production variability quite different from those anticipated for repetitive, standardized, high-volume production. Efficiency measurement must incorporate the complexity of setups between different processing runs, variability related to the physical-chemical characteristics of materials employed, and learning effects associated with non-repetitive processing requiring continuous micro-adjustments see, for example, recent literature on measurement systems in high-mix/low-volume production, which highlights the role of metrics such as changeover times, first pass yield, and OEE per variant (Adrian Miqueo et al 2022).
2. The centrality of quality as a primary competitive factor requires more sophisticated metrics of statistical process capability, stability of production processes over time through control techniques, and customer satisfaction, in line with the extension of measurement systems toward process and quality dimensions proposed in the literature on manufacturing PMS (Neely et al., 1995).
3. The growing and inescapable regulatory pressure regarding environmental sustainability and material traceability relevant in the food & beverage sector, which represents one of the main outlet markets for self-adhesive labels, requires the integration into PMS of environmental indicators and regulatory compliance metrics.
4. The digital dimension of performance assumes peculiar characteristics in the B2B manufacturing context compared to consumer markets. Unlike B2C markets, where digital metrics focus on dimensions such as overall web traffic, immediate e-commerce conversion rates, and social media engagement metrics, in B2B manufacturing, digital indicators must capture the operational effectiveness of tools such as customized customer portals for managing recurring orders, online product configurators enabling automatic quotation of customized solutions, order tracking systems providing real-time visibility on progress status, and collaborative platforms for pre- and post-sales technical support. Measuring the B2B customer's

digital experience requires methodological approaches that involve multiple stakeholders and recognize the critical importance of qualified technical assistance and post-sales support in building lasting trust-based relationships.

For the company under consideration, Auroflex, the preceding analysis has highlighted both opportunities and criticalities connected to the development of a mature and digitally enabled PMS. On one hand, the significant investments in digital technologies already made by the company in recent years with particular reference to digital printing lines and the ongoing B2B e-commerce platform launch project constitute a potentially solid technological foundation for building advanced and data-driven measurement systems; on the other hand, the still significant fragmentation of information tools currently in use and the consequent organizational need to effectively integrate data from technologically heterogeneous sources represent significant operational obstacles that must be strategically addressed through a systematic integration plan.

The integrated measurement framework that will be developed and detailed in the following sections aims to concretely respond to these specific needs, articulating an organic system of economic-financial, operational, and digital metrics, methodologically aligned with the theoretical principles of the Digital Balanced Scorecard and explicitly designed to effectively support daily decision-making processes. Section 2.2 will examine traditional economic-financial metrics and their necessary evolution in digital terms, while Section 2.3 will focus on the most relevant digital KPIs for the self-adhesive label sector, thus completing the theoretical and methodological framework necessary for the in-depth empirical analysis that will constitute the substantial core of the thesis.

Operational design requires balancing methodological rigor and practical feasibility, recognizing that theoretical consistency and conceptual soundness of a measurement system area is valuable only if a system is effectively adopted, understood, and used daily by the organization in decision-making processes.

2.2 Economic-Financial Metrics and Business Performance

This section examines the main economic-financial indicators traditionally used to

evaluate business performance. The aim of this section is: to provide a structured overview of key financial indicators used to assess profitability and financial stability . At the same time, it highlights the limits of these metrics in capturing digital transformation dynamics, preparing the analysis of digital and non-financial KPIs to be discussed in the following section.

2.2.1 Foundations of Economic-Financial Measurement

Economic-financial measurement is rooted in management accounting and financial statement analysis, which for decades have been considered the primary tool for evaluating business performance. The annual financial statements and their mandatory accounting documents - Income Statement, Balance Sheet, and Cash Flow Statement - represent, according to Brigham & Houston (2019), the formal synthesis of business management and provide a quantitative representation of the company's economic, equity, and financial performance. Management accounting then enables the decomposition of aggregate results into operational units and cost centers, product lines, allowing for more detailed analysis of the determinants of economic performance (Brigham & Houston, 2019).

Financial metrics, thanks to their quantitative nature, inter-company comparability, and verifiability through external control systems such as accounting regulations and auditors, became the privileged tool for all stakeholders throughout the twentieth century.

However, in the 1980s, these considerations began to waver.

Thomas Johnson and Robert Kaplan, in their work "*Relevance Lost: The Rise and Fall of Management Accounting"* at Boston: Harvard Business School Press., demonstrated the inadequacy of traditional systems to support managerial decisions in increasingly complex competitive contexts. Indeed, in the 1990s and 2000s, several integrated measurement frameworks were developed, first among them the Balanced Scorecard (Kaplan, R. S., & Norton 1996).

It should be noted, however, that traditional financial metrics continue to maintain their relevance, as they are indispensable for defining value creation for shareholders, demonstrate the ability to generate positive cash flows, and often prove to be the only comparable indicators across companies from different sectors

(Brigham & Houston, 2019). The ideal approach is therefore to blend the two techniques in a multidimensional system that enhances their respective strengths.
In the self-adhesive label sector, investments in technologically advanced machinery such as rotogravure presses, offset equipment, embellishment systems, automated quality control systems, require substantial capital, efficiency in asset utilization and working capital management, as well as the ability to generate sustainable operating margins, become indispensable elements for competitiveness and market survival (Politecnico di Milano).

2.2.2 Fundamental Financial Metrics for B2B Manufacturing Companies

The first group of economic-financial indicators we will now analyze concerns return on invested capital.
Return on Equity (ROE) measures how effectively the company generates returns for its shareholders as the ratio between net income and equity. However, ROE does not distinguish between the effect of operating performance and financial structure. As a result, a highly leveraged company can have a high ROE because of l leverage, despite weak operating margins.
Return on Assets (ROA), which relates operating income to total assets, partially overcomes the limitations of ROE by focusing on the efficiency with which the company uses its resources, regardless of financing sources[8]. However, ROA could also be misleading due to partially depreciated assets or specific valuation policies.
Return on Invested Capital (ROIC) is particularly significant in the manufacturing context as it measures the profitability of capital invested in operational activity, independent of financial structure. ROIC is calculated by relating net operating profit after taxes (NOPAT) to operational invested capital, sum of operational working capital and net fixed assets[9]. This indicator is particularly meaningful for SMEs operating in capital-intensive sectors such as converting. In Auroflex's case, a company that has invested heavily over the years in high-end printing technologies such as hot stamping, embossing, or screen printing, ROIC constitutes an important measure to assess whether such investments are contributing to value

[8] According to Brigham & Houston in "Fundamentals on Financial management 12th edition"2019.
[9] Ibid 7.

creation or instead tying up capital inefficiently.

2.2.3 A second group of metrics pertains to margins and operational profitability.

Gross Margin is calculated as the ratio between gross contribution margin and revenues. It measures the company's ability to cover direct costs while maintaining a positive spread . In the converting sector, where raw material costs (specialty papers, plastic films, inks, or metallic laminates) or semi-finished products represent a significant share of turnover, gross margin is the principal indicator for understanding whether implemented pricing policies are effective and whether variable costs are well managed in the thesis of Ferraloro G. & Peradotto M. "*How design can boost competitive performancesin SMEs*" (Politecnico di Milano).

EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization) indicates operating profitability before depreciation, provisions, and financial expenses. It provides the company's ability to generate cash from core operations. In capital -intensive firms such as Auroflex, characterized by substantial investments in tangible fixed assets, EBITDA becomes particularly relevant because it provides a clearer view of evaluation of operational performance independently of depreciation policies and financial structure choices.

Operating Margin, obtained by relating operating income (EBIT) to revenues, instead measures profitability after also considering depreciation and provisions, thus providing a more complete picture of operational efficiency and the company's pricing power (Identità, innovazione e impatto dell'aziendalismo italiano. Dentro l'economia digitale).

Operating Cash Flow measures actual monetary flows generated from operational management. The ability to generate cash, for a B2B manufacturing company, is fundamental for financing working capital, supporting investments in technological innovations, and limiting dependence on external financing sources. Free Cash Flow, calculated by subtracting capital expenditures (CAPEX) necessary for production capacity from cash flow, constitutes the residual liquidity available for growth, shareholder remuneration, or debt reduction.

2.2.4 A third and final group of metrics concerns operational efficiency.

Asset Turnover, or the ratio between revenues and total assets, reflects how efficiently the company uses its asset base. A high ratio suggests that the company can generate a significant volume of revenues relative to the asset base employed, suggesting efficient resource use. A label converter operating hybrid Flexo-digital lines that reduces setup times and minimizing waste, will present good and competitive asset turnover.

2.2.5 DuPont Analysis as a Performance Decomposition Framework

A common analytical tool for interpreting economic-financial performance is DuPont Analysis. This framework, developed in the 1920s by the DuPont Corporation and subsequently refined by financial theory, uses structured decomposition of profitability indicators into underlying operational drivers, clarifying the contribution of different performance drivers.

In its most common form, DuPont Analysis breaks ROE into three multiplicative components: profit margin or sales profitability, asset efficiency or asset turnover, and financial leverage or equity multiplier.

In formula:

ROE = (Net Income/Revenue) × (Revenue/Total Assets) × (Total Assets/Equity)

This decomposition reveals how return on equity does not depend on a single factor but on the combination of three distinct dimensions: marginality, or how much the company earns on each euro of turnover; capital rotation, or how effectively it uses its assets to generate revenues; and financial leverage, or the extent to which it uses debt to multiply its operational capacity.

A company can therefore achieve high ROE through multiple different strategies. The value of DuPont Analysis emerges clearly when applied to specific contexts. Indeed, in our sector for example we can normally observe operating margins above the manufacturing average but relatively contained asset turnover due to high capital intensity. In this context, Auroflex positions itself in the “high margins, medium-low rotation” area of the DuPont framework, by virtue of a deliberate choice toward distinctive technical quality and excellence as well as personalized services.

This decomposition can also be extended to analyze second- and third-level drivers. Profit Margin, for instance, can be decomposed into its cost components (incidence of cost of goods sold, operating costs, financial charges, tax burden), while asset turnover can be detailed by separately analyzing the rotation of different asset classes (tangible fixed assets, inventories, trade receivables).

In Auroflex's specific case, we can estimate the potential impact of digital transformation on the different levers of the model. Investments in production process automation, for example, can improve asset turnover by reducing setup times and increasing output per unit of time. Conversely, the adoption of advanced CRM and B2B e-commerce platforms can positively influence profit margin through better understanding of customer needs and consequently strengthening pricing power. The implementation of digital supply chain management systems could then reduce working capital, improving asset composition and indirectly overall rotation.

In conclusion, DuPont Analysis not only enables diagnosis of the current state of economic performance but also offers a conceptual framework for hypothesizing how digital transformation initiatives can translate into improvements in financial drivers (Kaplan & Norton,1996).

2.2.6 Supply Chain Metrics and Impact on Economic Performance

Supply chain management, and more specifically operational working capital management, holds particular importance for manufacturing companies.

The Cash Conversion Cycle (CCC) represents the most widely used metric for evaluating working capital efficiency. It measures the average time, expressed in days, between when the company incurs monetary outlays for raw material purchases and when it collects revenues from sales[10] .

In formula:

$CCC = DIO + DSO – DPO$

Where DIO (Days Inventory Outstanding) represents the average days inventory is held in stock, DSO (Days Sales Outstanding) represents the average days for

[10]Brigham & Houston in “Fundamentals on Financial Management 12th edition”2019 pp.125-130.

collecting trade receivables, and DPO (Days Payables Outstanding) represents the average days for paying suppliers. Thus, a reduced CCC indicates that the company can rapidly convert investments in working capital into liquidity, thereby reducing financial requirements and improving return on invested capital.

In the converting sector, DIO is generally limited to standardized raw materials but can become substantially higher for special components requiring made-to-order procurement. Semi-finished products and finished goods, given their often-customized nature, may remain in inventory for very different durations. Labels for luxury brands are typically produced to order and delivered quickly, while stock production for standard items requires more careful inventory management. DSO largely depends on customer payment terms, generally between 30 and 90 days, occasionally prolonged for major customers with strong bargaining power. DPO, in turn, depends on terms agreed with suppliers of raw materials and services. Converters with greater financial soundness and significant purchase volumes enjoy more favorable negotiation terms, contributing to a more favorable CCC.

It is entirely evident that good working capital management is crucial for the company's wellbeing. First, it has a direct impact on financial requirements and their cost; second, it influences invested capital rotation - indeed, with equal operational profitability, a company with a shorter CCC will present superior ROIC. Third, good working capital management represents a key source of competitiveness in sectors where profitability is volatile, and a CCC significantly below average can reduce the impact of raw material cost swings and support aggressive commercial policies (Brigham & Houston, 2019).

In Auroflex's specific case, working capital management is influenced by the structural features of its luxury segment customer base, characterized by high bargaining power and creating tension between competitive payment terms and maintaining a sustainable DSO position. Production complexity can extend throughput times and consequently increase DIO. On the other hand, digital transformation, through tools such as ERP systems, can reduce stock imbalances, while digital receivables financing tools can accelerate receivables collection without damaging commercial relationships.

2.2.7 EBITDA and Operating Margins as Efficiency Indicators in Manufacturing

In the manufacturing sector, the utility of EBITDA and operating margins lies in providing a proxy of cash generation capacity from core operations, net of accounting choices related to depreciation schedules and prudential provisions[11]. For a company with significant investments in tangible fixed assets, depreciation can weigh heavily on the income statement but corresponds to monetary outlays that occurred in previous years. EBITDA, by purging the operating result from these non-cash components, offers an immediate view of current operational profitability and the financial sustainability of operations. EBITDA margins, calculated as the ratio between EBITDA and revenues, in the self-adhesive label converting sector tend to fall within a range reflecting the company's competitive positioning: generalist converters competing on price and volumes present EBITDA margins between ten and fifteen percent, while operators specialized in premium niches can achieve margins exceeding twenty percent. For Auroflex, the managerial challenge consists of maintaining high EBITDA margins despite increased operating costs related to production complexity and continuous innovation. Operating margin, as mentioned, also incorporates the effect of depreciation and therefore provides a more conservative measure of operational profitability. It proves more appropriate for long-term sustainability assessments, as it accounts for the fact that productive assets deteriorate over time and require periodic reinvestment. For a company that has recently made substantial investments in new machinery, EBITDA may be significantly higher than EBIT due to the high incidence of depreciation; conversely, for companies with partially depreciated assets, the difference between the two indicators narrows.

Comparing EBITDA and operating margins allows a better understanding of competitive dynamics and the company's strategic choices. A company presenting high EBITDA margins but containing operating margins reflects high capital intensity and is often linked to recent capital expenditure. By contrast, operating margins aligned with EBITDA indicate a more mature capital structure. In digital transformation processes, it is reasonable to expect that in the short-to-medium

[11] Brigham & Houston in "Fundamentals on Financial management 12th edition"2019 pp.72-78.

term, investments in automation, robotics, and advanced information systems may temporarily reduce operating margins even if EBITDA improves. However, in the long term, greater operational efficiency should translate into improved operating margins also due to the higher degree of depreciation.

2.2.8 The Gap Between Traditional Financial Metrics and Digital Transformation

When addressing the theme of digital transformation, a structural limitation of economic-financial metrics emerges clearly: their retrospective nature and their character as outcome measures make traditional metrics not fully suited to monitoring profound change processes.

This manifests across multiple dimensions (Ittner & Larcker, 1998), (Kaplan & Norton, 1996):

1. Financial metrics measure economic results that are the outcome of a long causal chain. Relying exclusively on traditional indicators to evaluate the success of digital transformation initiatives would mean having an inadequate early warning system, capable of detecting problems only when it is too late to intervene effectively.
2. Many effects of digital transformation emerge in areas that are only partially reflected in financial metrics. A relevant example is, the implementation of an advanced CRM system that enables detailed tracking of customer interactions, analysis of purchasing patterns, and personalization of commercial proposals. The benefits of such a system manifest in terms of customer satisfaction, retention rate, cross-selling and up-selling capability. The economic impact of these improvements unfolds gradually with different time horizons. In practice, improved retention reduces customer acquisition costs and stabilizes revenues in the medium term; personalization strengthens pricing power and could improve margins; sales cycle reduction accelerates cash generation. However, each of these effects tend to be diluted within aggregate accounting results that simultaneously reflect multiple external factors (demand trends, competitive pressure, raw material cost dynamics) and internal dynamics.

3. Financial metrics tend to privilege operational efficiency and short-term optimization, likely penalizing investments in innovation. Companies allocating significant resources to process digitalization, personnel training, and B2B e-commerce platform implementation will see operating costs and intangible capital investments increase in the short term, with probable deterioration of operating margins and ROIC. If we evaluate this company according to traditional short-term metrics, it might appear less performant compared to competitors that have postponed such investments. Only in the medium-to-long term can we appreciate the benefits, and the strategic choice will appear optimal due to improvements in economic-financial indicators.

Academic literature has extensively discussed this problem.expecially with Ittner and Larcker[12] in an empirical study, analyzing the relationship between non-financial performance measures and future economic results. In their studies in 1998 and in 2003 Harward Business Review articles, "*Coming up short on non-financial performance measurement*" they identify key mistakes a company makes, such as failing to link measures to strategy, or not validating cause-and-effect relationship, setting inappropriate performance targets.

They demonstrated that indicators related to customer satisfaction, process quality, and innovation constitute leading indicators of financial results, that is, they anticipate and predict future economic performance. The aforementioned (Kaplan & Norton,1996), through their Balanced Scorecard, created a logical-causal model in which performance in non-financial dimensions (learning, growth, customers) are the drivers that ultimately determine economic-financial results. In doing so, financial metrics are not eliminated or devalued but are repositioned as outcome measures, integrating them with non-financial indicators that explain operations and strategies.

In the context where the company under study operates, digital technologies impact numerous operational dimensions: production efficiency through automation, process optimization, product quality through real-time control systems and

[12] Ittner e Larcker are prominent researchers in managerial accounting and performance measurement.

predictive maintenance, customer relationships through e-commerce platforms and self-service portals. Each of these dimensions can be monitored through specific KPIs (defect rate, production lead time, Net Promoter Score, time-to-market for new products, etc.) that provide the company with timely signals on digitalization effectiveness, testing daily whether expected operational improvements are actually materializing (Kotarba,2017).

What has been stated thus far is extremely relevant for Auroflex, a company that, as already mentioned, has built its history and competitiveness on solid and deeply rooted operational and relational foundations. Strategic evolution toward integrated order management through B2B platforms, extensive production process automation, business intelligence applied to customer portfolio analysis, and IoT-based traceability and quality systems require significant investments and entails non-negligible risks. It is therefore essential to evaluate by flanking financial indicators with a system of digital and non-financial KPIs. Only by integrating these two measurement levels financial and digital operational will it be possible to construct a broad, complete, and balanced view of business performance (Neely, Gregory & Platts,1995) [13].

Based on these premises, the following section will analyze digital KPIs and operational indicators that are relevant for measuring and guiding the digital transformation process in the B2B manufacturing sector and at Auroflex. The target is to enrich economic-financial measurement through an integrated and multidimensional Performance Measurement System capable of capturing both economic results and the operational and strategic drivers that determine them.

2.3 Non-Financial Digital Metrics in B2B Manufacturing

According to recent literature, in B2B manufacturing contexts, simple economic-financial measurement is insufficient to capture the effects of digital transformation on competitiveness and value creation (Kotarba, 2017).

[13] Ibid nota 7.

As highlighted in the previous section, traditional metrics present structural limitations; indeed, their retrospective nature and final outcome character make them unsuitable for monitoring changes in real time.

More specifically, studies on digital servitization in SMEs have demonstrated how investments in connected technologies such as IoT, data services, or digital platforms initially impact intangible dimensions namely, service quality and response speed which anticipate variations in profitability and growth over time [14]. The mix of non-financial and digital metrics thus assumes the role of leading indicators of financial results, especially when the company evolves from pure manufacturer to provider of product-service solutions (Ittner &Larcker,1998).

2.3.1 Digital Servitization and Relational Value Propositions

Digital servitization, in this context, should be understood as the process through which manufacturing companies combine tangible goods, advanced services, and digital platforms to build continuous and relational value propositions . It represents a significant evolution from the traditional B2B transactional logic in which the relationship was exhausted with product delivery. In digital servitization, the product represents the starting point of an ongoing relationship in which digital services, remote support, predictive analytics, and personalization extend value creation well beyond the sales model[15].Empirical research on European SMEs highlights how this path requires metrics capable of measuring not only product performance (quality, OEE, defects) but also the effectiveness of digital services such as platform usage rate, frequency of customer interactions, and intensity of data use in decision-making processes (Rappaccini et al, 2020). These new indicators have proven useful for estimating companies' shift toward new approaches to capturing value over time (Kohtamaki et al., 2019).

[14] These topics have been addressed by Coreynen, Van Bockaven and Matthyssen, researchers who have collaborated extensively about digital transformation and innovation in manufacturing. Matthyssens is a professor of Global Business and Global Marketing at the University of Milano Bicocca.

[15] These topics have been addressed by. Reim, Sjodin and Parida prominent researchers in the fields of digitalization, business model innovation and sustainable industry, particularly focusing on product-service systems PSS.

Adopting servitized models implies changes in internal organization and competencies. In practice, this means monitoring indicators related to digital skills development, such as the adoption rate of new tools by personnel, the digital literacy level of the sales team, and the effectiveness of training programs on emerging technologies. Otherwise, digital investments may remain underutilized, compromising the expected growth result.

2.3.2 Digital KPIs for B2B Manufacturing E-commerce

In the specific case of B2B manufacturing with an e-commerce channel, the literature on digital KPIs proposes structural sets that flank traditional web traffic indicators - such as sessions, sources, bounce rate with more evolved behavioral measures (Shopify, 2025). We speak, therefore, of:

- Portal adoption rate by existing customers, when we want to measure what percentage of the customer base migrates toward the digital channel; Ratio between digital revenues and total turnover, to appreciate the degree of e-commerce penetration in the channel mix.
- Lead-to-customer conversion rate, to evaluate the effectiveness of the digital funnel in transforming prospects into actual customers.
- Repeat purchase rate, to measure the crucial capacity of the online channel to retain and stimulate repurchases (Routine Automation, 2025).

In several studies on U.S. and European manufacturer-distributors, the increase in the share of sales mediated by digital platforms can be correlated with greater revenue resilience and more stable operating margins, although such benefits become evident with time lags relative to the launch of technological investments. B2B e-commerce indeed requires, on average between 18 and 36 months before benefits can be appreciated in terms of measurable economic results in financial statements. During this maturation period, it is advisable to systematically monitor digital KPIs of adoption, engagement, and retention to potentially anticipate future growth trajectories and understand whether the adopted business model is moving toward greater dependence on online channels.

Platform adoption can be measured by the number of active customers on the platform and their growth rate, can be considerating the entry point of the digital

customer journey. Engagement, namely, login frequency, average time on platform, number of pages viewed per session, and interactions with advanced features constitute evidence of how much customers value the digital ecosystem (Shopify, 2025). Retention, finally tracked with metrics such as digital customer lifetime value, online channel churn rate, and net revenue retention, serves to measure how well the platform can grow customer value over time (Routine Automation, 2025). Research has indeed demonstrated the importance of segmenting these KPIs by customer type. Customers of different sizes, sectors, or geographies present significantly different digital behaviors. Large industrial accounts might prefer EDI or API integration for recurring orders, while SMEs appreciate intuitive web-to-print interfaces for customized orders Monitoring digital KPIs in aggregate thus risks hiding critical patterns and segment-specific optimization opportunities. In this sense, the literature suggests developing segmented dashboards that enable performance evaluation distinguishing at least among strategic key accounts, recurring transactional customers, and new prospects, consequently adapting engagement strategies and platform development roadmaps (Kotarba, 2017).

2.3.3 Specific Metrics for the Self-Adhesive Label Sector

For a self-adhesive label converter operating in high-mix/low-volume logic, non-financial measurement must pay particular attention to the peculiarities highlighted by packaging literature. Indeed, recent studies on the packaging value chain have shown how the spread of smart labels, end-to-end digital workflows, and regulatory traceability systems related to sustainability and safety make KPIs dedicated to the digital dimension of both product and process necessary (Smart Packaging Hub, 2025).

Specifically, regarding the product, it is important to monitor the share of jobs managed in fully digital flows, that is, orders in which the entire process from creative brief to final delivery occurs without media interruptions. Artwork-to-print lead time, measuring the time between final artwork approval and production start, is a key indicator of operational agility in a market where speed represents a competitive factor (Sessions, 2025). The percentage of orders configured online directly by customers then constitutes a good signal for testing the effectiveness of

web-to-print systems and the digital maturity of the customer/supplier relationship. If we analyze the process, we can speak of line automation index, which quantifies the percentage of production operations managed by automated systems versus manual ones; machine-management data integration, to measure the percentage of machines connected to ERP and MES systems and real-time data synchronization (Smart Packaging Hub, 2025). Lastly, we can also speak of the usage rate of web-to-print interfaces by customers, an indicator reflecting both the use of digital tools and the degree of autonomy acquired by customers for managing repetitive orders (Sessions, 2025).

To the above, indicators must also be added indicators made necessary in light of regulatory pressures on sustainability and traceability themes, think of material recyclability, environmental labeling, and traceability throughout the supply chain. This requires the development of specific metrics such as the percentage of labels certified for recyclability, the adoption rate of sustainable materials at customer request, and the ability to provide digital compliance documentation in real time .

2.3.4 Digital Customer Experience and B2B Touchpoints

The importance acquired over time by B2B e-commerce platforms has prompted the inclusion in measurement systems of digital customer experience metrics. While in B2C contexts customer experience often concentrates on the purchase interface, in B2B manufacturing firms, customer experience spans the full relationship lifecycle: from discovery and evaluation (online catalogs, configurators, simulators) to purchase through self-service portals, extending to logistics and after sales support (shipping, technical support, automatic reorders), (Kotarba, 2017).

The Net Promoter Score (NPS) of online channels, that is, the measure of customers' propensity to recommend the company's digital platform to other potential buyers is certainly among the most significant metrics (Routine Automation, 2025). Where timeliness can represent the cornerstone of production relationship continuity, average response time on digital touchpoints such as email and ticketing is also fundamental, as is the quality customers perceive regarding remote technical support via video calls or online knowledge bases (Shopify, 2025). It is also useful to measure digital effort, or the effort required of the customer to

complete specific tasks through digital channels. Indeed, in the B2B system, high digital effort often translates into abandonment of the online channel and return to traditional methods with evident financial burdens and nullifying investments in digital. We speak of metrics such as the number of clicks to complete an order, cart abandonment rate at different funnel stages, and the percentage of transactions completed without need for human support (Routine Automation, 2025).

Another metric of non-negligible importance concerns measuring the quality of data exchanged at digital touchpoints. Incomplete specifications and errors in order data can indeed generate production defects, incorrect deliveries, or commercial disputes. It therefore appears fundamental to monitor the error rate on digitally transmitted order data, the percentage of requests for post-order clarifications, and time-to-resolution for technical problems reported through online channels (Sessions, 2025).

2.3.5 Integration into the Performance Measurement System

To bridge the gap between ex-post measurement of economic-financial results and real-time monitoring of the mechanisms that contribute to their generation, we resort to integrating the non-financial and digital metrics within a Performance Measurement System (Neeely, Gregory & Platts, 2005). All this, as already stated, is solely to correct the course in progress before negative events can occur.

This entails designing a set of KPIs capable of connecting three levels of analysis:

1. The first concerns the effectiveness of digital channels in generating demand. Here we can place metrics such as qualified traffic to the e-commerce site, conversions from visitor to lead and from lead to customer, repeat purchase rate, and customer lifetime value of digitally acquired customers. These serve to measure how effectively the company's digital presence can attract, convert, and retain customers to drive revenue growth.
2. The second level concerns the efficiency of production and logistics processes in satisfying demand generated by digital. Here we speak of KPIs such as overall lead time from order to delivery, distinguishing between order processing time, production time, and shipping time; waste and defect rate; utilization of digital and traditional printing equipment with particular

attention to saturation of the most modern machines; and efficiency of the artwork management process measured through the average number of interactions before final approval and overall artwork-to-print time. The above with the purpose of understanding whether the production organizational machinery is truly capable of managing the volumes and complexity generated by the digital channel and whether bottlenecks exist that limit efficiency.

3. The third level, finally, pertains to the translation of these dynamics into margins, growth, and financial soundness. It is precisely at this stage that operational and digital metrics merge with more traditional economic-financial indicators.

   Analyzing revenue growth by channel helps identify the contribution made by e-commerce; evolution of gross margins and EBITDA segmented by customer type and channel reveals whether digital initiatives are increasing margins; working capital metrics, particularly DSO and DIO, reflect the effectiveness of digital processes in accelerating the operational cycle; while indicators of equity soundness and investment capacity help understand whether the transformation path is sustainable.

A critical element not to be underestimated is the frequency and method of detection of different KPIs. By their nature, digital metrics allow high measurement frequencies-web traffic is available in real time, e-commerce conversions are trackable daily, and touchpoint interactions are recorded automatically. However, operational indicators generally follow weekly or monthly production cycles, while financial metrics have monthly or quarterly cadence. This asymmetry must be managed through the design of a multi-level dashboard with:

- Daily operational dashboards for line management.
- Weekly reports to coordinate production and sales.
- Integrated monthly reviews for general management.

Only with this temporal articulation can the predictive potential of digital metrics be fully exploited, using them to seize every concrete opportunity.

2.4 Adapting the Balanced Scorecard to digital B2B manufacturing

As previously described, the advent of Industry 4.0 and digital transformation has made the integration of economic and financial metrics with operational and digital indicators indispensable in order to capture the complexity of competitive dynamics and value creation mechanisms. This is why the Balanced Scorecard, or more briefly BSC, has established itself as one of the most suitable frameworks for managing integration thanks to its capacity to connect strategic objectives with operational drivers and financial results through an articulated system of cause-and-effect relationships.

This paragraph proposes to study how the BSC can be adapted to the specific context of digital B2B manufacturing and in particular to the self-adhesive label sector. The objective is twofold: to show how the four traditional perspectives of the BSC Financial, Customer, Internal Processes, Learning & Growth can be reinterpreted in light of the drivers of digital transformation, and to highlight the practical implications and organizational challenges connected to the implementation of a Digital Balanced Scorecard in an SME context.

The analysis will conclude with a reflection on the Auroflex case, anticipating the proposal that will be presented in the following chapter.

The original BSC model, introduced by Robert Kaplan and David Norton in the early 1990s, is to be considered a turning point in the evolution of corporate performance measurement systems. As described in their article "*The Balanced Scorecard - Measures that Drive Performance*" published in the Harvard Business Review in 1992, the BSC takes its start from the observation that traditional financial indicators prove insufficient to provide a complete picture of corporate performance (Kaplan & Norton,1992). The authors proposed a measurement system articulated in four fundamental perspectives:

- The financial perspective, which reflects the economic objectives of the enterprise.
- The customer perspective, which monitors the capacity to create value for customers.
- The internal business processes perspective, which evaluates operational efficiency and effectiveness.

- The learning and growth perspective, which focuses on the development of skills, information systems and organizational culture.

Highlighting cause-and-effect relationships, overcoming the fragmented vision typical of traditional systems, is the strong point of the BSC. As argued by Kaplan and Norton in their 1996 work, "*The Balanced Scorecard: Translating Strategy into Action*" the BSC is configured as a strategic management system, capable of translating the corporate vision into a coherent set of objectives, measures, targets and initiatives. Using strategy maps, the BSC allows graphical visualization of how investments in intangible resources (skills, systems, culture) translate into the improvement of internal processes which in turn lead to greater value for customers and finally produce superior economic results (Kaplan & Norton, 1996).

The original framework must in any case be adapted to contemporary contexts; the advent of Industry 4.0 and the diffusion of digital technologies along the entire value chain have in fact introduced new value creation mechanisms. As highlighted by Wolniak and Grebski (2024) in their study on the use of the Balanced Scorecard under Industry 4.0 conditions, digital transformation requires the integration of new categories of indicators, capable of measuring the digital readiness of the organization, the level of data maturity, the quality of the customer's digital experience and the degree of servitization of B2B business models (Wolniak & Grebski, 2024).

Having regard to the four perspectives highlighted by Kaplan and Norton, they must be revisited considering what has been said; in doing so, the internal processes perspective must incorporate indicators relating to automation, machine connectivity, workflow digitization and the capacity to manage data in real time. The customer perspective must then be expanded to include digital customer experience metrics, such as interactions through e-commerce platforms or customer relationship management systems. The financial perspective must contemplate investments in digital technologies and their capacity to generate measurable economic returns through indicators such as ROI or the share of revenue generated through online channels. Finally, the learning and growth perspective must pay particular attention to the development of digital skills of personnel, data governance and the diffusion of a data-driven culture within the organization

(Wolniak & Grebski, 2024).

Unlike the B2C sector, where digitization concentrates on the purchasing interface and the final consumer's experience, in B2B digital transformation involves the entire commercial relationship cycle. This means that the Digital BSC for B2B manufacturing must comprise both operational and relational and transactional dimensions. As emphasized by Frederico (2020) in his study on performance measurement in the supply chain of the industry 4.0 era, integration requires a systemic approach in which KPIs are organized in a way that makes explicit the cause-and-effect relationships between digital investments, operational improvements and economic and financial results (Frederico, 2020).

As highlighted by W.S. Shin[16] (Shin, 2018) in his master's thesis on "*A Quality Scorecard for the Era of Industry 4.0*" the digital BSC acts as an early warning system, capable of signaling in time any critical issues in digitization processes, before they can translate into revenue losses or margin erosion.

Moving to the analysis of the four perspectives, this paragraph will highlight for each of them how the drivers of digital transformation and the empirical gaps emerged for the self-adhesive label sector must provide for the introduction of new KPIs and new methods of interpreting results.

2.4.1 Financial perspective: beyond traditional economic and financial indicators

The financial perspective represents the point of arrival and the ultimate synthesis of the value generated by the enterprise for shareholders and stakeholders. From a more traditional point of view, this perspective focuses on indicators such as Return on Sales (ROS), EBITDA margin, revenue growth and return on invested capital indices. In truth, as emphasized by Performance Metrics and Strategy Execution in SMEs (2025), an effective BSC for SMEs must include in the financial perspective not only aggregate results, but also the breakdown of such results in order to understand the specific contribution of digital channels to revenue volume

[16] A QSC quality scorecard is a performance measurement model designed to evaluate quality aspects within organizations operating in the contest of Industry4.0 developed by Shin, Dalgard and Kim in 2018. It was created by refining existing quality measures, particularly those related to the cost of quality, by eliminating redundancies.

(Performance Metrics and Strategy Execution in SMEs, 2025).

Beyond the share of revenue generated by digital channels, it is also important to calculate the ROI of digital investments, comparing costs sustained (e-commerce site development, CRM, MES, ERP-MES-web-to-print integration systems) with economic results. In this context, a particularly relevant indicator is the payback period of digital investments; it serves to measure the time necessary for the cash flow generated by investments to compensate for the initial costs sustained. As highlighted by Wolniak and Grebski (2024), in manufacturing SMEs the payback period of Industry 4.0 investments tends to be longer compared to sectors characterized by lower capital intensity and shorter production cycles (Wolniak & Grebski, 2024).

Another important aspect of the digital financial perspective pertains to the analysis of profitability divided by customer segment, this for the purpose of understanding whether the digital channel is effectively channeling high-margin customers toward the company. The metrics then of working capital, and in particular Days Sales Outstanding (DSO) and Days Inventory Outstanding (DIO), reflect the effectiveness of digital processes in accelerating the operational cycle, reducing collection times and optimizing warehouse management through more accurate forecasting systems integrated with digital sales platforms.

2.4.2 Customer perspective: digital experience and relationship performance in B2B

In the digital context, differently from the traditional approach based on customer satisfaction and loyalty, the experience is articulated along a multiplicity of online touchpoints, ranging from the discovery and evaluation phase (digital catalogs, product configurators, quote simulators) to the purchase phase (e-commerce portals, recurring ordering systems) up to the final post-sales phase (shipment tracking, remote technical support, online knowledge base, automatic reorders); each of these touchpoints generates behavioral traces that will be useful for future relationship management.

Among these, the Net Promoter Score (NPS) of online channels certainly stands out; it measures the customer's propensity to recommend the company's digital

platform to other potential buyers. Unlike the traditional NPS which focuses on overall customer satisfaction, the B2B NPS of digital channels allows understanding whether investments in e-commerce and CRM are effectively improving the relationship with the customer. In fact, a high NPS on digital channels is predictive of retention and share of wallet growth, insofar as customers perceive digitization as added value that simplifies and improves the purchasing process.

A high retention rate, moreover, is understood as the percentage of customers who over time continue to place orders, in a sector where switching costs can be high, represents a fundamental asset. The share of wallet then, that is, the share of the customer's total spending in the reference sector that is captured by the company, if increasing suggests that digitization is strengthening the commercial relationship and increasing the customer's dependence on the supplier.

Another crucial KPI is the digital Customer Lifetime Value (CLV), which measures the expected value of cash flows generated by a customer over the entire duration of the commercial relationship, weighted by the probability of retention. Digital CLV tends to be higher among customers who have fully adopted digital channels, thanks to higher purchase frequency and greater propensity to evaluate the entire range of products offered through online channels and configurations (Successful Performance Measurement in SMEs, 2012).

To measure the percentage of leads generated through digital channels such as search engines, online advertising campaigns, content marketing, B2B social media, that transform into actual orders, we use the lead-to-order conversion rate. In the presence of a high rate, we can affirm that the platform is working well as a tool capable of attracting customers. In the specific self-adhesive label sector where product configuration is complex, optimization of the lead-to-order conversion rate must be integrated with web-to-print systems that allow real-time visualization of the final result and autonomous modifications.

2.4.3 Internal processes perspective: digital operational efficiency and automation

The perspective pertaining to internal processes is the area in which the adaptation

of the BSC to the Industry 4.0 context proves most evident and necessary. As highlighted by Shin in his *"Quality Scorecard for the Era of Industry 4.0"* the internal processes perspective must function as a bridge between investments in digital technologies (learning & growth perspective) and the operational results that such investments should generate in terms of cost reduction, quality improvement and increased production flexibility.

A central indicator in this perspective is Overall Equipment Effectiveness (OEE), a metric used in lean manufacturing that serves to measure the overall efficiency of plants having regard to three dimensions: availability, that is, the percentage of time in which the plant is effectively operational; performance, namely production speed compared to the standard; and quality, or the percentage of compliant production compared to the total.

In summary, OEE serves to monitor the state of machinery in real time, predict downtime for maintenance and optimize job allocation on production lines. Therefore, a high OEE guarantees competitiveness and waste reduction. In this perspective, the digital BSC should monitor not only the overall OEE but also study its breakdown for each individual job, distinguishing between traditional and digital ones.

As mentioned, in the sector that concerns us it is fundamental to pay attention to waste and therefore to scrap management; they must be avoided starting from the artwork approval phase. A lower scrap rate for digital jobs is in fact ascertained, given that integration between order management systems, approval and production control reduces errors in fact and allows tracing back quickly to any causes of defects.

A further indicator to cite here is the artwork-to-print lead time, that is, the measure of the time elapsed between the final approval of the artwork by the customer and the start of production. To be competitive in a particularly competitive market, speed is everything. Reducing lead time, as highlighted by Frederico (2020), in the supply chain of the industry 4.0 era does not originate from the acceleration of individual process phases, but rather from the elimination of dead times and non-value-added activities through end-to-end digital integration (Frederico, 2020).

The percentage of jobs managed end-to-end in digital workflow is a direct indicator

of the degree of digitization of operational processes. We are talking about those orders that are entirely managed without media interruptions and without manual interventions; high values of this indicator signal great digital maturity of the organization. As emphasized by Wolniak and Grebski (2024), the percentage of digitized processes is one of the best predictors of the enterprise's capacity to scale the adoption of Industry 4.0 technologies and fully benefit from digitization investments.

Last, but not least, we discuss indicators relating to ERP-MES-web-to-print integration, which measures the percentage of transactions and information flows that occur automatically between the various systems, without the need for manual data entry. A high degree of integration reduces process times, minimizes errors, improves traceability and provides an updated view of the state of orders.

2.4.4 Learning & growth perspective: digital competencies and data governance

Digital transformation, as repeatedly stated, cannot disregard the learning perspective. In the Industry 4.0 context, it is necessary to develop a data-driven culture that allows the use of analytics and artificial intelligence. As highlighted by Szóka (2018)[17] In his work on "*The appearance of digitization among the viewpoints of the Balanced Scorecard*", the learning & growth perspective must function as a leading indicator of future performance.

The first indicator concerns the digital skills of personnel and the percentage of subjects who use the applications made available by the company (CRM, MES, collaboration platforms, advanced reporting systems). A low adoption rate implies a certain cultural resistance to change or inadequate training. The level of digital literacy of the commercial team, namely the latter's capacity not only to use CRM well to manage customer relationships but also to encourage and guide customers themselves toward the digital channel. As highlighted in Successful Performance Measurement in SMEs, 2012, in manufacturing SMEs, the success in the adoption of digital platforms by customers depends to a large extent on the commercial staff's capacity to act as change facilitators, supporting customers in the transition toward

[17] Szóka University of Sopron.

new interaction methods and valorizing the benefits of digital channels in terms of speed, traceability and autonomy.

A second indicator in the learning & growth perspective pertains to data governance and the quality of the information infrastructure. In fact, in the industry 4.0 context, not only the availability of data collection and analysis technologies but also the capacity to generate value from these is fundamental. We are talking about indicators such as the percentage of complete data, the number of automatic integrations between different systems and the frequency of data updates. As emphasized by Wolniak and Grebski (2024), in the absence of effective data governance, investments in Industry 4.0 technologies risk generating fragmented, contradictory or unusable information, nullifying the potential benefits of digitization.

A further relevant indicator is the level of use of analytics in decision-making processes to appreciate to what extent operational and strategic decisions are made based on quantitative data analysis rather than on experience or intuition. In the self-adhesive label sector, this indicator can be applied to multiple areas: examining customer purchasing patterns to refine marketing campaigns and cross selling strategies; monitoring line performance supports more efficient job allocation; tracking defect rates supports preventive interventions; and cost data examination may highlight opportunities for margin improvement.

To complete the picture of the learning & growth perspective, we must include indicators relating to the effectiveness of training programs on emerging technologies. The capacity to train personnel and update skills in line with technological evolution proves of primary importance and is a critical factor of competitiveness. The percentage of training hours dedicated to digital topics compared to the total will therefore be measured, then monitoring the effectiveness of training programs in terms of increased use of tools or through post-learning tests. As highlighted by Szóka (2018), in manufacturing SMEs digital training often represents an undervalued investment, with negative consequences on the organization's capacity to fully exploit implemented technologies and to sustain the digital transformation process over time (Szóka, 2018).

2.4.5 Digital and non-financial KPIs within the Balanced Scorecard framework

As discussed in previous paragraphs, recent literature has narrated the existence of a wide range of digital metrics, each destined to capture specific dimensions of online performance: qualified traffic toward e-commerce platforms, conversion rates, cost per lead, digital customer lifetime value, engagement metrics, and operational KPIs of automation and integration.

However, the simple juxtaposition of these indicators with traditional economic and financial KPIs is not sufficient to construct an effective measurement system. What differentiates a well-designed Digital Balanced Scorecard from a mere set of dashboards is the capacity to make explicit the causal relationships between the various indicators, showing how digital KPIs function as leading indicators of economic and financial performance and how investments in the various perspectives learning & growth, internal processes, and customer translate into measurable financial results.

The taxonomy of digital KPIs proposed by authors such as Kotarba and Saura[18] already discussed previously, can be mapped onto the four perspectives of the BSC, organizing the indicators in a way that makes cause-and-effect relationships explicit. In doing so, the KPIs of the learning & growth perspective represent the inputs of the system since insufficient investments in skills, data governance and data-driven culture will translate into internal process inefficiency and the impossibility of providing quality digital experiences to customers. The KPIs of the internal processes perspective constitute the transformation processes insofar as skills and technologies such as OEE, artwork-to-print lead time, jobs managed in digital workflow, are translated into operational efficiency and quality of service rendered. The KPIs of the customer perspective such as NPS, conversion rate, CLV and retention, represent the immediate outputs of the system and in fact measure the enterprise's capacity to create added value for customers through digital channels. Finally, the KPIs of the financial perspective ROI of digital investments, margins by channel and digital revenue share are the final outcomes, that is, the

[18] Saura tenured professor at Rei Juan Carlos University of Madrid.

economic synthesis of the value generated through digital transformation.

The causal chains generated by the integration of the various KPIs do not operate in a deterministic and immediate manner. As highlighted by Frederico (2020), in supply chains of the 4.0 industry era the benefits of digital investments often manifest with significant time delays, which vary from several months to several years depending on the complexity of the investment and the level of digital maturity of the organization.

In the sector that concerns us, it is legitimate to expect that the benefits of investment in web-to-print platforms or in ERP-MES integration systems will manifest fully only after 18-36 months from implementation. In the intervening time, defined as maturation, systematic monitoring of digital KPIs in the learning & growth, internal processes and customer perspectives become crucial to detect any problems that could compromise the financial result.

A certainly relevant topic in the integration of digital and operational KPIs within the BSC pertains to the management of sustainability and traceability.

As previously argued in paragraph 2.3, the growing regulatory pressures on issues of environmental sustainability and traceability along the entire supply chain have made the development of ad hoc metrics necessary, such as for example the percentage of labels certified for recyclability, the adoption rate of sustainable materials at customer request, the carbon footprint per label produced and the capacity to provide digital compliance documentation in real time. Obviously, without an adequate digital traceability system capable of automatically generating the compliance documentation required by customers and regulatory authorities, it becomes extremely onerous and risky to operate in sectors subject to stringent regulatory constraints.

The integration of these KPIs into the Digital BSC can occur primarily through the internal processes perspective, placing them alongside traditional efficiency metrics. However, it must be said that offering high levels of sustainability and traceability represents a competitive driver in the customer perspective, where final customers such as food & beverage brand owners are subjected to growing pressures from customers and authorities to guarantee environmental sustainability and product safety.

2.4.6 Implications for SMEs and self-adhesive label converters

The adoption of a Digital Balanced Scorecard in manufacturing SME contexts presents specific peculiarities that it is appropriate to analyze. Manufacturing SMEs and in particular self-adhesive label converters operating in high-mix/low-volume logic are characterized by a series of structural constraints that often condition the capacity to implement complex performance measurement systems: limited human resources, often with digital skills to be developed; financial constraints that limit the possibility of simultaneous investments in production technologies, digital platforms, and business intelligence systems; decision-making processes concentrated in the hands of a restricted number of people, typically founders or partners who often manifest resistance to change and difficulty with a data-driven approach; and organizational culture often centered on experience and intuition rather than on systematic data analysis.

As highlighted in (*Successful Performance Measurement in SMEs through Simplified Balanced Scorecard, 2012)*, the main risk in BSC adoption in SMEs is constructing an excessively complex system that requires disproportionate efforts compared to available resources and that generates an excessive quantity of data without providing clear operational indications for management.

To avoid this risk, it is necessary to follow some guidelines:

- The number of monitored KPIs must be contained, focusing only on indicators deemed strategic.
- The frequency of measurement and reporting must be calibrated on actual organizational capacities, avoiding imposing excessively tight schedules.
- The indicators must be easily understandable by management and operational personnel, avoiding sophistication that would make the approach worthy of advanced skills.
- The system must be implemented in an incremental manner, starting from a restricted number of essential KPIs and progressively adding new indicators parallel to the subjects' capacity to use them to make informed decisions.

In the self-adhesive label sector, the adoption of a Digital BSC is characterized by high management complexity deriving from the wide number of SKUs, the

variability of technical specifications and the different printing technologies. This leads to a large administrative and coordination burden that involves a large number of available personnel. However, the implementation of an effective Digital BSC in SMEs requires overcoming some organizational barriers. According to what is highlighted by Costantini, Landi and Bonazzi in 2019 in their analysis of the factors that influence the use of the Balanced Scorecard, in Italian SMEs the lack of advanced managerial skills and measurement culture represents one of the main obstacles to the effective adoption of Performance Measurement systems (Costantini et al, 2019). To overcome this barrier, it is necessary to invest in training programs that increase technical skills and improve the interpretive capacity of data. Another limit to overcome concerns data governance, which must be cleaned and standardized before implementation in the BSC. The latter then requires the definition of clear processes for the collection, validation and updating of information. According to what is emphasized by Wolniak and Grebski (2024), in the absence of effective data governance, investments in business intelligence systems and dashboards risk producing unreliable or misleading information, undermining the credibility of the measurement system and generating resistance to its adoption.

A further barrier is represented by the IT investments necessary to implement an integrated data collection and analysis system. The high costs, as previously stated, push toward an incremental and modular approach to equipment capable of building the technological infrastructure step by step, starting from the most critical components for the business and adding more advanced functionalities. According to what is written by ECI Connect in their article on Manufacturing companies on the road to Industry 4.0 – Digital with Balanced Scorecard potential, the incremental approach allows SMEs to distribute investments over time, to learn from experience and to progressively adapt the technological architecture to actual operational needs, reducing the risk of oversized or poorly calibrated investments (ECI Connect, 2018, p.3).

The final barrier concerns resistance to change, especially in family-owned firms or in organizations with a long history, where decision-making is concentrated in the hands of founders who are resistant to formalized control systems. According

to Costantini, in Italian SMEs the success in BSC implementation largely depends on the commitment of owners and top management, who must not limit their role to formal approval but also change how decisions are made, dedicating time to KPI analysis andembedding data analysis in daily routines.

However, in the face of the aforementioned barriers, we can find some advantages in favor of adopting a Digital BSC. The organizational leanness and reduced hierarchies typical of SMEs, as well as the use of modern BI dashboards equipped with intuitive interfaces and advanced visualization capabilities (spider charts, traffic-light dashboards, performance maps), facilitate use by management and allow communicating in an immediate manner the state of corporate performance, with obvious benefits in terms of time spent.

2.4.7 Towards a Digital Balanced Scorecard for Auroflex Srl

The analysis conducted in previous paragraphs highlighted how the Balanced Scorecard, appropriately adapted to the Industry 4.0 context and to the specificities of B2B manufacturing, represents the most suitable framework for integrating economic and financial metrics, operational indicators and digital KPIs into a system oriented towards strategic management.

The present paragraph proposes to apply these concepts to the specific case of Auroflex Srl, an Italian SME operating in the self-adhesive label sector, showing how a Digital Balanced Scorecard can be designed to respond to the concrete needs of measuring digital performance in a context characterized by high operational complexity, limited resources and a digital transformation journey still in its initial phases.

As discussed in the introductory chapter, Auroflex currently finds itself in a critical phase of its digitization journey: the e-commerce site of the Insuperlabel brand is in the launch phase, but the company still operates without a structured customer relationship management system, the artwork approval processes are not completely digitized, and a systematic framework for measuring the return on digital investments is lacking. In this context, a well-designed Digital Balanced Scorecard can perform multiple functions: serve as a strategic planning tool, making explicit the objectives of digital transformation and the expected

relationships between investments, operational improvements and economic results; operate as a monitoring system, allowing management to verify in real time whether the digitization journey is proceeding in the expected direction; and constitute an internal communication mechanism, aligning the various corporate functions (production, commercial, administration) around a shared set of objectives and priorities.

The design of a Digital BSC for Auroflex must start with the identification of the strategic objectives that the company intends to pursue through digital transformation.

Based on the available information and the analysis of the competitive context of the self-adhesive label sector conducted in Chapter 1, it is possible to identify some priority objectives: to increase the share of revenue generated through digital channels, reaching within 24-36 months from the start of e-commerce a significant percentage of the total (for example, 15-20%); to improve operational efficiency through workflow digitization, reducing artwork-to-print lead time and minimizing waste on digital jobs; to increase retention and customer lifetime value of customers acquired through digital channels, exploiting cross-selling and up-selling opportunities made possible by online platforms; and to develop digital skills within the organization, creating a data-driven culture that allows fully exploiting the potential of implemented technologies.

Starting from these strategic objectives, it is possible to articulate a Digital BSC for Auroflex structured along the four classic perspectives, each of which incorporates a balanced set of financial, operational and digital KPIs. In the financial perspective, the key indicators should include: overall Return on Sales (ROS) and EBITDA margin, which measure the company's operating profitability; the share of revenue generated through the Insuperlabel e-commerce channel, which provides a direct measure of the success of the commercial digitization strategy; the evolution of gross margins by channel (e-commerce vs. traditional channels), which allows verifying whether digital is effectively contributing to improving profitability or is mainly attracting low-margin customers; the ROI of cumulated digital investments, which measures the capacity of investments in platforms, CRM and MES systems to generate measurable economic returns; and financial solidity indicators (debt

ratio, Days Sales Outstanding) that verify the sustainability of the transformation journey in the medium-long term.

In the customer, the Digital BSC should emphasize indicators that capture both customer acquisition through digital channels and the retations and development of existing clients. Key indicators include: the B2B e-commerce conversion rate; digital customer lifetime value; the retention rate of e-commerce customers; the Net Promoter Score of digital channels; and the share of existing customers who have adopted the e-commerce platform (adoption rate).

As discussed in paragraph 2.4.2, these indicators should be segmented at least by customer type (key accounts vs. transactional customers vs. new prospects) in order to identify specific patterns and optimization opportunities for each segment. In the internal process's perspective, the Digital BSC for Auroflex must integrate traditional production efficiency KPIs with specific indicators of workflow digitization. Key indicators include: overall Overall Equipment Effectiveness (OEE) and segmented by machine type (traditional printing vs. digital printing); artwork-to-print lead time, a critical indicator in a market where speed represents a competitive factor; the percentage of jobs managed end-to-end in digital workflow; the scrap rate on digital jobs vs. traditional jobs; the degree of ERP-MES-web-to-print integration, measured through the percentage of transactions that occur automatically between systems, and traceability and sustainability indicators, such as the percentage of labels certified for recyclability and the capacity to provide digital compliance documentation in real time, in line with the requirements of EU Regulation 2024/825 on packaging and packaging waste.

From the learning and growth standpoint, the Digital BSC for Auroflex should focus on the development of digital skills and the strengthening of the information infrastructure. Priority indicators include: the level of digital literacy of the commercial team; the percentage of training hours dedicated to digital topics compared to the total; the adoption rate of new digital tools by personnel (CRM, MES, BI dashboards); the quality and completeness of data in management systems, measured through periodic audits; and the level of use of analytics in decision-making processes, which verifies whether the company is effectively developing a data-driven culture.

As emphasized in paragraph 2.4.2, insufficient investments in this perspective would lead to inefficiencies in the other perspectives, compromising the overall success of digital transformation.

It is important to emphasize that this Digital Balanced Scorecard for Auroflex must not be conceived as a static system, but as an evolutionary framework that can and must be adapted over time in response to changes in the competitive context, the evolution of available technologies and the learning deriving from operational experience. In manufacturing SMEs on a digital transformation journey, the BSC must function as a living system, which is periodically reviewed and updated to reflect new strategic priorities and to incorporate learnings deriving from the analysis of collected data.

A crucial aspect of the Digital BSC for Auroflex concerns the method of visualization and communication of information. As discussed in paragraph 2.4.4, in manufacturing SMEs management time is a scarce resource, and the Performance Measurement system's capacity to effectively synthesize a large quantity of information becomes a critical success factor. For Auroflex, it is proposed to implement a single-page managerial dashboard, visually organized according to the four perspectives of the BSC, which allows management to grasp at a glance the overall state of performance and to rapidly identify areas that require attention. This dashboard could use a traffic-light system to immediately signal indicators in the red zone (performance significantly below target), orange (performance slightly below target or with worsening trend) and green (performance in line with or above target), accompanied by trend graphs that show the evolution of key indicators over the last 12 months and spider charts that visualize the company's overall positioning compared to predetermined targets.

In summary, the Digital Balanced Scorecard for Auroflex represents an integrated framework that allows connecting investments in digital transformation (e-commerce platforms, CRM, MES, digital training) to expected operational improvements (lead time reduction, efficiency increase, quality improvement) and to desired economic and financial results (digital revenue growth, margin improvement, retention increase). Through the explication of cause-and-effect relationships between the various perspectives and the systematic monitoring of a

balanced set of financial, operational and digital KPIs, the Digital BSC provides management with a concrete tool to guide and monitor the digital transformation journey, reducing the risks associated with investments and maximizing the probability of success.

The following chapter will present an operational proposal for a Digital Balanced Scorecard for Auroflex, detailing the specific KPIs to be monitored, the targets to be pursued, the data collection methods and the managerial dashboards to be implemented to translate the theoretical framework discussed in this chapter into a concrete and immediately usable tool for the company.

CHAPTER 3

Methodology

3.1 Research Design

This study adopts a research–intervention design centered on a single-case investigation of Auroflex S.r.l., a small manufacturing enterprise operating in the self-adhesive label converting sector. The methodological approach is deliberately configured to serve a dual purpose: on the one hand, it seeks to generate rigorous empirical evidence regarding the relationship between digital maturity and organizational performance in a niche B2B manufacturing context; on the other hand, it aims to co-design and validate a Performance Measurement System specifically, a Digital Balanced Scorecard that is both theoretically grounded and operationally viable within the constraints and opportunities of a digitally transforming SME. This dual orientation aligns the research with the tradition of action-oriented case studies in management accounting and organizational research, where the researcher does not merely observe phenomena from the outside but actively participates in shaping the organizational artifact under investigation.

The research intervention design is particularly appropriate for addressing the three research aims articulated in Chapter 1. The first aim to design and validate an integrated performance measurement framework adapted to the needs of self-adhesive label converters requires a deep engagement with the specific operational and strategic context of the case company, which a traditional observational case study would struggle to achieve. The second aim to quantify the impact of digital maturity on economic and operational performance demands access to longitudinal financial and operational data, as well as the analytical capacity to reconstruct time series and identify lagged causal relationships between technological investments and performance indicators. The third aim to define operational guidelines for the implementation of performance measurement systems necessitates direct involvement in the organizational processes through which such systems are conceived, negotiated, and embedded in managerial routines. In each instance, the research intervention logic enables a form of inquiry that is simultaneously

analytical and generative, diagnostic and design oriented.

Within the single-case design, the researcher occupies a hybrid position inside Auroflex. This embedded role affords privileged access to internal data sources, decision-making processes, and organizational actors, while simultaneously introducing methodological obligations around reflexivity, transparency, and the management of potential biases. The researcher's analytical function involves systematically measuring the company's digital maturity and economic-operational performance, reconstructing multi-year time series of key financial indicators such as Return on Investment (ROI), Return on Sales (ROS), and Earnings Before Interest, Taxes, Depreciation, and Amortization (EBITDA)[19], and mapping the evolution of digital capabilities through operational and digital metrics drawn from Enterprise Resource Planning (ERP) systems, Manufacturing Execution Systems (MES), business intelligence dashboards, web analytics platforms, and internally developed automation tools. The researcher's design-oriented function, by contrast, involves collaborative engagement with company founders and functional managers to iteratively conceptualize, test, and refine the architecture of the Digital Balanced Scorecard, including the identification of strategic perspectives, the definition of causal linkages among objectives, and the selection and calibration of financial and non-financial Key Performance Indicators (KPIs).

A critical methodological consideration concerns the temporal positioning of the research relative to Auroflex's digital transformation trajectory, which directly shapes the nature and availability of empirical data**.** At the time of data collection (Q4 2025), the company had already implemented a range of digital tools and systems across production, administration, and business intelligence functions, generating a substantial archive of historical financial and operational data spanning the period 2020–2025. However, the company's B2B e-commerce platform designed to enable online product configuration, quotation, and order placement by existing and prospective customers while technically operational, had not yet generated any commercial transactions. The platform remained in an extended

[19]EBITDA captures the firm's ability to generate cash from its core operating activities. For companies such as the case study examined, which are characterized by significant investments in tangible fixed assets, EBITDA is particularly relevant .

optimization and testing phase, with the company prioritizing technical refinement, user experience validation, and internal workflow integration over commercial launch. This temporal asymmetry in digital infrastructure maturity has profound implications for the research design and the structure of the Digital Balanced Scorecard itself. It means that while comprehensive retrospective data are available for the financial and internal process perspectives, enabling rigorous quantitative analysis of historical performance trajectories and operational efficiency metrics, no empirical baseline exists for the customer perspective KPIs that depend on e-commerce activity such as conversion rates, digital Customer Lifetime Value (CLV), retention rates, or Net Promoter Scores (NPS) specific to online channels. Consequently, the Digital Balanced Scorecard developed in this research is necessarily hybrid in its temporal orientation: it functions as a retrospective diagnostic tool for assessing past financial and operational performance, while simultaneously serving as a prospective strategic instrument for defining targets and monitoring future performance in the digital customer domain once the e-commerce platform becomes commercially active.

This hybrid design aligns with the original intent of the Balanced Scorecard as originally proposed by Kaplan and Norton (1996), who conceived the framework not only as a measurement system but as a strategic management framework capable of translating strategic vision into operational action and guiding organizational transformation. In the context of Auroflex, the absence of current e-commerce transaction data does not undermine the validity of the Digital Balanced Scorecard; rather, it positions the framework as a *forward-looking governance mechanism* that will enable the company to monitor the effectiveness of its digital customer engagement strategy from the moment of commercial activation. The targets and KPIs defined for the customer perspective are grounded in sector benchmarks drawn from industry reports (FINAT RADAR 2024, Shopify B2B Commerce Report 2024) and academic literature on B2B e-commerce adoption in manufacturing contexts (Saura et al., 2017; Routine Automation, 2025), thereby ensuring that the performance expectations are realistic and contextually appropriate even in the absence of internal historical data. This approach aligns with established practices in strategic performance measurement, where leading

indicators metrics that anticipate future outcomes are deliberately incorporated alongside lagging indicators that reflect past results (Ittner & Larcker, 1998).

The research design combines quantitative and qualitative data sources. From a quantitative perspective, the analysis is based on five consecutive annual financial statements covering the period 2020–2024, supplemented by provisional data for the first semester of 2025. These statements, sourced from official financial reporting channels and internal accounting records, enable the calculation of a comprehensive set of financial indicators, including revenue dynamics, equity evolution, profitability ratios, and working capital management metrics (Days Inventory Outstanding, Days Sales Outstanding, Days Payables Outstanding, Cash Conversion Cycle). The time horizon of the dataset allows for analyzing delayed effects of digital investments on economic outcomes, as noted in previous studies, where the benefits of technological adoption often materialize with a delay of eighteen to thirty-six months (Frederico, 2020; Wolniak & Grebski, 2024). In parallel, operational data are drawn from the company's digital systems, including production management software (CERM), business intelligence tools (Power BI), and internally developed automation solutions such as boots designed to accelerate data collection, raw material procurement workflows, and scrap management processes. These data sources yield metrics such as the automation index of printing lines, the percentage of production jobs managed in digital mode, and operational efficiency indicators derived from time-stamped production records. Importantly, the dataset does not include transactional or behavioral data from the e-commerce platform, as the platform has not yet entered commercial operation. This constraint is explicitly acknowledged in the research design and addressed through the incorporation of sector-derived benchmarks and the prospective specification of digital customer KPIs that will become empirically measurable once the platform is launched.

Together, these quantitative indicators populate the four traditional perspectives of the Balanced Scorecard: financial, customer, internal process, and learning and growth while also feeding the extended dimensions introduced by the Digital Balanced Scorecard framework, namely digital readiness, data maturity, and digital customer experience. The financial perspective is comprehensively operationalized

through historical time-series data, enabling robust retrospective analysis. The internal process perspective benefits from partial operational data extracted from MES and production reports, though gaps remain in the systematic tracking of certain digital workflow metrics, which are reconstructed through interviews and observational data. The customer perspective, as noted, relies on prospective target-setting informed by external benchmarks rather than internal baselines, reflecting the current absence of e-commerce activity. The learning and growth perspective is populated through a combination of qualitative assessments (derived from interviews and self-assessment questionnaires administered to employees) and quantitative proxies (such as the adoption rates of digital tools, measured through system access logs, and the percentage of training hours dedicated to digital competencies, extracted from HR records).

From a qualitative perspective, the study relies on semi-structured interviews with key company members, including the company's founders and the heads of production, sales, and administration. These interviews, designed in accordance with Yin (2018) for case study research, aim to explore strategic priorities, perceptions of digital transformation challenges and opportunities, and the informal decision-making patterns around technology adoption and performance measurement. In addition, the researcher conducts direct observation of operational processes on the shop floor and in back-office functions, participates in internal meetings focused on digital initiatives and performance monitoring, and reviews internal documents such as strategic plans, investment proposals, and ad hoc performance reports. This qualitative material serves multiple functions: it contextualizes the quantitative patterns observed in the financial and operational data, it surfaces tacit knowledge and organizational routines that are not captured in formal information systems, and it provides the relational and interpretive basis for the co-design of the Digital Balanced Scorecard. Crucially, the qualitative data also enables the researcher to understand the organizational rationale behind the delayed commercial launch of the e-commerce platform, a decision driven by the company's commitment to ensuring technical robustness, user experience quality, and internal process readiness before exposing the platform to customers and risking reputational damage through premature deployment.

The research is structured into three interconnected phases, each characterized by distinct analytical and intervention activities. The first phase, labeled the *diagnostic phase,* involves a structured assessment of Auroflex's current economic and digital maturity profile. Using financial data from 2020-2024, examines trends in key indicators (ROI, ROS, EBITDA, working capital components, revenue growth, cost ratios, and equity dynamics) and analyzes their trajectories in relation to significant digital investments and key technological upgrades. Concurrently, the researcher assesses the company's digital maturity by mapping existing digital tools and systems, evaluating their degree of integration and functional coverage, and identifying gaps or redundancies in the technological architecture. This diagnostic work establishes the reference point against which subsequent changes will be measured and provides the factual grounding for strategic discussions in the co-design phase. Critically, this phase also reveals the absence of customer-facing digital transaction data, which led to a methodological decision to design the customer perspective of the Digital Balanced Scorecard in a prospective rather than retrospective mode.

The second phase, the *design and co-creation phase*, represents the core intervention component of the research. Working collaboratively with company managers, the researcher facilitates the joint definition of the Digital Balanced Scorecard's architecture. This process begins with the articulation of strategic objectives across the scorecard's perspectives, followed by the construction of causal maps (or strategy maps) that visually represent the hypothesized linkages among objectives for instance, how improvements in data maturity are expected to enhance internal process efficiency, which in turn should drive customer satisfaction and ultimately financial performance. The causal mapping phase daw on both the theoretical frameworks reviewed in Chapter 2 and the managers' practical knowledge of the company's competitive dynamics and operational constraints. In a second step, the group identifies KPIs linked to each strategic objective, defining data sources and measurement procedures, setting targets, and assigning responsibility. For the financial and internal process perspectives, targets are calibrated based on observed and historical trends. For the customer perspective, targets are defined through a combination of sector benchmarks and strategic

ambition, acknowledging that these will be validated and adjusted once empirical data become available following the e-commerce platform's commercial launch. This collaborative design process is iterative, with preliminary versions of the scorecard being reviewed, challenged, and refined through multiple rounds of discussion.

The last stage of the process consists of *testing and refinement*, the application of the Digital Balanced Scorecard to both historical and recent data, with the aim of assessing its coherence and practical usefulness. The scorecard is populated with available data, calculates the selected KPIs for relevant time periods, and checks whether observed trends align with the causal hypotheses embedded in the strategy map. For perspectives where historical data are available, namely, financial and internal process this validation relies on quantitative analysis. For the customer perspective, validation is mainly conceptual, relying on the strength of the proposed linkages and the alignment of targets with industry norms. Discrepancies between expected and observed relationships in the retrospective perspectives prompt further reflection and may lead to adjustments in the choice of indicators, the specification of targets, or the interpretation of causal linkages. Importantly, this phase also assesses the feasibility of sustaining the performance measurement system over time, considering factors such as data availability, organizational capacity for data collection and analysis, and the alignment of the scorecard with existing managerial routines and governance structures. The outputs of this phase include a refined version of the Digital Balanced Scorecard, a set of empirical findings regarding the relationship between digital maturity and performance in the case company (focused primarily on the financial and operational dimensions where longitudinal data exist), and a series of operational guidelines for implementing similar systems in comparable organizational contexts - including guidance on how to manage the temporal misalignment between digital infrastructure deployment and data availability that is common in SMEs undergoing digital transformation.

A distinctive feature of this research design is the explicit acknowledgment of a *hybrid human-AI arrangement* in the analytical workflow. While human actors, the researcher and the company’s managers retain ultimate responsibility for problem

framing, strategic interpretation, and decision-making, artificial intelligence tools are employed to support data processing, time-series analysis, pattern recognition, and scenario modeling. Specifically, AI-assisted techniques are used to explore potential lagged effects between digital investments and financial outcomes, to identify non-linear relationships or threshold effects in the data, and to simulate alternative configurations of the Digital Balanced Scorecard under different strategic assumptions. This hybrid approach reflects a pragmatic recognition that contemporary organizational research increasingly leverages computational capabilities to manage data complexity, while simultaneously maintaining that the interpretive and normative dimensions of inquiry remain fundamentally human endeavors. The use of AI is particularly valuable in this research given the uneven data availability across the scorecard's perspectives; AI tools enable the exploration of plausible scenarios for the customer perspective based on synthetic data and benchmark-derived parameters, thereby supporting strategic planning even in the absence of empirical transaction histories.

The choice of a single-case design warrants explicit justification, particularly given ongoing debates in management research about the generalizability of findings derived from idiosyncratic organizational settings. In this study, the single case is not selected as a representative sample of a broader population but as a *revelatory instance* that embodies theoretically significant characteristics (Yin, 2018). Auroflex operates in a niche B2B manufacturing sector self-adhesive label converting that is under-represented in the academic literature on Performance Measurement Systems and digital transformation, yet strategically important within the broader packaging value chain. The company is currently navigating a fragmented digital transition, having adopted multiple digital tools and systems without yet achieving full integration or systematic performance measurement, and is on the cusp of launching a customer-facing digital platform that will fundamentally alter its commercial model. This positioning makes Auroflex an information-rich case for exploring the practical challenges and opportunities associated with designing and implementing a Digital Balanced Scorecard in a context characterized by high technological complexity, significant capital intensity, and the coexistence of traditional and digital operational logics. The depth

of analysis enabled by the single-case design, coupled with the longitudinal financial and operational data available, provides a level of contextual detail and temporal granularity that would be difficult to replicate in a multi-case or survey-based study. Furthermore, the research–intervention logic prioritizes the generation of actionable knowledge and the co-creation of a functional organizational artifact, outcomes that are inherently context-dependent and require sustained engagement with a specific organizational setting.

The temporal positioning of the research occurring at the inflection point between digital infrastructure deployment and commercial activation introduces both constraints and opportunities. The constraint lies in the impossibility of empirically validating the customer-facing components of the Digital Balanced Scorecard within the timeframe of the study, as the e-commerce platform has not yet generated the transactional and behavioral data necessary for calculating conversion rates, digital CLV, or other customer-centric KPIs. This limitation is explicitly acknowledged and will be discussed further in Chapter 5 (Limitations). The opportunity, however, lies in the ability to observe and document the process of designing a performance measurement framework prospectively, that is, in anticipation of a major organizational and commercial shift rather than in response to it. This prospective stance is methodologically valuable because it captures the forward-looking, anticipatory dimension of strategic performance measurement that is often obscured in retrospective case studies. It reveals how organizations define success criteria, set targets, and establish monitoring mechanisms for initiatives that have not yet been empirically tested, thereby shedding light on the role of benchmarks, analogies, and managerial judgment in performance measurement design. Moreover, it aligns the research with the normative purpose of the Balanced Scorecard as a tool for strategic guidance and organizational learning, rather than merely retrospective accountability.

In summary, the research design balances empirical analysis and organizational intervention. The researcher plays a dual role, integrates quantitative and qualitative data, and unfolds through a phased process of co-creation and testing. Crucially, it recognizes and responds to different availability of historical and digital data on one hand, and prospective digital customer metrics on the other by designing a Digital

Balanced Scorecard that functions both as a retrospective diagnostic tool and as a prospective strategic instrument. This methodological architecture is deliberately constructed to address the three research aims: it enables the design and validation of a sector-specific Digital Balanced Scorecard adapted to the realities of SME digital transformation, it supports the quantification of relationships between digital maturity and performance through longitudinal data analysis where such data exist, and it generates operational insights grounded in the lived experience of implementing a performance measurement system at the critical juncture between digital infrastructure deployment and commercial activation. The resulting knowledge is intended to be both locally useful providing Auroflex with a functional tool for strategic management as its e-commerce platform enters operation and transferable to other organizations operating in similar contexts, thereby contributing to the broader scholarly and practitioner conversation on digital transformation and performance measurement in B2B manufacturing.

3.2 Criteria for the choice of the company under investigation

The selection of Auroflex S.r.l. as the empirical locus for this research–intervention was guided by a deliberate and theoretically informed set of criteria rather than by convenience or opportunistic access. Four interrelated dimensions structured the selection logic: sectoral relevance, phase of digital transition, access to economic-financial and operational data, and operational accessibility afforded by the researcher's organizational role. Each of these dimensions is examined in turn, demonstrating how Auroflex constitutes an appropriate and information-rich case for designing, testing, and refining a Digital Balanced Scorecard tailored to the needs of a digitally transforming SME in the self-adhesive label converting industry.

The first dimension concerns the sectoral relevance of the company. Auroflex operates in the self-adhesive label converting sector, a specialized segment of the packaging industry that plays a strategically important role in the European manufacturing value chain. Industry reports estimate the European market for self-adhesive labels at more than seventeen billion euros, with consolidated growth rates driven by demand from sectors such as food and beverage, pharmaceuticals,

personal care, and logistics (Finat Radar, 2024); (Deep Market Insights, 2024). Notwithstanding its economic relevance, the self-adhesive label sector remains scarcely addressed in the academic literature on Performance Measurement Systems, particularly in relation to digital transformation contexts. The existing literature on the Balanced Scorecard and its digital extensions has concentrated on larger enterprises and high-technology industries, leaving a evident gap regarding how performance measurement frameworks can be adapted to the unique characteristics of specialized B2B manufacturing industries.

Self-adhesive label converters face a unique combination of challenges and opportunities: they operate in capital-intensive production environments characterized by high technological complexity (including processes such as flexographic printing, hot stamping, embossing, screen printing, and die-cutting); they serve diverse customer segments with highly customized product requirements; they must balance pressures for operational efficiency with the need for continuous innovation; and they are increasingly compelled to integrate digital tools across the value chain, from design and production planning to customer relationship management and supply chain coordination. Auroflex embodies these sectoral characteristics and can therefore serve as a paradigmatic case for exploring how digital-operational alignment and performance measurement can be conceptualized and enacted in a context that has received limited scholarly attention. By focusing on this sector, the research addresses a substantive empirical gap while simultaneously generating insights that are likely to resonate with other small and medium-sized enterprises operating in similarly specialized and capital-intensive manufacturing niches.

The second-dimension concerns Auroflex's current phase of digital transition, which positions the company as a suitable case for applied research on Performance Measurement Systems. Auroflex has already introduced multiple digital tools, having implemented a range of technological tools and systems aimed at enhancing operational efficiency with decision-making capabilities, and customer engagement. These include business intelligence platforms such as Power BI, which enable the visualization and analysis of performance data; production management software such as CERM, which supports the planning and tracking of

manufacturing workflows; and a suite of internally developed automation solutions designed to accelerate routine processes and reduce manual intervention. Among the latter are several bots developed in-house and given personified names such as Clair, Ugo, and Kate that automate tasks related to data collection, raw material procurement, and scrap management. These initiatives indicate a clear approach to digitalization and a willingness to experiment with emerging technologies.

However, the company's digital landscape is characterized by fragmentation rather than integration. The various digital tools and systems operate largely in isolation, with limited interoperability and no overarching framework for coordinating their outputs or aligning them with strategic objectives. Moreover, Auroflex lacks a structured Customer Relationship Management (CRM) system capable of systematically capturing and analyzing customer interaction data across the full lifecycle of the commercial relationship. Most critically, while the company has developed a B2B e-commerce platform to enable online product configuration, quotation, and order placement, this platform remains in an extended optimization phase and has not yet entered commercial operation. At the time of data collection (Q4 2025), the platform had generated zero transactions, as management prioritized technical refinement, user experience validation, and internal workflow integration over premature commercial launch. This situation exemplifies a pattern commonly observed in the digital transformation literature: many SMEs pursue scattered technological initiatives in response to immediate operational needs or competitive pressures but struggle to develop the integrative mechanisms necessary to translate these initiatives into sustained performance improvements (Naeem & Garengo, 2022); (Digital Maturity in Europe, 2022).

Auroflex thus represents a revelatory case of *fragmented digitalization at a critical inflection point*. The challenge is not to initiate digital adoption that has already occurred but rather to measure, integrate, and strategically align existing digital capabilities while simultaneously preparing to monitor and optimize the performance of a nascent digital commercial channel. This positioning makes the company an ideal setting for co-designing a Digital Balanced Scorecard that can serve both as a diagnostic tool to assess the current state of digital maturity across retrospective data (financial, operational) and as a prospective governance

mechanism to guide future investments and performance monitoring once the e-commerce platform becomes commercially active. The temporal positioning at the threshold of e-commerce activation is methodologically valuable, as it enables the research to capture the anticipatory, forward-looking dimension of strategic performance measurement, how organizations define success criteria and establish monitoring frameworks for major initiatives before empirical validation is possible. The third-dimension concerns access to economic-financial and operational data, fundamental for reliable empirical analysis in research–intervention designs. The availability of data in this case is characterized by a structural difference that reflects the company's current stage of digital transformation. The researcher has obtained access to five consecutive annual financial statements covering the period from 2020 to 2024, supplemented by provisional data for the first semester of 2025. These statements, sourced from established financial information providers (Cerved and Centrale dei Bilanci) and verified against internal accounting records, provide a comprehensive view of the company's economic evolution during a period influenced by external shocks (including the COVID-19 pandemic and subsequent supply chain disruptions) and internal changes (such as significant investments in production technologies and digital infrastructure). From these statements, the researcher has extracted and calculated key financial indicators including:

- Revenues (measured as the value of production),
- Operating costs and Equity,
- Return on Investment (ROI),
- Return on Sales (ROS), and Earnings Before Interest,
- Taxes, Depreciation and Amortization (EBITDA),

as well as working capital management metrics such as:

- Days Inventory Outstanding (DIO),
- Days Sales Outstanding (DSO),
- Days Payables Outstanding (DPO),
- Cash Conversion Cycle (CCC).

The availability of time-series data spanning multiple years is particularly valuable for modeling the temporal dynamics of digital transformation, as the academic

literature consistently demonstrates that the economic benefits of digital investments often materialize with a lag of eighteen to thirty-six months (Frederico, 2020); (Wolniak & Grebski, 2024). This temporal granularity enables the researcher to investigate whether observed improvements in financial performance can be causally attributed to prior digital initiatives, or whether the relationship is spurious or mediated by other factors. The financial dataset thus provides a robust foundation for populating the *financial perspective* of the Digital Balanced Scorecard with empirically grounded historical baselines and for establishing realistic improvement targets based on observed trends.

Beyond financial data, the study relies on operational data generated by the company's information systems. Production reports extracted from the Manufacturing Execution System (CERM) provide data on manufacturing efficiency, machine utilization patterns, and process quality indicators, although the data are incomplete and unevenly recorded. Digital analytics platforms and internally developed automation tools provide indicators of process automation, such as the percentage of data collection tasks handled by bots and the proportion of production jobs that involve digital workflow components. Time-stamped production records, visible in internal dashboards and illustrated in operational reports, make it possible to calculate throughput times and help in the identification of bottlenecks. These operational data, while less comprehensive and systematic than the financial data, nevertheless provide enough detail to populate key indicators within the *internal process perspective* of the Digital Balanced Scorecard.

However, the dataset does not include transactional or behavioral data from the e-commerce platform, as the platform has not yet entered commercial operation and has generated no customer transactions. This absence effects the *customer perspective* of the Digital Balanced Scorecard. Metrics such as e-commerce conversion rates, digital Customer Lifetime Value (CLV), online customer retention rates, Net Promoter Scores (NPS) specific to digital channels, and the adoption rate of the platform by existing customers cannot be calculated from empirical data. Consequently, the customer perspective must be designed in a *prospective mode*, with KPIs defined and targets established based on sector

benchmarks drawn from industry reports (Finat Radar, 2024), (Shopify B2B Commerce Report, 2024), academic literature on B2B e-commerce adoption in manufacturing contexts (Saura et al., 2017); (Routine Automation, 2025), and strategic ambitions articulated by company management during the co-design process. This prospective orientation is not a methodological limitation per se, but rather a reflection of the temporal reality of the case: the Digital Balanced Scorecard is being designed at the inflection point between digital infrastructure deployment and commercial activation, positioning it as a forward-looking strategic tool that will guide performance monitoring once empirical data become available.

Similarly, while some data are available for the *learning and growth perspective* such as system access logs that proxy the adoption rates of digital tools by employees, and HR records documenting the allocation of training hours to digital competencies these data are incomplete and require supplementation through qualitative methods. Employee surveys and semi-structured interviews, conducted following the guidelines proposed by Yin (2018) for case study research, are used to assess the digital literacy of commercial and operational staff, to gauge the organizational culture's receptivity to data-driven decision-making, and to identify barriers to the effective use of digital systems. The combination of partial quantitative data and rich qualitative insights enables a holistic assessment of the organizational capabilities that underpin digital transformation.

In summary, the data availability profile is asymmetric but methodologically sufficient for the research aims. The financial perspective benefits from comprehensive, longitudinal, empirically robust data that enable rigorous quantitative analysis of historical performance trajectories and their relationship to digital investments. The internal process perspective is supported by partial operational data that, while fragmented, are adequate for identifying efficiency patterns and establishing baseline metrics for key processes. The customer perspective, by necessity, relies on prospective target-setting informed by external benchmarks and strategic judgment, reflecting the current absence of transactional e-commerce data. The learning and growth perspective is populated through a mixed-methods approach that combines quantitative proxies with qualitative assessments. This asymmetry is not merely acknowledged but is actively integrated

into the research design, shaping the dual temporal orientation of the Digital Balanced Scorecard as both a retrospective diagnostic tool and a prospective strategic instrument.

The fourth dimension relates to the insider position, which is central to the feasibility and validity of the research–intervention design. The researcher works inside Auroflex, a positioning that provides several critical methodological advantages while also introducing awareness of potential bias and ethical implications. This embedded role enables a high level of organizational access that would be difficult or impossible to achieve as an external observer. Concretely, the researcher can conduct semi-structured interviews with the company's founders and with the heads of key functional areas, including production, sales, and administration. These interviews facilitate uncovering informal knowledge, strategic priorities, and interpretive frameworks that shape how organizational actors understand performance, digital transformation, and competitive dynamics.

The researcher also engages in direct observation of operational processes, both on the shop floor where production activities such as printing, finishing, and quality control occur and in back-office functions, where administrative, commercial, and planning tasks are performed. This observational access allows the researcher to discern the gap between formal procedures and actual practices, to identify ad hoc solution, and to appreciate the organizational context in which digital tools are embedded. Additionally, the researcher participates in internal meetings focused on digital initiatives, performance monitoring, and strategic planning, thereby gaining real-time insight into decision-making processes and the political and cultural dimensions that influence organizational change. This level of immersion is characteristic of research–intervention approaches in management accounting and organizational studies, where the objective is not merely to describe organizational phenomena but to collaboratively shape them (Jönsson & Lukka, 2006).[20]

The embedded position enables the co-design of the Digital Balanced Scorecard to be genuinely collaborative, with the researcher and company managers engaging as partners who bring complementary forms of expertise theoretical and

[20] They are central figures in the development and application of interventionist research within management accounting and organizational studies. "*There and back again: Interventionist research in management accounting*" 2006.

methodological knowledge on one side, and contextual and operational knowledge on the other. This partnership is essential for ensuring that the resulting performance measurement framework combines theorical rigor with practical feasibility, responding to the third research question, defining operational guidelines for system implementation. Moreover, being an insider allows access to financial data, strategic documents, and candid discussions about organizational challenges that might not be disclosed to external researchers. At the same time, the embedded role necessitates careful reflection on potential bias; the researcher must remain conscious of how their combined role as analyst and practitioner may shape data collection, interpretation, and the co-design process. This reflexivity is maintained through regular documentation of methodological decisions and transparent reporting of data sources with analytical procedures, and ongoing dialogue with academic supervisors who provide an external critical perspective.

The embedded role also provides direct insight into the organizational rationale behind the delayed commercial launch of the e-commerce platform, a factor that is central to understanding the data availability profile**.** Through participation in internal discussions and access to strategic planning documents, the researcher understands that management's decision to keep the platform in an extended optimization phase stems from a deliberate risk-management logic: the company is unwilling to expose a customer-facing digital channel to market until it is confident that the platform’s technical performance, user experience, and integration with back-end systems (order processing, production planning, inventory management) meet the high standards expected by its premium clientele in the wine, spirits, and craft beverage sectors. This cautious approach reflects the company’s broader competitive strategy, which emphasizes quality, reliability, and personalized service over first-mover advantage or rapid scaling. The researcher's access to this contextual knowledge enriches the interpretation of the data availability asymmetry and underscores that the absence of e-commerce transaction data is not a methodological accident but a strategic choice that shapes the temporal positioning of the research.

Taken together, these four dimension show that, Auroflex was selected not merely for convenience, but because the company brings together sectoral relevance, an

ongoing and fragmented digital transition at a critical turning point, access to longitudinal financial records coupled with prospective access to emerging digital customer data, and the organizational accessibility required for a research–intervention approach.

Sectoral *relevance* ensures that the research addresses a substantive gap in literature and generates findings that are pertinent to an economically significant but understudied industry. Self-adhesive label converting embodies the characteristics of niche B2B manufacturing high capital intensity, product customization, technological complexity, and the imperative to integrate digital tools across operational and commercial functions making Auroflex a paradigmatic case for exploring digital performance measurement in contexts that have received limited scholarly attention.

The company's *phase of digital transition* positions it as a relevant case of the challenges faced by SMEs seeking to move from disconnected technological initiatives to integrated strategic alignment. The presence of multiple digital tools and systems that operate in isolation, combined with the imminent commercial activation of a B2B e-commerce platform that has been developed but not yet deployed, creates a empirically relevant context for observing how organizations design performance measurement frameworks prospectively in anticipation of major organizational and commercial shifts.

The availability of data, while asymmetric across the four perspectives of the Balanced Scorecard, is methodologically sufficient and strategically informative. Comprehensive financial and partial operational data enables rigorous retrospective analysis of the relationship between digital investments and economic-operational performance, addressing the second research aim. The absence of e-commerce transactional data, far from being a limitation, positions the research at the critical juncture between infrastructure deployment and commercial activation, enabling the study to capture the prospective, anticipatory dimension of strategic performance measurement. This stage reflects that the Digital Balanced Scorecard as a tool for strategic guidance and organizational learning, not merely retrospective accountability.

The researcher's *embedded organizational role* provides the depth of engagement

and collaborative access necessary for co-designing a performance measurement framework that is both grounded in theory and workable in practice. It enables the collection of qualitative data that contextualizes quantitative patterns, support gradual refinement of the Digital Balanced Scorecard through sustained dialogue with managers, and ensures that the resulting framework is aligned with the organizational realities and strategic priorities of the case company.

Overall, these criteria indicate that Auroflex as an information-rich case through which support both theorical reflection and managerial application regarding the design and deployment of Performance Measurement Systems in digitally transforming B2B manufacturing SMEs. The link between these criteria and the research aims is clear: Auroflex provides the organizational context, data infrastructure and collaborative environment necessary to design and refine a Digital Balanced Scorecard that is adapted to the distinctive needs and constraints of self-adhesive label converters managing digital transformation at the threshold of major commercial and operational change.

3.3 Data-collection protocol and analysis techniques

The *data collection protocol* for this research was designed to ensure systematic, comprehensive, and methodologically rigorous gathering of empirical evidence across the four perspectives of the Digital Balanced Scorecard - financial, customer, internal processes, and learning and growth - while also capturing the qualitative insights necessary to contextualize quantitative patterns and inform the co-design process. The protocol reflects the mixed-methods logic articulated in Section 3.1, integrating structured quantitative data extraction from archival sources and information systems with semi-structured qualitative inquiry through interviews, observation, and documentary analysis. Throughout the *data collection process*, explicit attention was paid to issues of data quality, reliability, and triangulation, consistent with established standards for case study research in *management accounting and organizational studies* (Yin, 2018; Eisenhardt, 1989).

The data collection phase took place between November 2025 to January 2026, a period during which the researcher had direct access to the company's internal data sources, personnel, and decision-making processes. This timeframe enabled the

reconstruction of past performance trends (2020–2025) while also capturing real-time strategic discussions and organizational dynamics related to the expected launch of the e-commerce platform. The protocol was designed to align with the four perspectives of the *Balanced Scorecard*, facilitating the subsequent population of KPIs and the validation of causal linkages embedded in the strategic map.

Financial data constitutes the main empirical basis of the retrospective component of the *Digital Balanced Scorecard*, providing longitudinal evidence of economic performance and the makes it possible to study delayed impacts between digital investments and financial outcomes. The financial data collection strategy prioritized the use of *official, externally verified sources* to ensure data reliability, supplemented by internal accounting records where necessary to access more granular information or to obtain provisional data for the most recent fiscal period.

Primary data source: Official annual financial statements (2020–2024)

The core financial dataset comprises five consecutive annual financial statements (Bilancio d'Esercizio) covering fiscal years 2020, 2021, 2022, 2023, and 2024. These statements were obtained from two established financial information providers serving the Italian market: **Cerved**[21] and **Centrale dei Bilanci**.[22] Both sources draw on official filings deposited with the *Italian Business Register* (Registro delle Imprese) and apply standardized quality controls to ensure data accuracy and completeness. The use of externally verified statements, rather than relying solely on internal management accounts, enhances the external validity and auditability of financial analysis.

From these statements, the following data elements were systematically extracted and compiled into a longitudinal dataset:

- **revenue**: total production value as reported in the reclassified income statement, representing the comprehensive measure of operating turnover including sales of goods and services, inventory variation, and capitalized internal costs.
- **Operating costs**: aggregated operating expenses including cost of raw

[21] https://www.cerved.com/
[22] https://sirc.centraledeibilanci.it/sirc/

materials, cost of services, cost of use of third-party goods, and other operating expenses, but excluding depreciation, amortization, financial charges, and taxes.

- **Shareholders'equity:** book value of equity as reported in the reclassified balance sheet, reflecting cumulative retained earnings and capital contributions.
- **Operating income (EBIT)**: *Earnings Before Interest and Taxes*, calculated as the difference between revenue and total operating costs, serving as the primary measure of operating profitability.
- **Depreciation and amortization**: annual charges reflecting the systematic allocation of capital expenditure on tangible and intangible assets over their useful economic lives.
- **Working capital components (2022–2024 only)**: For the three most recent fiscal years, detailed balance sheet data were extracted to support the calculation of working capital management metrics:

  - **inventory**: Value of raw materials, work-in-progress, and finished goods held at year-end.

  - **Trade receivables**: amounts owed by customers for goods and services delivered but not yet paid.

  - **Trade payables (Debiti verso Fornitori)**: amounts owed to suppliers for goods and services received but not yet paid.

  - **Cost of goods sold** approximated as the sum of raw material costs, service costs, and costs for the use of third-party goods, consistent with standard practices for manufacturing SMEs in the absence of a formally reported COGS figure.

  These data allowed the calculation of *Days Inventory Outstanding (DIO), Days Sales Outstanding (DSO), Days Payables Outstanding (DPO)*, and the resulting *Cash Conversion Cycle (CCC)*, offering a view of working capital management and its evolution over time.

Provisional data for first half 2025 (H1 2025) to extend the temporal coverage of the analysis into the current fiscal year and capture the most recent financial performance, *helping evaluate the first semester of 2025* were obtained from the

company's internal accounting system. These data, while not yet audited or officially filed, were compiled following the same accounting criteria applied to the annual statements and were reviewed by the company's administrative manager to confirm data coherence. The provisional H1 2025 data include revenue, operating costs, shareholders' equity, and the key profitability indicators (ROI, ROS, EBITDA) recalculated on a semi-annual basis.

3.3.1 Calculation of financial KPIs

From the raw financial data, the following performance indicators were systematically calculated for each fiscal year (and provisionally for H1 2025):

1. **Return on Investment (ROI)**: calculated as (EBIT / Total Operating Capital Invested) × 100, where operating capital invested is approximated by the sum of shareholders' equity and net debt. ROI measures the efficiency with which invested capital generates operating profit, independent of financing structure. This indicator is particularly relevant for capital-intensive industries such as label converting, where substantial investments in machinery and technology are required.
2. **Return on Sales (ROS)**: calculated as (EBIT / Revenue) × 100. ROS measures the operating profitability margin, showing the share of operating profit per euro of sales after covering all operating costs. It provides insight into pricing and cost management.
3. **EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization)**: calculated as Revenue − Operating Costs (excluding depreciation and amortization), or equivalently as EBIT + Depreciation + Amortization. EBITDA is often used as an approximation of operating cash flow, excluding non-cash items.
4. **EBITDA Margin**: calculated as (EBITDA / Revenue) × 100. The EBITDA margin indicates the percentage of revenue that translates into operating cash flow before the impact of depreciation, interest, and taxes. It is commonly used in manufacturing contexts to assess operational efficiency and compare firms with different capital structures and depreciation policies.

5. **Working capital management metrics (2022–2024)**:
    1. **DIO = (Inventory / Cost of Goods Sold) × 365**
    2. **DSO = (Trade Receivables / Revenue) × 365**
    3. **DPO = (Trade Payables / Cost of Goods Sold) × 365**
    4. **CCC = DIO + DSO − DPO**

These calculations followed standard financial analysis methodologies (Brealey, Myers, & Allen, 2020) and were verified against sector benchmarks reported in industry publications (FINAT RADAR 2024).

3.3.2 Digital investment data

To support the analysis of the relationship between digital transformation and financial performance, and to calculate the *ROI of digital investments*, data on cumulative digital capital expenditures were extracted from invoices, purchase orders, and internal investment tracking documents. The following categories of digital investment were identified and quantified:

- **e-commerce platform development and ERP-CERM-web-to-print integration**: €120,832.40 in 2024 and €58,712.50 in 2025, for a cumulative total of **€179,544.90**. These expenditures covered the design, development, and technical integration of the Auroflex B2B e-commerce platform with the company's existing ERP (CERM, Automation Engine, ARCA) and production management systems.
- **Power BI licenses**: €1,684.80 per year for 2024 and 2025, totaling **€3,369.60**. These costs supported the implementation of business intelligence dashboards for performance monitoring and data visualization.
- **CRM Auroflex MBO**: the customer relationship management system was developed internally by the company's IT team and does not have an explicit external cost associated with its development. This reflects a relevant internal effort not captured in financial statements in terms of internal working hours.
- **Digital training and competency development**: no specific tracking was available during the period under study. Training expenditures are aggregated in the company's HR budget but are not disaggregated by

thematic area (digital vs. non-digital).

- **Total traceable digital investment (2024–2025)**: Approximately **€183,000**, concentrated in the e-commerce platform and integration work. This figure can be considered a minimum estimate for calculating the ROI of digital transformation, though it likely underestimates the actual overall investment when internal labor, opportunity costs, and training are included.

The customer perspective of the *Digital Balanced Scorecard* presented a unique data collection challenge due to the temporal misalignment between the research period and the commercial activation of the e-commerce platform. As documented in Section 3.1, the platform, while technically operational, had generated zero transactions at the time of data collection (Q4 2025). Consequently, transactional and behavioral metrics typically associated with digital customer engagement - such as conversion rates, digital Customer Lifetime Value (CLV), retention rates, and Net Promoter Scores specific to online channels - could not be calculated from empirical data.

The data collection process for the customer's perspective therefore followed *two complementary approaches*:

1. capturing baseline perceptions, satisfaction, and frequency and volume of orders of the existing traditional customer base through a structured survey,
2. extracting descriptive analytics on the customer portfolio from the company's ERP and CRM systems to define future KPI targets.

3.3.3 Customer satisfaction survey

The company administered a customer satisfaction survey to a sample of Auroflex's existing client base in December 2025. The questionnaire aimed to gather ratings of customer perceptions across multiple dimensions of service quality, product performance, and reaction time to customer requests, using a 5-point Likert scale (1 = very dissatisfied, 5 = very satisfied) for closed-ended items, with additional open-ended items to capture qualitative feedback. The survey was based on established practices in customer satisfaction measurement (Parasuraman, Zeithaml, & Berry, 1988) and adapted to B2B dynamics typical of the label

converting sector.

The survey comprised *27 closed-ended questions* organized into the following thematic areas:

1. **Customer Service and support**: speed and effectiveness of response to requests, courtesy and availability of customer service team, preparedness of technical personnel (7 questions).
2. **Technical quality and product performance**: preparedness of technical staff, consistency of technical advice with delivered products, quality of materials used (4 questions).
3. **Lead times and responsiveness**: speed of response to quotation requests, order confirmation issuance, production turnaround, reprint requests, and delivery times (9 questions).
4. **Communication effectiveness**: clarity and informativeness of communication on products and services, effectiveness of order status updates, engagement and humanization of online and offline communication channels (5 questions).
5. **Product conformance and materials**: conformance of delivered labels to specifications (print quality, color accuracy, finishing), performance of materials used (paper, adhesive, foil) (2 questions).
6. **Digital awareness and engagement**: awareness of the Insuperlabel initiative, interest in receiving a monthly digital newsletter on label printing trends (2 questions).

The questionnaire also included spaces for qualitative comments to share additional impressions on each thematic area. It was distribuited by email to a sample of active customers, with a response collection period of three weeks.

The survey yielded responses from an unspecified number of customers (the exact response rate and sample size were not documented in the data file provided, representing a limitation in the data collection protocol. The aggregated results across the 27 quantitative items revealed the following patterns:

- **overall mean satisfaction score**: **4.66 out of 5.00** (93.2% on a percentage scale), indicating very high levels of customer satisfaction.
- **Median satisfaction score**: 4.75 out of 5.00.

- **Standard deviation**: 0.22, indicating low dispersion and consistently high ratings across questions.
- **Range**: Minimum score 4.06, maximum score 4.88, with no question falling below the "satisfied" threshold.

Top-rated service attributes (scores ≥ 4.88):

- effectiveness and speed of customer support (4.88).
- Preparedness of technical personnel (4.88).
- Courtesy and availability of customer service team (4.88).
- Speed of response to reprint requests (4.88).

Lowest-rated attributes (scores ≤ 4.44):

- interest in receiving a monthly digital newsletter (4.06).
- Awareness of the Insuperlabel contest (4.12).
- Speed of quotation issuance (4.44).
- Speed of order confirmation issuance (4.44).
- Humanization and engagement of online and offline communication (4.44).

Thematic breakdown by category (mean scores):

- customer service: 4.81/ 5.00 (96.2%).
- Technical quality: 4.78/5.00 (95.6%).
- Timeliness and speed: 4.66/5.00 (93.2%).
- Product quality: 4.62/5.00 (92.4%).
- Communication effectiveness: 4.59/5.00 (91.8%).
- Digital awareness and engagement: 4.31/5.00 (86.2%) notably the lowest-rated dimension.

While the survey did not explicitly employ the standard Net Promoter Score question format (“How likely are you to recommend Auroflex to a colleague or business partner?” on a 0-10 scale), the high mean satisfaction score can be converted to an approximate NPS-equivalent metric by scaling the 1-5 Likert responses to a 0-10 scale. Using a linear transformation (meaning Likert score × 2), the satisfaction data translate to an **NPS proxy of approximately 9.3 out of 10**, placing the majority of respondents in the "Promoter" category (scores ≥ 9).

This suggests a highly loyal and satisfied customer base, with an estimated **66–75%**

of customers functioning as active promoters of the brand.
The survey results establish a strong baseline of customer satisfaction in the traditional (non-digital) service model, with particular strength in customer service responsiveness, technical expertise, and product quality.

However, the relatively lower scores in digital awareness and interest in digital communication channels (newsletter) highlight a potential adoption barrier for the e-commerce platform: customers are highly satisfied with the current personalized, relationship-driven service model and may require education, incentives, or demonstrated added value to transition to self-service digital channels. This insight is critical for informing the customer perspective targets in the Digital Balanced Scorecard and for designing the commercial activation strategy for the e-commers.

To complement the survey data and to establish the sizing and segmentation of the customer base for prospective digital adoption modeling, transactional data were extracted from the company's ERP system for fiscal years 2024 and 2025 (year-to-date). The dataset comprised 323 customer records in the master customer database, of which 230 customers were commercially active in 2024 (defined as having generated at least one transaction during the year).

Key descriptive statistics on the customer portfolio

- **Total revenue 2024**: €5,536,115
- **Total revenue 2025 YTD**: €5,666,775
- **Number of active customers 2024**: 230
- **Number of active customers 2025 YTD**: 254
- **Average revenue per active customer 2024**: €24,076

Customer Concentration Analysis (Pareto distribution)

The revenue distribution across the customer base exhibited strong concentration consistent with the Pareto principle:

- **top 10 customers (Key Accounts)**: generated **68.6%** of total revenue in 2024 (€3,800,396), with individual account revenues ranging from €97,182 to €1,442,399.
- **Top 20 customers**: Generated **79.5%** of total revenue in 2024.
- **80% revenue threshold**: Reached with only **20 customers**, representing

8.7% of the active customer base.

This concentration suggests that digital adoption strategies should be differentiated by customer segment: key accounts, which contribute the majority of revenue, may require personalized onboarding, training, and enhanced support to encourage platformadoption, while lower-revenue customers may be more amenable to self-service digital channels if the user experience is user-friendly.

Top 10 Key Accounts (2024 revenue, with 2025 YTD comparison):

Rank	Customer Name	2024 Revenue (€)	2025 YTD Revenue (€)
1	CANTINE SETTESOLI Soc.Coop. Ag	1,442.4	1,442.4
2	CANTINE ERMES - SOC. COOP.	739,40	739,40
3	CARLO PELLEGRINO & C. S.p.A.	430,713	430,713
4	MANFREDI BARBERA & FIGLI S.p.A	272,839	272,839
5	KEMECO s.r.l.	220,038	220,038
6	DELIZIE DI CALABRIA srl	189,369	189,369
7	GIACINTO CALLIPO CONSERVE ALIMENTARI SRL	175,637	175,637
8	AZ. AGRICOLE SCYAVURU SRL	132,538	132,538
9	DAMIANO S.p.A.	100,276	100,276
10	DUCA DI SALAPARUTA S.P.A.	97,182	97,182

Figure nr.2: Top 10 Key Accounts .Source: own elaboration

These accounts are predominantly concentrated in the wine, spirits, and premium food sectors, consistent with Auroflex's strategic positioning in high-quality, customized label production. The year-over-year revenue variability observed in some accounts reflects the project-based nature of label orders, where large campaigns or product launches drive significant revenue spikes followed by periods of lower activity.

3.3.4 E-commerce platform analytics: absence of transactional data

As noted earlier, the Insuperlabel e-commerce platform was technically functional during the data collection period but had not yet started commercial activity and therefore recorded no completed transactions. The platform remained in a refinement and testing phase, with management prioritizing technical robustness, user experience refinement, and internal workflow integration over early commercial launch.

A single test campaign conducted via Meta (Facebook/Instagram) advertising in Q4 2025 generated the following engagement metrics:

- **total visitors**: 70.
- **Engaged visitors** (defined as users who interacted with product configuration or quotation features): 3.
- **Engagement rate**: 4.3% (3 / 70).

While these data are insufficient to establish reliable baseline conversion rates or customer acquisition costs, they provide a preliminary indication of low organic engagement in the absence of targeted onboarding, education, or incentive mechanisms. This underscores the need for a deliberate commercial activation strategy that combines digital marketing, customer education, and relationship-driven promotion to key accounts.

3.3.5 Implications for Digital Balanced Scorecard

The absence of transactional e-commerce data necessitates that **all customer perspective KPIs related to digital channels** (conversion rate, digital CLV, retention rate, digital NPS, adoption rate) be specified in a **prospective mode**, with targets derived from:

1. **sector benchmarks**: B2B e-commerce conversion rates in manufacturing (Shopify B2B Commerce Report 2024; Routine Automation 2025).
2. **Strategic ambition**: Management's articulated goal of achieving 15–20% digital revenue share within 24–36 months.
3. **Customer base sizing**: Using the 360-customer total base and 230-active-customer baseline to model adoption trajectories.
4.

These prospective KPIs will be included in the Digital Balanced Scorecard specifying that they represent targets rather than historical benchmarks and will require review and adjustment once the platform enters commercial operation.

The internal process perspective of the Digital Balanced Scorecard focuses on operational efficiency, production performance, workflow digitalization, and

systems integration dimensions that are critical to turning digital investments into measurable improvements, quality, and cost control. Data collection for this perspective drew on multiple internal information systems, including the Manufacturing Execution System (MES), production reports, workflow management tools, and interviews with production and operations managers. However, the availability and standardization of data collection were uneven, reflecting the fragmented nature of Auroflex's digital infrastructure and the absence of a unified data warehouse or integrated reporting platform. This unevenness required the researcher to adopt a mixed approach, combining quantitative extracts from automated systems with qualitative analysis of workflows and manual estimation where automated data capture was incomplete.

The company's primary production management system, CERM MES, serves as the central repository for job tracking, machine scheduling, and production reporting. From CERM and supplementary ERP modules (Automation Engine, ARCA), the following aggregate production metrics were extracted for fiscal year 2024:

- **total production jobs (commesse)**: 8,554 jobs completed during the year.
- **Average jobs per month**: Approximately 713 jobs, indicating sustained high-volume, high-variety production consistent with the label converting business model.
- **Scrap rate**: Around 15% of total production volume, according to internal quality control reports and production waste records. This figure represents the percentage of produced material that did not meet quality standards and was discarded, including setup waste, printing defects, die-cutting errors, and finishing failures.

Limitations: the scrap rate figure is a rough estimate not systematically tracked in the MES. The company does not have a real-time scrap tracking system at job-level, which limits detailed scrap analysis, production line, or digital vs. traditional workflow. This gap was identified during the diagnostic phase of the research and highlighted as a key area for future improvement.

An important operational indicator in the label converting industry is the artwork-

to-print lead time, indicates the elapsed time from artwork approval by the customer and the initiation of the physical production run. This metric captures the efficiency of pre-production workflows, including artwork file processing, pre-flight checks, plate preparation (for flexographic jobs), setup configuration, and material allocation. During interviews with the production manager and commercial staff, the average artwork-to-print lead time for 2024 was reported at around 15 days, with considerable variation depending on job complexity, customer responsiveness, and production queue congestion.

Within this 15-day total lead time, the customer artwork approval phase was estimated to consume approximately 1 day on average, representing the time required for the customer to review and sign off on digital proofs. The remaining 14 days reflect internal processing time, a portion of which could be compressed through further workflow digitalization, automation of pre-flight checks, and dynamic scheduling algorithms integrated with the MES.

Limitations: these lead time figures are estimates based on interviews and sampled job records rather than automatically recorded data. The CERM system does not provide automatic lead time reporting, and time stamps data for key workflow stages (artwork receipt, approval, production start, completion) are not consistently recorded for all jobs. This reveals a further limitation in the company's data infrastructure gap that makes it difficult to define precise baselines and monitor performance evolution.

Workflow digitalization: Percentage of jobs in end-to-end digital workflows

A key aim of the Digital Balanced Scorecard is to monitor the move toward more digital, end-to-end workflows in which job data move without interruption from customer order placement through production scheduling, machine execution, quality control, and shipment without manual intervention or non-digital steps (e.g., paper-based work orders, email-transmitted artwork files, manual data re-entry). During interviews, production and IT staff estimated that approximately 100% of current production jobs involve some degree of digital workflow components, such as digital tools, automated scheduling via CERM, and digital machine setup parameters.

However, this figure is highly optimistic and requires critical qualification. While

nearly all jobs involve digital tools at some stage, very few jobs traverse the entire value chain without manual interventions, data re-entry, or non-integrated handoffs between systems. Examples of workflow interruptions include:

● manual email-based artwork file transmission from customers (not integrated with web-to-print).

● Manual re-entry of customer specifications from email into CERM job tickets.

● Paper-based production work orders printed from CERM and carried to the shop floor.

● Manual quality inspection logs are not digitally integrated with MES.

A more conservative and methodologically defensible estimate, informed by the researcher's direct observation and corroborated by IT staff, is that fewer than 30% of jobs currently execute in truly end-to-end digital workflows where data flow is automated and uninterrupted from order placement to delivery. The "100%" figure reflects the presence of digital tools rather than the absence of manual intervention, and this distinction is critical for establishing realistic baselines and targets in the Digital Balanced Scorecard.

In addition to aggregate production statistics, the researcher obtained visual reports and sample job records extracted from CERM in the form of graphical dashboards and tabular reports. These materials, shared as screenshot files, show how production volumes are distributed over time - Scaling, Scarichi (waste/offcuts), and GMG (color management) - and provide detailed breakdowns of time spent on each job.

Example: Monthly production volume charts, Scrap and correction costs



Figure nr. 3: monthly production volumes. Source: CERM

The first set of visual data depicts monthly production volumes (measured in euros of output value) for three representative jobs over a 12-month period.

The charts reveal:

- seasonal variability in production, with peak volumes in certain months (e.g., October, November) and troughs in others (e.g., August, traditionally a low-activity period in Italian manufacturing due to summer closures).
- Job-specific patterns reflecting customer campaign timing and order batching behavior.

Scrap and correction costs

One of the visual analytics charts provided aggregates monthly scrap and rework costs across the year, distinguishing between:

- **Corrections**: Costs associated with reworking jobs that did not meet specifications on the first pass.
- **Unregistered scrap:** Material waste that was not formally logged in the MES but was reconstructed from inventory reconciliation.

The aggregate annual figures indicate:

- **Total value of scrap and corrections:** €28,026.13.
- **Associated material cost (negative, representing waste)**: -€120,654.10. The large negative material cost reflects the embedded value of raw materials (paper, adhesive, foil) that were consumed in defective production and could not be recovered or resold. This figure underscores the financial impact of quality failures and setup inefficiencies, reinforcing the strategic importance of reducing scrap rates through improved process control, training, and digital pre-flight checks.
- **Limitations**: these scrap cost figures are aggregated at the annual level and do not distinguish between job types, production lines, or root causes. The absence of systematic, real-time scrap tracking represents a significant data infrastructure gap.

Example: Job-level time breakdown

CombinedColumn	Avviamento_on_jobOreMinuti	Avviamento_globaleOreMinuti	Stampa_on_jobOreMinuti	StampaOreMinuti	Totale_on_jobOreMinuti	TotaleOreMinuti
629419 - Retro Le Tufare Bouvigne	00:20	00:13	00:03	00:00	00:23	00:13
632206 - Etichette neutre 150x100	00:15	00:25	00:26	04:16	00:41	04:41
632533 - Ionico 2 file	02:35	05:06	03:08	10:30	05:43	15:36
633085 - FR - OLIO KAREN AKUNOWICZ	02:10	00:16	00:35	02:06	02:45	02:22
633086 - FR - HERITAGE - BIANCOLILLA - FORTE 500 ML	02:20	02:18	01:31	03:44	03:51	06:02
633087 - RT - HERITAGE - BIANCOLILLA - FORTE 500 ML	01:10	00:29	00:42	00:49	01:52	01:18
633124 - FR - BELEDA 150 cl (H)	01:25	01:14	00:26	00:10	01:51	01:24
Totale	**3894:48**	**3513:19**	**4281:31**	**5502:38**	**8176:19**	**9015:57**

Avvio Commessa, Preparazione, Avviamento, Stampa, Pulizia, Fine Commessa e Altro per wn_naam_

Figure nr. 4: Job-level time breakdown. Source: CERM

A second visual dataset provides a detailed time-phase breakdown for a sample of completed jobs, showing the allocation of time across key production stages:

- **avviamento (Setup/Startup)**: Time required to configure the printing line, load materials, and perform initial test runs.
- **Stampa (Printing)**: Actual printing run time.
- **Pulizia (Cleaning/Changeover)**: Time required to clean the line and

prepare for the next job.

- **Totale (Total job duration)**: Sum of all phases.
  For example, one job record (Commessa 629419) shows:
  - setup time: 00:20 (20 minutes).
  - Printing time: 00:13 (13 minutes).
  - Additional phases contributing to a total job duration of approximately 00:41.

These detailed data help identify inefficiencies in setup and changeover processes, which make up a large part of total production in short-run, high-variety manufacturing. However, the data were provided as selected examples rather than systematically extracted for all 8,554 jobs completed in 2024, limiting their statistical representativeness.

Understanding the degree of integration among the company's digital systems is critical for assessing the maturity of the digital infrastructure and for identifying opportunities to eliminate data silos and manual handoffs. During interviews with IT staff and through direct observation of system interfaces, the researcher compiled an inventory of the company's primary IT systems:

1. **ERP systems:**

- **CERM**: Primary MES and production planning system.
- **Automation Engine**: Legacy ERP module used for certain administrative and financial workflows.
- **ARCA**: ERP module used for accounting and financial reporting.

2. **Business Intelligence and Analytics:**

- **Power BI**: A data visualization and dashboard platform. In 2024, the company developed 10 dashboards covering financial performance, production KPIs, and commercial metrics. Currently, 6-7 users actively access these dashboards, primarily management and functional heads, corresponding to roughly 30-35% adoption among the potential users group (approximately 20 individuals in roles that would benefit from BI access).

3. Customer Relationship Management:

- **Auroflex MBO** (custom CRM): An internally built CRM platform. All 30 employees are registered in the system, but only 15 employees (50%) use it actively monthly, defined as regular login and updates customer records or interaction histories. The 50% adoption rate suggests partial integration into routine activities of CRM, with parallel use of email and informal communication channels outside the formal CRM.

4. E-commerce and Web-to-Print:

- **platform**: B2B e-commerce platform developed externally and integrated (partially) with ERP systems. As documented, the platform is not yet commercially operational.
- **Web-to-print functionality**: No dedicated web-to-print system is currently deployed. Customers cannot independently upload artwork files, configure products, or generate quotes online. This represents a major gap in the digital customer journey.

5. Process Automation (RPA - Robotic Process Automation via Power Automate):

The company has implemented three automation bots using Microsoft Power Automate selected back-office tasks:

- **Kate**: Automates ordering consumable supplies (e.g., office supplies, maintenance materials) from vendors.
- **Clair**: Automates raw material procurement workflows, including generating purchase orders based on inventory thresholds and forwarding them for approval.
- **Ugo**: Monitors job status in CERM, flagging jobs that are delayed or missing key data elements, and sending alerts to production managers.

These bots represent a practical, low-code solution to process automation and show management's openness to digital experimentation with emerging technologies. However, automation remains limited in scope : fewer than 30% of repetitive back-

office processes are currently automated, and the bots are not interconnected without integration into a coordinated automation architecture.

An important but hard-to- measure aspect of digital maturity is the extent to which these different systems communicate and transfer data automatically, rather than requiring manual export, and re-import of data. Based on interviews with IT staff and the researcher's observation of data flows, the overall **integration** level was estimated at below **50%**, meaning that fewer than half of the data flows between systems currently occur without manual intervention.
Examples of integration gaps include:

- Customer order data captured in email are manually re-entered into CERM job tickets rather than being pulled automatically from the CRM or e-commerce platform.
- Financial data in ARCA must be manually exported to Excel and then re-imported into Power BI for dashboard updates, rather than being pulled directly via API.
- Production completion data in CERM do not automatically trigger updates to inventory records in the ERP, requiring periodic manual reconciliation.

As part of the internal process perspective, particularly relevant in the context of increasing regulatory pressure (e.g., EU Regulation 2024/825 on packaging recyclability and traceability), the researcher collected data on the company's sustainability practices and material source:

- **percentage of sustainable materials used: 50% of total raw materials** by value are considered sustainable, of which 100% of paper-based substrates are sourced from FSC-certified or equivalent sustainable forestry sources.
- **Percentage of labels certified as recyclable: 0%** as of 2024. The company has not yet obtained formal recyclability certifications for its finished products, though many labels are technically recyclable based on substrate and adhesive composition. This represents a possible area for future development.

- **Environmental certifications**: The company achieved EcoVadis Silver Medal status in 2025, positioning Auroflex within the top 25% of companies assessed globally for corporate social responsibility and environmental performance. This certification reinforces the company's credibility with sustainability-conscious customers and supports marketing positioning around responsible manufacturing.

The learning and growth perspective of the Digital Balanced Scorecard focuses on the organizational capabilities that underpin digital transformation—specifically, the adoption of digital tools by employees, the development of digital literacy and competencies, the quality and accessibility of data within information systems, and the organizational culture's receptivity to data-driven decision-making. Data collection for this perspective combined quantitative system usage logs, HR records, and a structured internal questionnaire designed to assess employees' self-perceived digital maturity and training needs.

As of Q4 2025, Auroflex employed around 30 full-time equivalent (FTE) employees. The functional distribution of personnel, based on the internal digital maturity questionnaire (see below) and confirmed in interviews with the HR manager, includes:

- **Founders / General Management**: 4 individuals.
- **IT / Information Systems**: 2 individuals.
- **Production / Operations**: Approximately 15-18 individuals (shop floor operators, production supervisors, quality control).
- **Commercial / Customer Service**: 2-3 individuals.
- **Administration / Finance**: 2 individuals.
- **Marketing / Graphics / Communication**: 2-3 individuals.

This distribution is typical of the structure of a manufacturing SME with a large production workforce, small administrative and commercial teams, and limited dedicated IT resources - a profile that aligns with findings in the literature on digital transformation in SMEs (Naeem & Garengo, 2022).

To systematically assess employees' perceptions of the company's digital maturity, their own digital competencies, and the organizational culture's orientation toward data-driven decision-making, the researcher designed and administered a structured

digital maturity questionnaire in January 2026. The questionnaire was distributed electronically to all employees via email, with a data collection window of three days (January 17-19, 2026). The instrument comprised 46 questions organized into the following thematic sections:

1. **Respondent profile**: Role in the company, years of tenure, frequency of interaction with digital tools (3 questions).
2. **Internal perceptions of digital maturity**: Clarity and communication of digital strategy, level of digitalization across business processes, data accessibility for decision-making, systems integration, presence of digital skills within the organization, managerial and employee support for innovation, regular evaluation of digital ROI (8 questions, 5-point Likert scale).
3. **Tool adoption and usage**: Frequency and intensity of use for specific tools - e-commerce platform, Power BI, Google Analytics, CERM, BMO CRM (5 questions, 6-point scale where 6 = maximum usage).
4. **Self-assessed digital competencies**: Ability to use management software (CERM, BMO), data analysis and reporting (Power BI, Excel), digital marketing and web analytics, e-commerce platform management, process automation, cybersecurity (6 questions, 5-point Likert scale).
5. **Organizational enablers and barriers**: Perceived support from management for adopting new tools, identification of barriers to digital adoption (2 questions, 5-point scale + open-ended).
6. **Strategic priorities for Digital BSC**: Perceived urgency of implementing a structured digital performance measurement system (1 question, 10-point scale) and open-ended reflections on digital transformation priorities (1 question).

The questionnaire yielded 12 complete responses, representing a **40% response rate** relative to the total workforce of 30 employees. The respondents' role distribution was as follows:

- founders / General Management: 4.
- IT / Information Systems: 2.

- Commercial / Customer Service: 1.
- Production: 1.
- Administration / Finance: 1.
- Marketing / Graphics: 3.

The sample is mainly composed of weighted toward management and functional heads, with few shop floor operators participated and junior administrative staff. This reflects the design of the questionnaire that was designed with a focus on digital tool usage and strategic awareness, areas where management and knowledge workers have greater responsibility in digital decision-making. However, the underrepresentation of production personnel limits the generalizability of findings to the organization as a whole and should be considered as a sampling bias.

Tenure distribution of respondents:

- more than 15 years: 7 respondents (58%).
- 8-15 years: 1 respondent.
- 4-7 years: 2 respondents.
- 1-3 years: 1 respondent.
- Less than 1 year: 1 respondent.

The high proportion of long-tenured employees (58% with >15 years) indicates that the respondents is composed primarily of staff with extensive experience within organization experience with the company's evolution. This has the advantage of providing experience-based insights into how digital transformation has developed over time but may also lead to more conservativeviews.

Key findings: Organizational digital maturity perceptions (5-point Likert scale, mean scores)

Dimension	Mean Score	Interpretation
Frequency of digital tool interaction	4.92 / 5	Very high daily usage
Clarity of digital strategy	3.17 / 5	Moderate; strategy not fully articulated or communicated
Degree of process digitalization	3.75 / 5	Good; processes partially digitalized but gaps remain
Data accessibility for decisions	3.33 / 5	Moderate; data exist but access/usability issues persist
Systems integration (interoperability)	4.00 / 5	Good; systems communicate but not seamlessly
Availability of digital competencies	3.42 / 5	Moderate; skills gaps identified
Culture of digital innovation	4.42 / 5	Very good; openness to experimentation and change
Systematic measurement of digital ROI	3.00 / 5	Low; ad hoc rather than structured measurement

Figure nr. 5: Key findings. Source: own elaboration

Interpretation: the results reveal a paradoxical profile: employees interact with digital tools intensively (4.92) and perceive the organizational culture as supportive of innovation (4.42), yet rate the clarity of digital strategy (3.17), data accessibility (3.33), and systematic ROI measurement (3.00) as moderate to low. This pattern suggests tactical digital adoption without strategic coherence, a common pattern in SME digital transformation where tools are implemented reactively to address immediate operational needs, but without an overarching framework for integration, capability development, or performance measurement. This finding directly motivates the need for the Digital Balanced Scorecard as a governance and alignment mechanism.

Tool adoption and usage intensity (6-point scale, mean scores):

Tool	Mean Score	Interpretation
Google Analytics 4	5.92 / 6	Near-universal, intensive usage
E-commerce platform (Insuperlabel)	5.58 / 6	High awareness/readiness but no commercial transactions
MBO CRM (internal)	5.50 / 6	Very good adoption
CERM (production)	4.75 / 6	Good adoption
Power BI	4.50 / 6	Moderate adoption; room for expansion
Power BI	4.50 / 6	Moderate adoption; room for expansion

Figure nr. 6: Tool adoption and usage intensity. Source: own elaboration

Interpretation: the very high scores for Google Analytics (5.92) and the e-commerce platform (5.58) reflect the fact that digital marketing and e-commerce

development were strategic priorities in 2024–2025, resulting in concentrated attention and training. However, the e-commerce platform score is somewhat misleading, as the platform is not yet in commercial use; the high score likely reflects awareness and preparedness among commercial and IT staff rather than daily operational usage. The moderate Power BI score (4.50) suggests that BI dashboards have not yet become embedded in routine decision-making workflows for a majority of managers, a finding consistent with the low adoption rate (6–7 active users) .

Self-assessed digital competencies (5-point Likert scale, mean scores):

Competency Area	Mean Score	Interpretation
Process automation and workflow digitalization	3.92 / 5	Good; reflects bot deployment and CERM usage
Use of management software (CERM, BMO)	3.42 / 5	Moderate; training gaps persist
Data analysis and reporting (Power BI, Excel)	3.17 / 5	Moderate; limited analytical depth
Cybersecurity and data protection	2.67 / 5	Low; identified as a skill gap
Digital marketing and web analytics	2.25 / 5	Low; specialized competency concentrated in marketing team
E-commerce platform management	1.42 / 5	Very low; platform not yet operationalized

Figure nr. 7: Self-assessed digital competencies. Source: own elaboration

Interpretation: the competency self-assessments reveal a bifurcated skill profile: employees feel relatively competent in operational tasks directly tied to their daily work (process automation, management software), but lack confidence in more specialized or strategic digital domains (e-commerce, digital marketing, advanced analytics, cybersecurity). The very low e-commerce management score (1.42) is particularly concerning given the imminent commercial launch of Insuperlabel, suggesting a critical training gap that must be addressed to ensure successful platform activation.

Perceived urgency of implementing a Digital Balanced Scorecard (10-point scale): respondents were asked: "On a scale from 1 to 10, how urgent do you consider it to implement a formal digital performance measurement system (Digital Balanced Scorecard)?" The average score was **7.58 out of 10** (range: 6-10, standard deviation: 1.38). This suggests a generally **high sense of urgency** among

management and staff, reflecting awareness that the company's fragmented digital initiatives require a more integrated measurement system and governance framework. However, the fact that no respondent rated urgency above 10 (maximum) and two respondents rated it as low as 6 may reflect some hesitation or competing priorities.

The researcher attempted to extract quantitative data on training hours allocated to digital competencies during fiscal year 2024 from the company's HR records. However, no systematic tracking of training hours by thematic area (digital vs. non-digital) was maintained. Training expenditures are aggregated in the HR budget, and some training sessions (e.g., software vendor-provided onboarding for CERM, Power BI workshops) were documented in meeting calendars and invoices, but no centralized training log exists. This represents a data infrastructure gap that constrains the ability to establish baselines for the learning and growth perspective of the Digital Balanced Scorecard.

In addition to the self-reported competency survey, the researcher extracted system access logs from the company's IT infrastructure to verify actual usage levels:

- **BMO CRM**: 30 registered users (all employees), **15 active monthly users** (50% adoption rate), considered as users who logged in and performed at least one data entry or query operation in a typical month.
- **Power BI**: 10 dashboards deployed, accessed by **6-7 regular users** (30–35% adoption rate among the ~20-person target user population of managers and analysts).
- **Bot automation (Power Automate)**: 3 bots operational (Kate, Clair, Ugo), automating **fewer than 30%** of processes identified as suitable for automation. This data is consistent with the questionnaire results, with pockets of intensive usage alongside significant non-adoption.

An important factor in enabling data-driven decision-making is the **quality, completeness, and accessibility of data** within the company's information systems. During the diagnostic phase, the researcher conducted informal **data quality audits** by reviewing records from the ERP, CERM, and CRM systems and assessing:

- completeness: Percentage of mandatory fields populated.

- Accuracy: Absence of obvious errors, inconsistencies, or duplicates.
- Timeliness: Currency of data relative to real-world events.

No formal, systematic data quality scoring framework was applied, but qualitative assessments based on interviews with IT staff and direct inspection of data exports suggested the following approximate quality profiles:

- **ERP (ARCA)**: **75-80%** data quality, with high completeness for financial transactions but inconsistencies in cost center allocations and inventory classifications.
- **CERM (MES)**: **70-75%** data quality, with good coverage of production volumes and machine utilization but incomplete logging of scrap, rework, and setup times.
- **MBO (CRM)**: **60-70%** data quality, reflecting the system's recent deployment and uneven adoption; customer records exhibit missing contact details, outdated interaction histories, and duplicates. These estimates will inform the data quality KPI in the learning and growth perspective of the Digital Balanced Scorecard, with targets set for systematic improvement through data governance protocols and user training.

While the quantitative data sources provide the empirical bases for populating the Digital Balanced Scorecard, qualitative analysis was essential for *contextualizing* quantitative patterns, *eliciting tacit knowledge and strategic priorities, and co-designing* the architecture of the Digital Balanced Scorecard in collaboration with company management. The qualitative data collection strategy combined *semi-structured interviews* with key company representatives and *direct observation* of operational processes and management meetings.

The interview protocol was designed following the guidelines articulated by Yin (2018) for case study research, prioritizing open-ended questions that allow respondents to articulate their perspectives in their own terms while maintaining thematic coherence across interviews. A *standardized interview guide* was developed, comprising four thematic modules:

1. **strategic objectives and digital transformation priorities**: How do you define success for Auroflex in the next 3–5 years? What role does digital

transformation play in achieving those objectives? What are the most important digital initiatives currently underway or planned?

2. **Performance measurement and decision-making practices**: How do you currently monitor the performance of your area? What indicators do you look at, and how frequently? What information do you need but currently lack? How are data used in decision-making? Is it intuition-driven, experience-driven, or data-driven?
3. **Barriers and facilitating factors in digital adoption**: What are the main obstacles preventing deeper or faster digital transformation? What organizational or cultural factors support or hinder the use of new digital tools? What skills or resources are missing?
4. **Expectations for the Digital Balanced Scorecard**: What do you expect from a structured performance measurement system? What questions should it help to answer? What would make it useful in your daily work rather than just a reporting burden? The interviews were audio-recorded with participants' consent and subsequently fully transcribed. Each interview lasted approximately 90 minutes, consistent with the researcher's report in the methodology section.

   The interviews were conducted in Italian, the native language of all participants, to ensure fluency and reduce linguistic barriers. Transcripts were thematically coded following a template analysis approach (King, 2012), with codes derived deductively from the Digital Balanced Scorecard framework (financial, customer, process, learning) and inductively from emergent themes in the data.

Interview sample: the researcher conducted interviews with the following organizational actors:

- **founder/CEO** (1 interview): Strategic vision, investment priorities, commercial strategy.
- **Founder/Operations Director** (1 interview): Production strategy, technology investments, quality management.
- **Production Manager** (2 interviews): Shop floor operations, workflow

efficiency, integration of digital tools in production.

- **Commercial Manager** (1 interview): Customer relationships, sales processes, expectations for e-commerce platform.
- **Administrative Manager** (1 interview): Financial reporting, cost control, administrative digitalization.

The interviews cover multiple functional areas across the main departments, ensuring that the Digital Balanced Scorecard reflects a holistic view of strategic priorities rather than the perspective of a single function. The absence of interviews with shop floor operator is a limitation, though direct observation partially mitigated this gap.

Key qualitative insights from interviews: analysis of the interviews highlighted several recurring themes that influenced the design of the Digital Balanced Scorecard:

- **Fragmentation of digital initiatives**: All interviewees acknowledged that digital tools had been adopted in a reactive, and fragmented way rather than according to a master plan. The lack of a unifying digital strategy was identified as a source of inefficiency and missed integration opportunities.
- **Measurement fatigue vs. measurement hunger**: Managers expressed ambivalence about performance measurement: on one hand, they felt overwhelmed by the volume of data generated by disparate systems and the time required to extract, reconcile, and interpret it; on the other hand, they articulated a strong desire for synthesized, actionable insights that would guide decision-making. This tension underscores the need for a Digital Balanced Scorecard that prioritizes a contained set of high-impact KPIs rather than comprehensive data collection.
- **Customer resistance to self-service**: The commercial manager expressed concern that key accounts, accustomed to personalized, relationship-driven service, would resist migrating to the e-commerce platform if it were perceived as a cost-cutting measure that reduced service quality. This insight informed the customer perspective targets, emphasizing the need for

hybrid service models that combine digital convenience with human touchpoints for high-value accounts.

- **Digital Skills gaps and training needs**: All interviewees pointed to gaps in digital literacy, particularly in advanced data analysis and e-commerce platform management, as a key limiting factor to digital transformation. The administrative manager specifically noted that training investments had been ad hoc and unstructured, with no formal competency development framework.

Direct observation: operational processes and decision-making forums

Beyond interviews, the researcher carried out direct observation of operational processes and internal meetings over a four-month period (November 2025-February 2026). Observational access encompassed:

- **shop floor operations**: The researcher spent multiple days observing production workflows, machine changeovers, quality control inspections, and data capture practices (or lack thereof) at the point of production. This revealed gaps between written procedures and everyday practices, such as the use of paper-based work orders alongside digital job tickets, manual transcription of machine parameters, and informal shortcuts used to bypass MES data entry.
- **Internal coordination meetings**: The researcher attended weekly production planning meetings, monthly management review meetings, and ad hoc project meetings focused on the e-commerce platform rollout. These observations highlighted internal power dynamics of decision-making, whether data were used in discussions (rather than experience or intuition), and the tensions regarding resource allocation for digital initiatives.
- **Systems usage observation**: By observing managers and staff interacting with Power BI dashboards, CERM interfaces, and the BMO CRM, the researcher could assess usability issues, user proficiency, and the integration (or lack thereof) of digital tools into daily workflows. For example, it was observed that many managers opened Power BI dashboards infrequently and relied primarily on static Excel reports emailed by the administrative

staff, indicating that BI adoption had not displaced legacy practices.

The researcher also reviewed a range of internal documents to supplement interview and observational data:

- strategic plans and investment proposals: Documented the company's stated digital transformation objectives and the business cases for major technology investments (e.g., e-commerce platform, CRM).
- Financial reports and budgets: Provided context for interpreting financial KPIs and understanding cost pressures and margin dynamics.
- Marketing and commercial materials: Illustrated the company's external positioning and the messaging used to communicate digital capabilities to customers.
- Meeting minutes and internal communications: Offered a longitudinal view of how digital initiatives were discussed and prioritized over time.

To increase the robustness of the findings, the data collection protocol incorporated multiple forms of triangulation (Denzin, 1978);(Jick, 1979), consistent with best practices in mixed-methods case study research:

1. Data source triangulation: Financial indicators were compared between external sources (Cerved, Centrale dei Bilanci) and internal accounting records. Production metrics were cross-referenced between CERM exports, manager interviews, and direct observation.
2. Method triangulation: Quantitative patterns (e.g., low Power BI adoption rates from system logs) were supported with qualitative evidence (e.g., manager statements in interviews that they preferred Excel reports; researcher observations of infrequent dashboard usage).
3. Investigator triangulation: While the researcher was the primary data collector, preliminary findings were reviewed and validated in iterative discussions with company founders and functional managers, who provided corrective feedback and alternative interpretations where discrepancies emerged.
4. Theory triangulation: The interpretation of data was informed by multiple

theoretical lenses - financial performance analysis (ROI, EBITDA), operations management (OEE, lead time reduction), digital transformation frameworks (digital maturity models), and strategic performance measurement (Balanced Scorecard) - ensuring that findings were not artificially constrained by a single disciplinary perspective.

3.4 Reliability, Validity, and Ethics

The methodological rigor of research--intervention design rests not only on the structured data collection and analysis but also on the explicit attention paid to the *reliability* of measurement procedures, the *validity* of interpretive claims, and the *ethical conduct* of inquiry. This section addresses each of these dimensions in turn, demonstrating how the research design incorporates methodological precautions to enhance credibility, trustworthiness, and ethical considerations of the study.

In mixed-methods case study research, reliability is pursued through the standardization of data collection protocols, the documentation of analytical procedures, and the use of multiple data sources to cross-validate findings.

In this research, several strategies were employed to enhance reliability:

Standardized financial data extraction: All financial indicators (ROI, ROS, EBITDA, working capital metrics) were calculated using consistent formulas applied uniformly across the five-year dataset (2020-2024) and provisional H1 2025 data. The source data were obtained from externally audited financial statements (Cerved, Centrale dei Bilanci), ensuring that the underlying accounting information adhered to standardized Italian GAAP reporting requirements. By relying on official, third party-verified sources rather than exclusively on internal management accounts, the research minimizes the risk of idiosyncratic accounting treatments or selective data reporting that could compromise comparability across time periods.

Structured survey and questionnaire instruments: The customer satisfaction survey and internal digital maturity questionnaire employed *standardized response scales* (5-point and 6-point Likert scales, respectively) with clearly defined anchor points. Question wording was held constantly across all respondents, and the electronic administration format ensured that all participants received identical

prompts. While the questionnaire was designed specifically for this study and thus lacks external validation through prior empirical testing, its structure drew on established frameworks in customer satisfaction measurement (Parasuraman et al., 1988) and digital maturity assessment (Kotarba, 2017), lending construct validity to the instrument design.

Consistence of the interview protocol: Semi-structured interviews followed a *standardized interview guide* comprising four thematic modules, ensuring that all key participants were asked comparable core questions while retaining flexibility to probe emergent themes. Interviews were audio-recorded and transcribed verbatim, creating a permanent record that allows the review of coding and interpretation. The use of a template analysis approach (King, 2012) with both deductive and inductive coding ensured that thematic analysis was structured.

Documentation of analytical steps: All data transformations, calculations, and interpretive decisions were documented in research logs and working papers, allowing the analytical process to be retraced. For example, the calculation of the Cash Conversion Cycle (CCC) is fully traceable from the source balance sheet line items (inventory, receivables, payables) through the intermediate metrics (DIO, DSO, DPO) to the final CCC value, with formulas and assumptions made explicit.

Limitations on reliability: Despite these safeguards, certain data sources presented *reliability challenges* that must be acknowledged. Operational metrics extracted from the CERM MES system were *incomplete and non-systematic*, with gaps in scrap tracking, lead time logging, and workflow timestamp capture. Consequently, some internal process KPIs (e.g., the 15-day artwork-to-print lead time, the 15% scrap rate) are based on *managerial estimates and sampled records* rather than comprehensive automated data logs. These estimates, while informed by the best available evidence and corroborated through interviews and observation, introduce measurement uncertainty that limits the precision of baseline calculations. Similarly, the absence of longitudinal e-commerce transaction data means that customer perspective KPIs are *prospectively defined rather than empirically observed*, limiting the possibility of buildings historical baselines for digital customer behavior.

Validity concerns the accuracy and credibility of the research findings, the extent to which the study measures what it purports to measure (construct validity), the degree to which findings can be generalized beyond the case (external validity), and the strenght of causal inferences drawn from the data (internal validity).

Construct validity was pursued through multiple strategies:

- **Triangulation of data sources:** Financial performance claims were validated by cross-referencing external financial statements, internal accounting records, and qualitative assessments from interviews. For instance, the observed improvement in EBITDA margin between 2023 and 2024 was corroborated not only by the official income statement figures but also by production managers' reports of efficiency gains and the administrative manager's confirmation of cost containment initiatives. This convergence of evidence from independent sources strengthens confidence that the financial indicators accurately reflect underlying economic realities.
- **Methodological triangulation:** Quantitative patterns were systematically cross-checked against qualitative insights. The low Power BI adoption rate (30-35%) observed in system access logs was corroborated by interview statements in which managers expressed a preference for traditional Excel reports, and by direct observation of dashboard usage during internal meetings. This consistency across methods enhances the construct validity of the “adoption rate” metric as a meaningful indicator of actual system integration into decision-making workflows.
- **Feedback from interviewees:** Preliminary findings from the diagnostic phase were presented to company founders and functional managers in iterative feedback sessions. These discussions provided opportunities for organizational insiders to discuss and refine the researcher's interpretations, like *member checking* (Lincoln & Guba, 1985). For example, management’s confirmation that the e-commerce platform’s delayed commercial launch reflected a deliberate quality-first strategy supported the researcher's interpretation of the data availability asymmetry as a strategic choice rather than a technical failure.

- **Use of established frameworks:** The Digital Balanced Scorecard architecture was broadly accepted in the literature (Kaplan & Norton, 1992, 1996; Fabac et al., 2022), ensuring that the performance measurement construct has established legitimacy in the academic literature. The four perspectives (financial, customer, internal process, learning and growth) and the causal logic linking them have been empirically tested across diverse organizational contexts, lending construct validity to their application in this case.

The strength of causal inferences is inherently constrained in single-case, non-experimental research designs. The study does not employ randomized controlled trials or quasi-experimental methods that would enable definitive causal attribution. Nevertheless, several design features enhance the plausibility of causal claims:

- **Longitudinal data analysis:** The five-year financial dataset enables the examination of *temporal precedence* between digital investments and performance outcomes. For example, the substantial e-commerce platform investment in 2024 (€120,832) temporarily precedes provisional performance improvements observed in H1 2025, consistent with the hypothesized lagged effect of digital transformation. While this temporal ordering does not prove causation, it is a necessary (if insufficient) condition for inferring causal relationships.
- **Mechanism tracing:** Qualitative data from interviews and observation provide insight into the *causal mechanisms* through which digital tools are assumed to influence performance. For instance, managers described how Power BI dashboards enabled faster identification of production bottlenecks, which in turn accelerated corrective action, a credible causal pathway linking BI adoption to operational efficiency. By identifying and tracing these micro-level mechanisms, the research strengthens the credibility of macro-level performance claims.
- **Rival explanation consideration:** The research explicitly acknowledges and explores *alternative explanations* for observed performance patterns. For example, the revenue growth observed in 2024-2025 could plausibly be attributed to exogenous market recovery following pandemic disruptions,

rather than to digital transformation. The analysis considers this rival hypothesis and argues that the combination of revenue growth, margin improvement, and qualitative evidence of operational efficiency gains together suggests a compound explanation in which both market recovery and digital capability enhancement play roles.

The transferability of findings to other contexts is limited by the single-case design and the idiosyncrasies of Auroflex's specific competitive positioning, customer base, and technological infrastructure. The research does not claim statistical generalizability to a broader population of label converters or manufacturing SMEs. However, *analytical generalization* (Yin, 2018) is pursued by explicitly linking the case findings to broader theoretical constructs (Digital Balanced Scorecard, digital maturity, performance measurement systems) and by articulating the *contextual boundary conditions* under which the findings are likely to hold.

The study's contribution lies not in producing universal laws, but in demonstrating how a theoretically grounded performance measurement framework can be *adapted and operationalized* in a niche B2B manufacturing context characterized by high capital intensity, product customization, and fragmented digital adoption conditions that are shared by many comparable SMEs in specialized manufacturing sectors across Europe.

CHAPTER 4

Findings

4.1 Company Profiles

Auroflex S.r.l. is a privately held company operating in the self-adhesive label converting sector, established in the Piedmont region of Italy. The company is positioned within a highly competitive market segment characterized by increasing customization demands, stringent quality requirements, and evolving technological standards. Auroflex specializes in the production of high-quality self-adhesive labels primarily targeting premium segments within the food, wine, spirits, beer, and cosmetics industries.

The company employs approximately 35-40 staff members across production, commercial, administrative, and technical functions. The organizational structure reflects the typical configuration of Italian manufacturing SMEs in the sector, with a limited managerial layers and founders are directly involved in daily operations. The production facility included modern printing equipment, including both conventional flexographic and digital printing systems, finishing lines equipped with hot stamping, embossing, and screen-printing capabilities, and automated quality control stations.

Auroflex operates within a business model marked by high product variability and relatively low volumes per order, typical of the premium label segment. The firm handles batch sizes ranging from small runs of a few hundred units to medium-sized orders of tens of thousands of labels, with strong emphasis on rapid turnaround times and technical flexibility. This position requires significant investments in both equipment versatility and operator skills, in contrast to high-volume commodity producers.

From a geographic perspective, Auroflex serves predominantly domestic clients, with the Italian market representing the core revenue base. However, the company has progressively developed relationships with selected international accounts, particularly within neighboring European markets. The client portfolio comprises established brands in the wine and spirits sector, boutique food producers, and

emerging cosmetics companies seeking differentiated packaging solutions.

4.2 History and governance

Auroflex was founded by four partners who brought different professional backgrounds to the enterprise. The ownership structure has not changed since inception, with the four founding members maintaining active involvement in both strategic and day-to-day operations. This governance model is typical of many Italian manufacturing SMEs, where ownership and management functions are closely integrated, enabling rapid decision-making but also may limit scalability.

The company's development has followed technological upgrading and market positioning refinement. Initial years focused on establishing operational reliability and building a client base through competitive pricing and service responsiveness. Subsequently, the strategic emphasis shifted toward quality differentiation and technical specialization, supported by selective investments in advanced printing and finishing equipment. The acquisition of digital printing capacity represented an important turning point, enabling the company to address shorter runs and more complex graphic requirements that conventional flexographic systems cannot economically accommodate.

In terms of governance mechanisms, decision-making authority is distributed among the four proprietors, with informal coordination mechanisms prevailing over formalized strategic planning processes. Regular meetings among owners address both operational issues and strategic priorities, though the absence of external board members or formal advisory structures means that strategic debates occur within a limited external involvement. This configuration facilitates consensus-building on core business matters but may limit exposure to alternative perspectives and external expertise.

The governance structure has implications for the digital transformation process examined in this study. The workshop conducted in December 2025 represented an attempt to formalize strategic objectives and translate them into measurable targets. As reflected in internal documents, this session required dealing with contrasting priorities among the owners, particularly concerning the tradeoff between expanding sales and improving profitability, and how quickly to launch the

platform. The final decision reflected a compromise of different viewpoints, emphasizing quality improvement and customer retention over aggressive cost reduction.

4.3 Product portfolio & market positioning (self-adhesive labelS)

The product portfolio of Auroflex encompasses a wide range of self-adhesive label solutions characterized by varying degrees of technical complexity, graphic sophistication, and material specifications. The core offers centers on labels for premium applications in food and beverage, with particular strength in wine and spirits sectors where aesthetic differentiation and material quality are critical competitive factors.

From a technical standpoint, Auroflex products are typically classified by production technology (flexographic versus digital), finishing processes (plain, embossed, hot stamped, screen printed), and material substrates (paper-based, film-based, specialty materials). The company has developed built solid expertise in multi-process labels that combine several finishing techniques on a single product, addressing clients who seek premium visual and tactile characteristics. Such products require involvement several production steps across lines anddemand close coordination and careful quality monitoring.

Market positioning is oriented toward the premium segment, where clients value technical capability, design collaboration, and service reliability overprice minimization. Auroflex competes primarily on quality consistency, technical versatility, and responsiveness rather than on scale economies. This positioning is consistent with broader patterns observed in the European self-adhesive label industry, where differentiation strategies based on specialized capabilities and customer intimacy coexist alongside volume-oriented commodity production by larger converters.

The wine label segment represents one of the main revenue sources. This market shows strong seasonal demand patterns, with concentrated demand periods preceding major wine releases and trade events. Customer requirements typically emphasize visual appeal, material authenticity (such as textured paper and metallic effects), and regulatory compliance (for example, traceability codes and recycling

symbols). Auroflex has built long-term relationship with established wineries as well as emerging producers seeking to differentiate their brands through packaging innovation.

The cosmetics segment, though smaller in volume terms, shows promising growth and higher margins. Cosmetics labels often require specialized materials (for instance, clear films and metallic finishes), complex die-cutting, and stringent adhesive performance under challenging application conditions. These requirements fit well with Auroflex's equipment capabilities and quality orientation.

Sustainability is becoming more relevant for market positioning. EU Regulation 2024/825 on packaging imposes minimum recyclability thresholds and traceability requirements, compelling label converters to source compliant materials and document product lifecycles. Based on available internal documents, approximately 50% of Auroflex's material inputs currently meet emerging sustainability standards, with medium-term goal of increasing to 70%.

4.4 Digital-maturity gap (website, CRM, analytics)

One of the main empirical findings of this study concerns the documentation of Auroflex's current digital maturity and the identification of capability gaps relative to sector benchmarks and strategic requirements. The assessment encompasses three main areas: web presence and e-commerce infrastructure, customer relationship management systems, and business intelligence and analytics capabilities.

Regarding web presence, Auroflex operates a corporate website that primarily serves informational functions, presenting the company profile, product categories, and contact details. However, the site lacks transactional functionality, interactive design tools, or customer self-service capabilities common among digitally mature converters. Benchmark data from the FINAT RADAR 2024 report indicate that approximately 62% of European label converters have implemented at least partial e-commerce capabilities, suggesting that Auroflex lags the sector median in this dimension.

The e-commerce platform is currently in development, with planned launch targeted for the near term. The primary goal of this initiative centers onreaching

new customer segments, particularly smaller clients (such as boutique wineries) who cannot meet traditional minimum order quantities and would benefit from web-based ordering systems. Internal forecasts suggest target of 15-20% of revenues generated through digital channels within 24 months of platform activation.

The CRM infrastructure represents a more significant gap. Auroflex currently lacks an integrated CRM system, relying instead on scattered data sources (including spreadsheets, email archives, and ERP transaction records) forcustomer data management. Survey evidence collected from employees indicates CRM adoption rates of approximately 50%, with considerable variation across functional areas. Sales personnel exhibit relatively higher usage, while production and administrative staff demonstrate lower engagement with customer data systems.

This fragmentation leads to operational inefficiencies in the form of duplicated data entry and limited visibility into client interaction history. More fundamentally, the absence of an integrated CRM constrains the capacity to analyze customer behavior patterns, segment accounts according to strategic criteria, or implement proactive retention strategies. The Digital Balanced Scorecard framework developed in this study includes CRM adoption as a key learning and growth indicator, with a target of 85% adoption within a defined implementation horizon.

Business intelligence and analytics capabilities display different levels of development. Auroflex has deployed Microsoft Power BI to visualize operational and financial data, improving on previous manual reporting methods. However, usage of these tools is still relatively low. Survey data indicates that approximately 30-35% of employees regularly use Power BI for decision support, with adoption concentrated among management and administrative functions. Production personnel, despite having potential to benefit from real-time performance dashboards, exhibit lower engagement.

Several factors limit analytics adoption. First, data quality issuesreduce trust in the data. Internal assessments suggest that data quality stands at approximately 70%, implyinga significant share of records contain errors. Second, digital competency levels among staff are modest. Employee surveys administered as part of this study yielded an average digital competency score of 2.98 out of 5.00, indicating that many personnel lack the skills necessary to independently formulate queries,

interpret visualizations, or troubleshoot data discrepancies.

The gap is also visible in systems integration. Auroflex operates multiple software platforms (ERP, MES, nascent e-commerce, Power BI) thatare only partially integrated. Empirical evidence gathered through interviews indicates that systems integration currently stands at approximately 50%, meaning that substantial manual intervention is required to reconcile data across platforms. The Systems Integration Index incorporated in the Digital Balanced Scorecard targets 80% integration, reflecting the key strategic objective to move toward a more unified digital architecture.

4.5 Financial performance

The analysis of Auroflex's economic and financial performance draws upon official balance sheet data for the period 2020-2024, supplemented by provisional figures for the first half of 2025. These data, sourced from Cerved and Centrale dei bilanci, provide a longitudinal perspective on the firm's profitability, operational efficiency, and financial structure.

	Revenue/value of Production	**Costs**	**Equity**
2020	3834	3519	1064
2021	4616	4296	1317
2022	5340	5100	1396
2023	5368	5041	1213
2024	5539	5495	1482
2025	3405	2908	1627

Table nr. 1: Revenue costs and equity. Source: Financial statements Auroflex srl

	Percentage variation in revenue	Percentage variation in costs on value production	Percentage variation in equity
2020	3.97%	91.78%	27.73%
2021	17.13%	93.07%	23.78%
2022	14.25%	95.51%	6.00%
2023	-4.74%	93.91%	-13.11%
2024	17.41%	99.21%	22.18%

Table nr. 2: Financial variations. Source: Financial statements Auroflex srl

	ROI
2020	2.5
2021	0.2
2022	NEGATIVO
2023	NEGATIVO
2024	12.44
2025 half year	10.73

Table nr. 3: ROI. Source: Financial statements Auroflex srl

EBIT / Sales revenue per 100 = ROS

	ROS
2020	9.50%
2021	10.80%
2022	5.20%
2023	3.70%
2024	9.73%
2025 half year	6.19%

Table nr. 4: ROS. Source: Financial statements Auroflex srl

	Employed sistems	**EBITDA**	**Percentage**
2020	**AUTOMATION ENGINE**	590	15.99%
2021	**AUTOMATION ENGINE**	559	12.93%
2022	**AUTOMATION ENGINE/CERM**	516	10.45%
2023	**SISTEMI DI AUTOMAZIONE PER RACCOLTA DATI**	544	11.56%
2024	**POWER BI e AUROFLEX BMO CRM**	933	16.89%
2025 half year	**POWER BI e AUROFLEX BMO CRM**	453	15.33%

Table nr. 5: Ebitda.Source: Financial statements Auroflex srl

Revenue evolution reflects considerable volatility. The value of production increased from €3.83 million in 2020 to €5.54 million in 2024, representing

aggregate growth of approximately 44.6% over the four-year period. However, this trajectory was non-linear, with substantial growth in 2021 (+20.4%) and 2022 (+15.7%), followed by marginal contraction in 2023 (-0.5%) and recovery in 2024 (+3.2%).

- Return on Investment (ROI) exhibits significant fluctuation across the observation period. For Auroflex, ROI stood at 2.5% in 2020, declined sharply to 0.2% in 2021, turned negative in 2022 and 2023, and rebounded to 12.44% in 2024. The first semester 2025 figure of 10.73% suggests sustained recovery. The negative ROI recorded in 2022 and 2023 reflects depreciation burdens from equipment investments and margin compression from input cost inflation. The recovery in 2024 ROI to 12.44% approaches the strategic target of ≥12% established during the December 2025 balanced scorecard workshop.
- Return on Sales (ROS), measuring operating profit as a percentage of revenues, provides complementary insight into operational efficiency. Auroflex's ROS ranged from a high of 10.8% in 2021 to a low of 3.7% in 2023, with a 2024 value of 9.73%. The provisional first semester 2025 ROS of 6.19% suggests some moderation relative to the full-year 2024 performance. The strategic target of ROS ≥10% has been achieved in certain years but not consistently sustained.
- EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization) represents an important performance indicator for capital-intensive industries, as it approximates cash-generating capacity from core operations. Auroflex’s EBITDA figures reveal relative stability compared to net profit metrics. EBITDA margins ranged from 10.45% in 2022 to 16.89% in 2024, with the first semester 2025 margin at 15.33%. The 2024 EBITDA margin of 16.89% closely aligns with the balanced scorecard target of ≥18%, representing substantial progress toward the strategic objective.

The progression of EBITDA margins over time also appears to be linked to tools for performance monitoring. Internal materials document that Auroflex transitioned from basic automation engines (2020-2021) to more sophisticated analytics

platforms (Power BI and CRM from 2024), during the same period in wich margins improved. A direct causal relationship cannot be statistically proven from observational data alone; the overlap in timing suggests that enhanced data visibility may have contributed to improved operational decision-making.

The Cash Conversion Cycle within the Balanced Scorecard and Financial Sustainability Perspective.

(data derived from financial statements)

Fiscal Year 2022

Data Used

- ➔ Inventories: €486,520
- ➔ Trade receivables: €2,084,125
- ➔ Trade payables (estimated within short-term liabilities): €2,748,781
- ➔ Sales revenues: €4,938,086
- ➔ Cost of goods sold (raw materials + services + use of third-party assets): €2,017,762 + €1,290,830 + €380,979 = €3,689,571

- DIO Calculation – 2022
  DIO 2022 = (486,520 / 3,689,571) × 365
  DIO 2022 = 48.1 days ≈ 48 days

- DSO Calculation – 2022
  DSO 2022 = (2,084,125 / 4,938,086) × 365
  DSO 2022 = 153.9 days ≈ 154 days

- DPO Calculation – 2022
  DPO 2022 = (2,748,781 / 3,689,571) × 365
  DPO 2022 = 272.0 days ≈ 272 days

- CCC Calculation – 2022

  CCC 2022 = 48 + 154 − 272

  CCC 2022 = −70 days

Fiscal Year 2023

Data Used

- ➔ Inventories: €470,000
- ➔ Trade receivables: €2,106,000
- ➔ Trade payables: €3,378,000
- ➔ Sales revenues: €4,704,000
- ➔ Cost of goods sold (raw materials + services + use of third-party assets): €3,960,000

- DIO Calculation – 2023

  DIO 2023 = (470,000 / 3,960,000) × 365

  DIO 2023 = 43.3 days ≈ 43 days

- DSO Calculation – 2023

  DSO 2023 = (2,106,000 / 4,704,000) × 365

  DSO 2023 = 163.3 days ≈ 163 days

- DPO Calculation – 2023

  DPO 2023 = (3,378,000 / 3,960,000) × 365

  DPO 2023 = 311.3 days ≈ 311 days

- CCC Calculation – 2023

  CCC 2023 = 43 + 163 − 311

  CCC 2023 = −105 days

Fiscal Year 2024

Data Used

- ➔ Inventories: €530,000

➔ Trade receivables: €2,683,000

➔ Trade payables: €3,088,000

➔ Sales revenues: €5,523,000

➔ Cost of goods sold (raw materials + services + use of third-party assets): €4,291,000

- DIO Calculation – 2024
  DIO 2024 = (530,000 / 4,291,000) × 365
  DIO 2024 = 45.1 days ≈ 45 days

- DSO Calculation – 2024
  DSO 2024 = (2,685,000 / 5,523,000) × 365
  DSO 2024 = 177.0 days ≈ 177 days

- DPO Calculation – 2024
  DPO 2024 = (3,088,000 / 4,291,000) × 365
  DPO 2024 = 262.6 days ≈ 262 days

- CCC Calculation – 2024
  CCC 2024 = 45 + 177 − 262
  CCC 2024 = −40 days.

Working capital management indicators

Fiscal Year	DIO (days)	DSO (days)	DPO (days)	CCC (days)
2022	48	154	272	-70
2023	43	163	311	-105
2024	45	177	262	-40

Table nr.6: Working capital management indicators. Source: own elaboration

Interpretation of the Cash Conversion Cycle (CCC)

The Cash Conversion Cycle (CCC) is a comprehensive indicator of a firm's ability

to manage operating working capital, measuring the number of days between the cash outflow for the acquisition of production inputs and the cash inflow generated from sales revenues.

The CCC is calculated as follows:

$CCC = DIO + DSO - DPO$

where:

- DIO (Days Inventory Outstanding) measures the average number of days inventory is held before being sold.
- DSO (Days Sales Outstanding) measures the average collection period of trade receivables.
- DPO (Days Payable Outstanding) measures the average payment period granted by suppliers.

The consistently negative CCC indicates that Auroflex benefits from favorable working capitalcondition, receiving payments from customers more slowly than it settles obligations to suppliers, but with supplier payment terms sufficiently extended to generate a net cash inflow from operations. This configuration provides natural cash-flow advantage, reducing the need for external working capital financing. However, the deterioration of CCC from -105 days in 2023 to -40 days in 2024should be monitored carefully, primarily driven by increased DSO and decreased DPO.

In the context of the Digital Balanced Scorecard, the CCC target was established at ≤120 days. While the current negative values are well within this threshold, the trend toward less favorable working capital positions warrants monitoring. The implementation of an integrated CRM system, proposed as a strategic initiative, could enhance receivables management through systematic tracking of payment patterns and proactive follow-up on overdue accounts.

Overall, the financial performance analysis reveals an enterprise that has navigated significant volatility, absorbing the impacts of capital investment cycles, input cost shocks, and competitive pressures, while progressively converging toward strategic profitability targets. The recovery in ROI, ROS, and EBITDA margins during 2024

provides support to the strategic priorities outlined in the Digital Balanced Scorecard.

4.6 Snapshot implementation roadmap

The development of a performance measurement framework for Auroflex requires careful planning of implementation steps. Based on empirical findings from interviews, workshop deliberations, and assessment of current capabilities, a phased roadmap is proposed to guide the digital transformation journey.

The first phase, designated as foundational infrastructure establishment (0-6 months), prioritizes addressing the most critical capability gaps. Key initiatives include: finalization and launch of the B2B e-commerce platform, with particular attention to user experience optimization and ERP integration; implementation of an integrated CRM system to consolidate customer data and enable systematic interaction tracking; enhancement of data quality through standardization of data entry protocols and validation rules; and establishment of baseline measurements for all key performance indicators incorporated in the Digital Balanced Scorecard.

The prioritization of these initiatives is based on empirical insights from the case study. Interview evidence indicates that data fragmentation and quality deficiencies constitute the main obstacles to operational decision. Without reliable, integrated data, even sophisticated analytics tools generate limited value. Similarly, the e-commerce platform represents both a diversification tool and a critical data source for understanding customer behavior in digital channels.

The second phase, characterized as capability building and process digitalization (6-18 months), shifts emphasis toward incorporating digital tools into regular workflows and developing organizational competencies. Main actions include: expansion of Power BI adoption from the current 30-35% to the target 70%, achieved through structured training programs and creation of role-specific dashboards; digitalization of core workflows, particularly order processing and artwork management; systematic collection and analysis of digital channel metrics (such as e-commerce conversion rates and customer acquisition costs); and initiation of integration projects aimed at reaching the 80% target.

Building these capabilities requires ongoing investment in employee training. The

digital competency assessment conducted as part of this study identified an average score of 2.98 out of 5.00. Raising competencies to the target level of 3.80 requires differentiated training paths: advanced analytics skills for management and administrative staff, and customer-facing digital tools proficiency for sales functions. The balanced scorecard framework establishes a target of ≥16 hours of digital training per employee per year.

The third phase, focused on optimization and continuous improvement (18-36 months), assumes that foundational infrastructure and basic competencies are in place. Key objectives include: achievement of financial performance targets (ROI ≥12%, ROS ≥10%, EBITDA ≥18%); attainment of digital revenue objectives (15-20% of total revenues from digital channels); realization of process efficiency gains, including artwork-to-print lead time reduction from 15 to ≤10 days and scrap rate reduction from 15% to ≤10%; and operationalization of the full Digital Balanced Scorecard with monthly performance reviews embedded in management routines.

This phased approach corresponds to patterns described in the digital transformation literature. Studies on Industry 4.0 adoption in SMEs suggest that the interval between initial technology investment and observable financial impact typically ranges from 18 to 36 months. The emphasis on leading indicators during early phases addresses the risk of losing momentum when lagging financial indicators have not yet responded.

Workshop discussions with Auroflex owners revealed several risks that need to be addressed. First, employee resistance to digital tools could undermine adoption rates. Mitigation strategies include explicit support from leadership and showing early results to build confidence. Second, integration complexity across legacy systems poses technical challenges. Gradual integration, starting with highest-priority data flows, can manage this risk. Third, the potential for e-commerce to disrupt relationships with strategic accounts prompted the decision to grant key account managers veto authority over automated ordering for their assigned clients.

4.7 Operational & digital KPI dashboard

The operational and digital KPI dashboard represents a synthesized performance

monitoring instrument designed to provide management with actionable insights into both traditional operational metrics and emerging digital capabilities. The dashboard structure emerged through an iterative process involving initial indicator brainstorming, AI-assisted analysis of interview transcripts, and validation workshops with ownership and functional managers.

The initial brainstorming session generated a candidate list of 45 potential KPIs spanning financial, customer, process, and learning dimensions. Recognizing that excessive indicators would create information overload and dilute managerial focus, a selection process followed. AI tools (specifically Deepseek and ChatGPT) were employed to analyze interview transcripts and identify recurring themes concerning digital barriers, transformation drivers, and measurement expectations. This analysis achieved approximately 85% concordance with manual validation, accelerating the pattern recognition process without compromising analytical reliability.

The AI-assisted analysis also proposed aggregated indicators. For instance, a Digital Adoption Index was suggested to synthesize Power BI usage rates, CRM adoption levels, and e-commerce proficiency scores into a single metric. During validation workshops, managers expressed concern that composite indices, while theoretically appealing, make it harder to identify the specific source of the problem. As one manager noted, if the Operational Excellence Index declines, actionable response requires knowing whether the problem stems from scrap rates, lead times, or integration shortfalls. Consequently, composite indices were retained selectively, primarily within the learning and growth perspective where they support skill development tracking, while process and customer metrics remained disaggregated to maintain clarity at the operational level.

Further refinement concerned adjustment of performance threshold. The initial prototype employed standardized threshold criteria: green status for achievement above 90% of target, yellow for 75-90%, and red below 75%. Managers challenged this uniformity, arguing that different KPIs exhibit varying degrees of difficulty and strategic significance. For example, reducing scrap rates from 15% to 10% requires years of systematic process improvement and capital investment, achieving 12% merits recognition rather than yellow-flag caution. Conversely, if 90% of

employees fail to adopt CRM, this constitutes a critical gap requiring immediate attention. The resolution involved establishing KPI-specific thresholds calibrated to sector benchmarks, internal capability assessments, and strategic priorities.

The dashboard incorporates both leading and lagging indicators. Leading indicators, such as CRM adoption rates, digital training hours, and data quality scores, anticipate improvements before financial results become visible. Lagging indicators, including ROI, ROS, and revenue growth, measure ultimate value creation but react more slowly to operational interventions. The balanced inclusion of both indicator types enables management to monitor progress along the transformation path while maintaining accountability for financial results.

Operational efficiency metrics within the dashboard focus on operational metrics essential for competitiveness in the premium label segment. Artwork-to-print lead time, currently estimated at 15 days, represents a key bottleneck constraining responsiveness to client requests. The target reduction to ≤10 daysdepends on further digitalization, particularly automation of artwork approval processes and integration between design systems and production planning. Scrap rate, reported at approximately 15%, is another critical area, with target reduction to ≤10% requiring enhanced quality control, operator training, and real-time monitoring of printing parameters.

Digital channel metrics address the e-commerce platform and web-based customer engagement. As the platform has not yet launched, baseline values for e-commerce conversion rate, customer acquisition cost, and digital customer lifetime value are unavailable. Targets were established through reference to external benchmarks (Shopify data for B2B platforms) and internal aspirations. The e-commerce conversion rate target of 3-5% reflects realistic expectations for initial platform performance, acknowledging that customer onboarding, user experience optimization, and marketing refinement require repeated adjustment.

The dashboard governance structure assigns ownership responsibility for each KPI to specific functional managers. The Production Manager holds accountability for scrap rates, lead times, and Overall Equipment Effectiveness (OEE). The Commercial Manager owns customer satisfaction, retention, and digital revenue metrics. The Administrative Manager oversees financial KPIs. The four proprietors

collectively monitor ROI of digital investments and strategic initiative progress. A monthly leadership meeting routine dedicates the initial 15 minutes to dashboard review, trend analysis, variance discussion, and corrective action planning, institutionalizing data-driven performance dialogue.

4.8 Customer-behaviour insights

Understanding customer behavior and satisfaction is a key aspect of performance measurement for premium label converters, where relationship quality and service perception directly affect client retention. Evidence regarding Auroflex customer behavior was gathered through two principal methods: structured satisfaction surveys administered to a sample of active clients, and analysis of transactional patterns derived from ERP data and sales records.

The customer satisfaction survey, conducted as part of this research initiative, yielded responses from a diverse group of client accounts. The survey employed a five-point Likert scale to assess satisfaction across multiple dimensions including product quality, delivery reliability, technical support, responsiveness, and value for money. The aggregate satisfaction score of 4.66 out of 5.00 indicates strong overall client approval, placing Auroflex within the upper range in the sector benchmarks where premium converters typically achieve scores in the 4.3-4.7 range.

Disaggregation of satisfaction scores reveals specific strengths and improvement opportunities. Product quality received the highest ratings, with clients particularly appreciating technical versatility in finishing processes and consistency in color reproduction. Delivery reliability scores were also elevated, reflecting the firm's emphasis on meeting committed deadlines despite the operational complexity of multi-process labels. Technical support and collaborative problem-solving received positive but slightly lower ratings, suggesting scope for enhanced pre-production consultation and design optimization services.

The survey also incorporated a Net Promoter Score (NPS) proxy question, inquiring about likelihood to recommend Auroflex to industry colleagues. The calculated NPS approximation of approximately 93 is well above typical benchmarks (which range from 30-50) and even surpasses premium service industry standards. This

exceptionally high score indicates strong client loyalty and suggests that Auroflex's customers perceive strong value in its target market. However, the strategic target of NPS ≥85 in the balanced scorecard framework acknowledges that maintaining this level requires continuous attention to quality, service, and relationship management.

Customer retention analysis, derived from ERP transaction histories, revealsclient relationships appear stable. While precise retention rate calculation awaits CRM system implementation to systematically track account status across fiscal periods, qualitative evidence from sales management indicates that core accounts exhibit multi-year tenure with Auroflex. This stability reflects both switching costs typical in label supply relationships (due to matching requirements, tooling investments, and approval process dependencies) and satisfaction with service delivery.

Transactional data shows concentration of revenues among a relatively small proportion of accounts, consistent with the Pareto principle common in B2B manufacturing. Approximately 20-25% of clients generate 70-75% of revenues, comprising established wineries, spirits brands, and cosmetics producers with substantial annual label requirements. These strategic accounts receive customized service including dedicated account management, priority scheduling, and collaborative product development. The remaining client base consists of smaller, more occasional buyers who value Auroflex's ability to accommodate short runs and specialized finishing but may exhibit less predictable ordering patterns.

The forthcoming e-commerce platformaims to attract lower-volume clients. Current minimum order quantities and manual quoting processes create barriers for boutique producers who require smaller label volumes but could generate profitable business if transaction costs were reduced through platform automation. The expected effects include broadening the client base, improving capacity utilization during off-peak periods, and capturing behavioral data (click-through patterns, cart abandonment rates, product preference signals) that manual sales processes cannot capture.

Customer-facing digital touchpoints beyond e-commerce are still limited. Auroflex lacks structured web analytics tracking of corporate website visitors, limiting visibility into prospecting behavior and information-seeking patterns. Email

marketing and digital content distribution are employed sporadically rather than as integrated components of client engagement strategy. Social media presence exists but receives minimal investment. These gaps offer room of development to enhance digital customer experience and generate data for relationship analytics.

The Digital Balanced Scorecard incorporates several customer-centric KPIs designed to monitor relationship health and digital engagement effectiveness. Beyond satisfaction scores and NPS, indicators include customer retention rate (targeting ≥85%), digital revenue percentage (targeting 15-20% within 24 months), repeat purchase rate via e-commerce (targeting ≥60%), and digital customer lifetime value. These metrics help assess whether digital transformation initiatives are enhancing customer value creation and relationship sustainability.

4.9 Digital balanced scorecard

The Digital Balanced Scorecard framework represents the main outcome of this empirical investigation, synthesizing strategic objectives, performance indicators, and implementation logic into structured performance systems. The development process, documented in the internal materials, proceeded through three phases: strategic objective formulation via workshop with ownership, AI-assisted indicator selection and structuring, and validation and refinement with functional management.

The December 2025 workshop with the four proprietors was used to clarify strategic priorities and define concrete targets across the four traditional Balanced Scorecard perspectives: financial, customer, internal processes, and learning and growth. The session lasted approximately three hours and employed facilitated discussion to gather different perspectives, reach agreement on contested priorities, and clarify cause-effect relationships. The discussion proved intensive, particularly regarding the balance between revenue growth and profitability enhancement, and the appropriate pace of e-commerce deployment.

On the financial perspective, the owners agreed on three primary objectives: achieving ROI of ≥12%, ROS of ≥10%, and EBITDA margin of ≥18%, alongside revenue growth of ≥8% annually. One proprietor emphasized that these targets must be achieved through quality and efficiency improvements rather than cost reduction

that compromises service levels, this condition shaped later process and capability targets. Additionally, a dedicated metric for digital investment ROI (targeting ≥15%) was included to monitor returns on technology investments.

The customer perspective emphasizes relationship quality and digital channel development. Strategic objectives include maintaining exceptionally high satisfaction and loyalty (NPS ≥85, satisfaction ≥4.7/5) while expanding revenue from digital channels to 15-20% of total within 24 months. The e-commerce platform is expected to open new market segments, particularly smaller producers unable to meet traditional minimum orders, without cannibalizing relationships with major accounts. To safeguard these relationships, key account managers retain veto authority over automated order processing for strategically important clients.

Internal process objectives focus on operational efficiency and digital workflow maturation. Key targets include: reduction of artwork-to-print lead time from 15 to ≤10 days through workflow automation; scrap rate reduction from 15% to ≤10% via enhanced quality control and operator training; achievement of ≥75% digital workflow coverage, minimizing manual paper-based processes; and systems integration reaching ≥80%, enabling seamless data flows among ERP, MES, CRM, and e-commerce platforms. These process improvements are recognized as prerequisites for achieving both customer satisfaction and financial performance targets.

The learning and growth perspective focusses on capability development for digital transformation. Objectives center on: expanding Power BI adoption from 30-35% to ≥70% of relevant staff; CRM adoption reaching ≥85%; digital competency scores improving from 2.98 to ≥3.8 out of 5; and provision of ≥16 hours of digital training per employee annually. Data quality improvement, from the current 70% to ≥85%, is a prerequisite for all analytics-dependent initiatives. One proprietordescribe the cultural shift: transitioning from intuition-based decision-making to data-informed insights, from assuming customer preferences to validating them through data.

A notable feature of the Auroflex Digital Balanced Scorecard is the distribution of KPIs across perspectives. The final scorecard includes 30 indicators: 6 financial (20%), 8 customer (27%), 8 internal process (27%), and 8 learning and growth (27%). This distribution, with 80% of indicators in non-financial perspectives,

reflects a conscious strategic choice. Interview evidence and workshop discussions revealed that Auroflex's primary challenges are not financial liquidity or capital access but rather operational and cultural: fragmented systems, modest digital competencies, manual processes, and inconsistent data quality.

The preponderance of non-financial indicators aligns with academic literature on digital transformation in SMEs. Studies by Vial (2019), Naeem and Garengo (2022), and Ghobakhloo (2021) consistently demonstrate that in early-stage digitalization, enterprises must prioritize infrastructure, competencies, and process redesign, with financial returns materializing 12-24 months subsequently. Focusing too heavily on short-term financial indicators during this capability-building phase risks discontinuing initiatives too early. Conversely, monitoring capability development provides tangible evidence of progress and sustains organizational commitment.

The cause-effect logic underlying the Balanced Scorecard links learning and growth investments to process improvements, process improvements to customer value, and to financial performance. For instance, digital training programs enhance employee competencies, enabling more effective use of Power BI and CRM; these tools provide visibility into production inefficiencies and customer interaction patterns; insights drive process optimization and relationship management improvements; operational efficiency and customer retention strengthen revenue stability and margin enhancement; and ultimately, financial targets are achieved. This causal structure was explicitly mapped during the workshop, with linkages represented graphically toto align the owners.

The January 2026 validation workshop expanded participation to include Production Manager, Commercial Manager, and Administrative Manager, ensuring that those responsible for KPI execution had a voice in final design. This sessionled two key adjustments. First, managers challenged the utility of composite indices, preferring visibility into disaggregated components to enable specific interventions. Second, they advocated for KPI-specific performance threshold calibration, recognizing differential improvement difficulty and strategic importance across indicators.

Implementation governance assigns clear ownership: Production Manager for

manufacturing efficiency metrics, Commercial Manager for customer relationship indicators, Administrative Manager for financial KPIs, and collective proprietor responsibility for strategic digital investment ROI. A monthly leadership meeting routine dedicates opening minutes to scorecard review, establishing systematic performance dialogue. Dashboard visualizations employ color-coded status indicators (green, yellow, red) calibrated to KPI-specific thresholds, enabling rapid identification of areas requiring attention.

The scorecard relies on data from multiple systems: financial KPIs from official balance sheets and internal accounting; operational metrics from ERP (lead times), production reports (scrap rates, OEE), and CERM workflow system; customer indicators from satisfaction surveys, CRM (post-implementation), and e-commerce platform analytics; and learning metrics from system usage logs (Power BI, CRM adoption), HR training records, and employee skill assessments. The fragmentation of data sources confirms the need for stronger systems integration, which not only reduces manual data reconciliation but also enhances data reliability and timeliness. Baseline measurements and targets for each KPI were set using historical performance (where available), sector benchmarks (FINAT RADAR, Shopify B2B data), and targets agreed during workshops. For indicators lacking historical baselines due to nascent initiatives (e.g., e-commerce metrics), targetsreflect realistic but ambitious expectations, with the understanding that targets will be revised. The Digital Balanced Scorecard framework developed for Auroflex thus represents both a strategic management tool and an empirical contribution to academic understanding of performance measurement in digitally transforming manufacturing SMEs. It demonstrates the feasibility and value of adapting established frameworks (Kaplan and Norton's Balanced Scorecard) to contemporary challenges (Industry 4.0, digital channels, data-driven decision-making) within resource-constrained yet strategically ambitious enterprises. The implementation roadmap, capability development priorities, and measurement architecture collectively provide a replicable model for similar firms navigating the digital transformation journey in specialized manufacturing sectors.

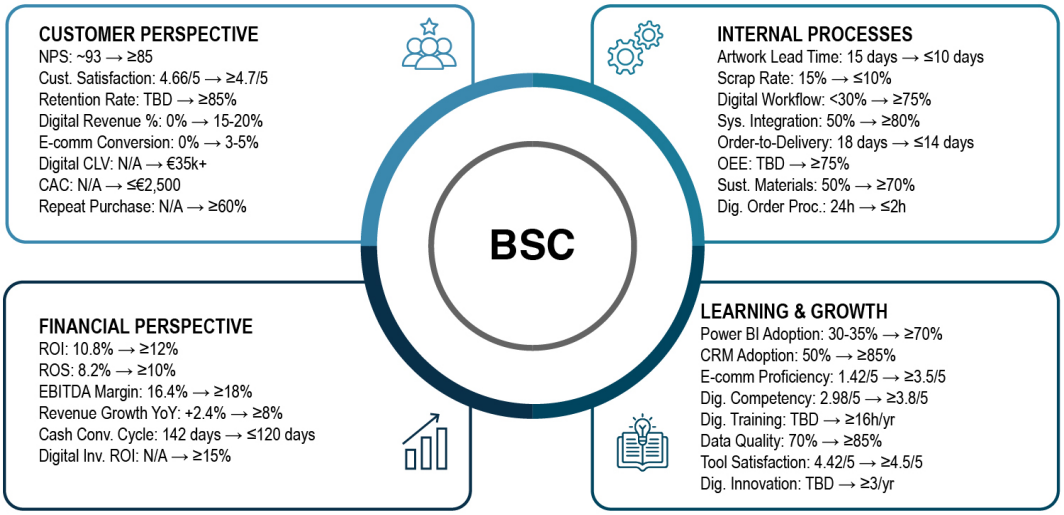


Figure nr. 8: BSC: own elaboration.

DISCUSSION AND CONCLUSIONS

This thesis investigated performance measurement in the context of digital transformation, focusing on the self-adhesive label manufacturing industry through a case study of Auroflex S.r.l., an Italian SME undergoing significant digital transformation. The research pursued four interconnected objectives: first, to define what "performance" means for a digital-first self-adhesive label manufacturer; second, to develop a comprehensive KPI set integrating classic financial ratios (ROI, ROS, EBITDA, Cash Conversion Cycle) with operational-digital metrics (website traffic, conversion rates, customer lifetime value, engagement indicators); third, to map Auroflex's digital maturity gaps and propose quick-win improvements; and fourth, to deliver a practical Digital Balanced Scorecard for strategic and operational decisionmaking. The methodological approach adopted an interventionist case study design (Jönsson & Lukka, 2006; Yin, 2018), characterized by the researcher's embedded organizational role within Auroflex.

This positioning provided exceptional access to longitudinal financial data, operational metrics from manufacturing execution systems, strategic documents, and semi-structured interviews with founders and managers. Data collection followed a triangulation strategy, integrating quantitative analysis of financial and operational KPIs, qualitative insights from digital competency surveys, and observational data from strategic meetings. The collaborative co-design methodology ensured the Digital Balanced Scorecard reflected both theoretical rigor and practical implementability, accommodating the asymmetric data availability profile characteristic of organizations at the threshold of major digital transitions. Given the interventionist nature of this study, it is appropriate to foreground practical operational contributions before theoretical advancements, reflecting engaged research's dual commitment to advance scholarly understanding while producing tangible organizational value.

The primary practical contribution is the development of a systemic, multidimensional vision of organizational performance through the Digital Balanced Scorecard framework. Prior to this intervention, Auroflex's performance monitoring was fragmented, emphasizing traditional financial metrics tracked through the ERP system while operational data from the manufacturing execution

system (CERM) and digital initiatives (Power BI, RPA bots, e-commerce platform) operated in isolation without systematic assessment of their collective impact.

The Digital Balanced Scorecard redefines this structure by establishing measurable relationships among the four perspectives. The financial perspective integrates classic indicators (ROI, ROS, EBITDA, CCC) with DuPont analysis, revealing how operational improvements translate into enhanced return on equity.

The customer perspective, primarily forward-looking given the e-commerce platform's pre-commercial status, establishes targets for digital channel adoption (15-20% revenue within 24 months), customer retention (85%), and NPS scores. The internal process perspective links operational efficiency indicators on-time delivery (baseline: 78%, target: 90%), production lead time, systems integration (baseline: 50%, target: 80%) to customer satisfaction and cost optimization. The learning and growth perspective connects organizational capabilities CRM adoption (baseline: 50%, target: 85%), digital competency (baseline: 2.98/5.00, target: 3.80/5.00), data quality (baseline: 70%, target: 90%) to operational and customer outcomes. This systemic vision enables management to understand performance as an connected framework where improvements in foundational capabilities impact sequentially through operational excellence to customer satisfaction and financial value creation. The BSC makes these causal relationships explicit, helps prioritize initiatives, and facilitating communication across organizational levels.

The BSC development process also catalyzed organizational learning across multiple levels, creating what can be characterized as a “beneficial data crisis” a structured impetus to formalize data collection, analysis, and sharing practices. At the strategic level, collaborative workshops necessitated explicit articulation of strategic priorities among the four founding partners, surfacing latent disagreements on revenue growth versus profitability, e-commerce deployment pace, and operational efficiency versus customer experience. The BSC framework provided a disciplined mechanism for negotiating trade-offs, producing consensus around objectives (15-20% digital revenue within 24 months while maintaining EBITDA >12%) that had been previously implicit or contested. At the managerial level, functional managers developed new capabilities: the financial manager learned

DuPont analysis, linking operational decisions to ROE; the production manager gained competency extracting MES efficiency metrics for proactive bottleneck identification; the commercial manager engaged with digital marketing metrics (conversion funnels, CAC, CLV) previously outside traditional B2B sales practices. These learning trajectories strengthen the organization's ability to make data-informed decisions. At the operational level, BSC data requirements led to more standardized procedures: production operators systematically logged times in MES; customer service documented interaction histories structurally; administrative staff developed protocols for ERP-to-Power BI data transformation, reducing errors and accelerating reporting. The BSC also fostered cross-functional communication through KPIs shared across departments, with regular review meetings creating formal occasions for information sharing and collaborative problem-solving.

While the Digital Balanced Scorecard was co-designed specifically for Auroflex, several insights may be relevant beyond this case for SMEs in traditional manufacturing sectors navigating digital transformation. First, managers should recognize that digital transformation performance cannot be fully assessed using financial metrics alone. Financial indicators provide only delayed signals; by the time underperformance becomes evident, issues may already be deeply rooted. The BSC approach, incorporating leading indicators (operational efficiency, customer satisfaction, organizational capabilities), enables earlier identification and faster corrective action. Second, digital maturity gaps often reside in organizational practices and competencies rather than technology availability.

Auroflex possessed sophisticated tools (ERP, MES, BI, RPA) but derived limited value due to low systems integration (50%), poor data quality (70%), and modest employee competency (2.98/5.00). "Quick win" initiatives targeting these organizational factors - data governance, training, cross functional communication - often yield greater near-term impact than additional technology investments. Managers should approach digital transformation as a set of initiatives that must be sequenced rather than scattered experimentation.

The BSC framework provided coherent logic ensuring basic prerequisites (data quality, system integration, competency) were addressed before customer-facing capabilities (e-commerce launch). structured roadmaps that sequence initiatives

according to dependencies and priorities prevent fragmented, under-coordinated investments.

Collaborative, interventionist approaches to performance system design yield frameworks with greater internal credibility and strategic alignment than off-the-shelf templates.

The co-design methodology balanced theoretical consistency with practical implementability, with the process itself becoming an occasion for organizational learning and strategic clarification.

Managers in niche B2B manufacturing should critically adapt digital transformation models developed for B2C or technology sectors. Sectoral characteristics-product customization, technical complexity, relationship-intensive sales, regulatory constraints - shape appropriate trajectories. Auroflex's cautious e-commerce approach, prioritizing robustness over rapid scaling, reflects strategic logic appropriate to its premium positioning. Finally, performance measurement during digital transformation requires explicit attention to temporal dynamics. The Auroflex BSC exhibits dual temporal orientation: retrospective (financial, partial operational) and prospective (customer). Managers should design frameworks accommodating both backward-looking accountability and forward-looking strategic guidance. From a theoretical perspective, this research contributes to two interconnected literature streams-performance measurement and digital transformation-while connecting these two research streams.

The research advances Balanced Scorecard literature by demonstrating how Kaplan and Norton's (1992; 1996) canonical framework can be adapted to capture digital transformation performance dimensions while preserving its fundamental four-perspective architecture. The Digital Balanced Scorecard integrates novel indicators - e-commerce conversion rates, digital revenue contribution, CRM adoption, systems integration indices, data quality, digital competency - into established perspectives without structural alteration.

This contrasts with literature proposing entirely new frameworks or additional "digital" perspectives, supporting the theoretical robustness. The research draws attention to a temporal issue not widely discussed: distinguishing retrospective diagnostic use from prospective strategic guidance. Traditional BSC

implementations rely on historical data across all perspectives; the Auroflex case reveals the necessity of asymmetric temporal orientations- extrospective where data permit (financial, operational) and prospective where transformation requires anticipatory target-setting (customer, partially learning/growth). This dual temporal configuration represents a crucial adaptation for organizations at phases of strategic and technological transition, a dimension rarely discussed in BSC studies.

The research also offers empirical insights on BSC implementation in SME contexts. While much BSC literature focuses on large corporations, this study demonstrates feasibility and pratical adjustments required by SMEs: deliberate KPI limitation, emphasis on simplicity and visual clarity, integration with existing data systems rather than presuming comprehensive infrastructure, and phased implementation roadmaps staging capability development over realistic timeframes.

The digital transformation literature (Vial, 2019; Kane et al., 2015) extensively examines digitalization's drivers, processes, and organizational consequences but devotes limited attention to measuring transformation success. Studies addressing measurement typically emphasize technology adoption metrics or operational efficiency indicators but rarely integrate these with customer outcomes or financial value creation within comprehensive performance architectures. Conversely, performance measurement literature developed sophisticated multidimensional frameworks (Neely et al., 1995, 2005; Van Aken et al., 2005) but much predates current digital transformation and inadequately addresses digitalization's specific measurement challenges.

Traditional PMS frameworks presume stable organizational boundaries and linear value chains, assumptions problematic for platform business models, ecosystems, and data-driven adaptation contexts.

This research offers a connection for demonstrating how an established performance framework (BSC) can systematically capture digital transformation's multidimensional performance implications, providing digital transformation literature with rigorous measurement methodology while contributing to the evolution into an increasingly strategic domain.

The Digital BSC provides measurable dimensions of "digital transformation

success", translating it into concrete indicators across financial, customer, operational, and organizational dimensions, establishing foundation for future empirical research. The research also extends performance measurement scholarship into self-adhesive label manufacturing, a context receiving limited academic attention despite economic significance.

Label converting combines traditional manufacturing (capitalintensive equipment, process control) with creative services (graphic design, client collaboration) and sophisticated digital capabilities (configurators, web-to-print, automated workflows). This hybrid profile generates distinctive measurement challenges: traditional manufacturing KPIs remain relevant but require complementation with design quality, customer co-creation, and digital channel performance metrics. By documenting performance system design tailored to this context, the research provides a foundation for examining digital transformation in niche manufacturing sectors. Findings suggest sectoral specificities - technical complexity, customer relationship nature, product customization degree, regulatory environment - substantively shaped appropriate digital trajectories and measurement priorities.

This sectoral contingency perspective adds nuance to existing studies, questioning universal model insufficiently attentive to industry dynamics. Moreover, the research demonstrates illutrates the potential of interventionist research (Jönsson & Lukka, 2006) for performance measurement system research. The embedded researcher role, collaborative co-design, and iterative refinement enabled deep insight into PMS design micro-processes, strategic priority negotiation, and organizational learning dynamics. These insights, accessible only through sustained engagement, enrich understanding of performance measurement systems as organizational tools shaped by context politically dynamics, cultural norms, and contextual constraints.

The research exhibits several limitations that limit the generalizability and depth of its contributions:

1). The single-case design constrains generalizability beyond Auroflex's specific context. The self-adhesive label sector's distinctive characteristics make it uncertain whether the Digital BSC framework requires substantial adaptation for other niche manufacturing sectors (flexible packaging, industrial coatings,

precision components). Comparative multi-sector case studies would identify sector-specific versus generalizable elements.

2). The research captures a specific temporal moment - the threshold between infrastructure deployment and commercial activation preventing empirical verification of whether the Digital BSC guides action effectively, facilitates learning, or improves outcomes once the e-commerce platform becomes operational. The proposed casual relationships (digital competency → systems integration → customer experience → financial results) require longitudinal analysis.

3). The study engages minimally with political and power dynamics shaping PMS design. The harmonious consensus-building may not represent more contested contexts where measurement becomes a site of political struggle over resources, priorities, or accountability.

4). The research accepts digitalization as a strategic imperative rather than interrogating whether digital transformation genuinely creates value versus representing institutionalized isomorphism driven by industry fads or consultant pressures. From an operational perspective, the absence of e-commerce transactional data prevented empirical analysis of customer perspective indicators (conversion rates, CAC, CLV, adoption rates), forcing reliance on prospective targets from industry benchmarks rather than organization-specific baselines.

Operational data quality from CERM is fragmented and non-systematic, limiting the precision of operational efficiency KPIs. Employee digital competency assessment via self-report questionnaire may suffer from social desirability bias. The single-organization focus does not allow systematic comparison with other organizations. Finally, the research timeline limited longitudinal analysis of performance indicator evolution or sustained BSC organizational impact.

These limitations suggest several directions for future resarch. Comparative multi-sector research examining digital performance measurement across niche manufacturing sectors would clarify which BSC elements are sector-specific and wich are generalizable, potentially propose classification models specifying how PMS should vary according to sectoral characteristics.

Longitudinal research tracking digital transformation performance over multi-year

periods would empirically test the proposed casual relationships and analyzing time gaps.

Research examining political and contested dimensions of digital performance measurement would provide important counterbalance to collaborative framings, exploring contexts where BSC design becomes a political struggle. Theoretical integration of performance measurement frameworks with emerging digital transformation concepts (digital maturity models, platform thinking, ecosystem orchestration, data-driven business models) would advance conceptual sophistication. Survey-based research examining digital performance practices across large manufacturing SME samples would establish measurement approach prevalence and identify common challenges.

Comparative implementation research across organizations at different digital maturity stages would reveal whether BSC frameworks should evolve fundamentally as digitalization progresses.

Action research co-designing digital performance systems in additional contexts - craft manufacturing, circular economy business models, family-owned businesses - would provide additional empirical evidence. Research on AI/ML integration into performance measurement systems represents an emerging area of research, examining how BSC frameworks should be adapted when algorithms monitor and respond to performance variations autonomously.

Longitudinal ethnographic research examining day-to-day practices through which PMS shape organizational attention, learning, and decision-making would reveal how managers actually use BSC data and how frameworks evolve informally through practice. In conclusion, this thesis demonstrated that digital transformation raises important questions about traditional performance measurement, requiring frameworks integrating financial accountability with operational efficiency, customer experience, and organizational capability development.

The Digital Balanced Scorecard developed collaboratively with Auroflex represents both practical management tools and theoretical contribution, showing how established frameworks can adapt to digitalization's different performance dimension while maintaining its core strategic structure.

For Auroflex, the research delivered systemic performance vision linking infrastructure investments to operational improvements, customer relationships, and financial value; structured implementation roadmap sequencing digital initiatives; one - page dashboard enabling transformation monitoring; and organizational learning catalyzed across strategic, managerial, and operational levels. For scholarly discourse, the research bridges performance measurement and digital transformation literatures, demonstrating how multidimensional frameworks operationalize digital transformation success, extends BSC research into underexplored sectoral contexts, and contributes methodologically by demonstrating interventionist scholarship value. As manufacturing industries globally navigate profound technological and organizational changes, the imperative for rigorous, multidimensional performance measurement will intensify. This research contributes one element - grounded in self-adhesive label converting but informed by broader theoretical principles - to the broader discussion of understanding how organizations understand and manage digital transformation outcomes.

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